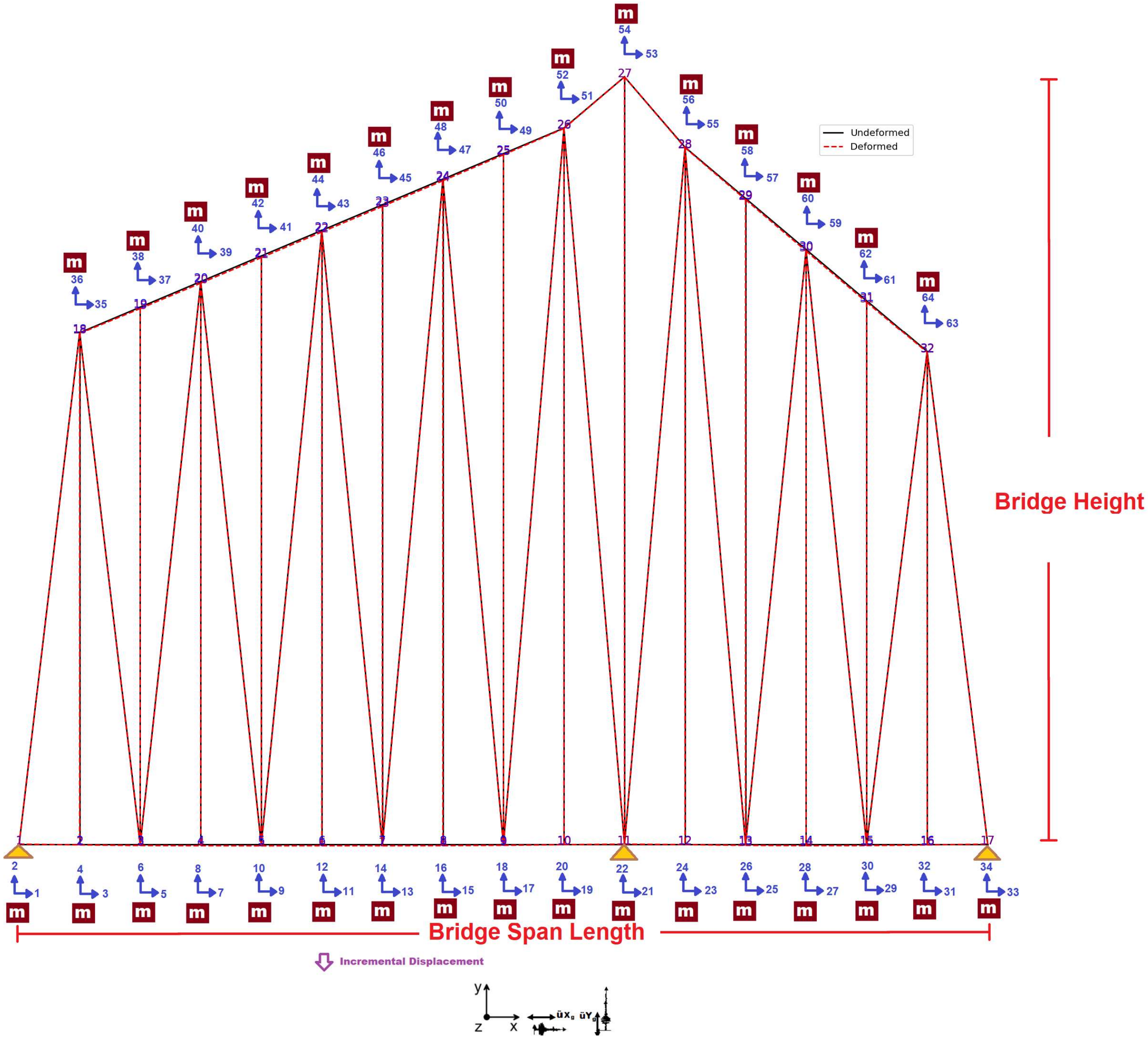
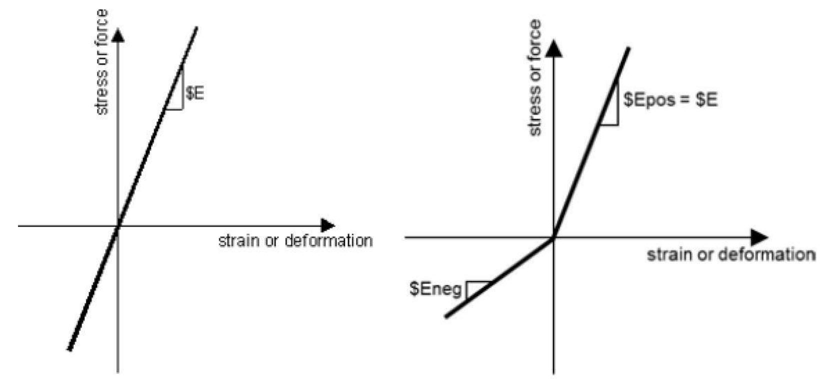
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

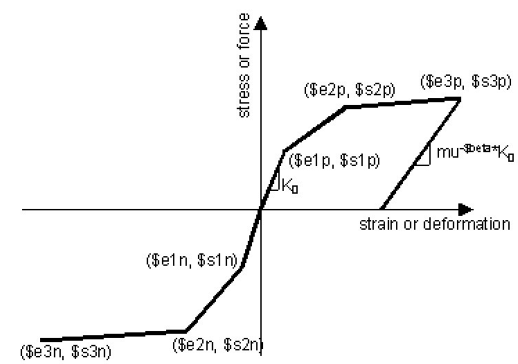
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL TRUSS BRIDGES: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)





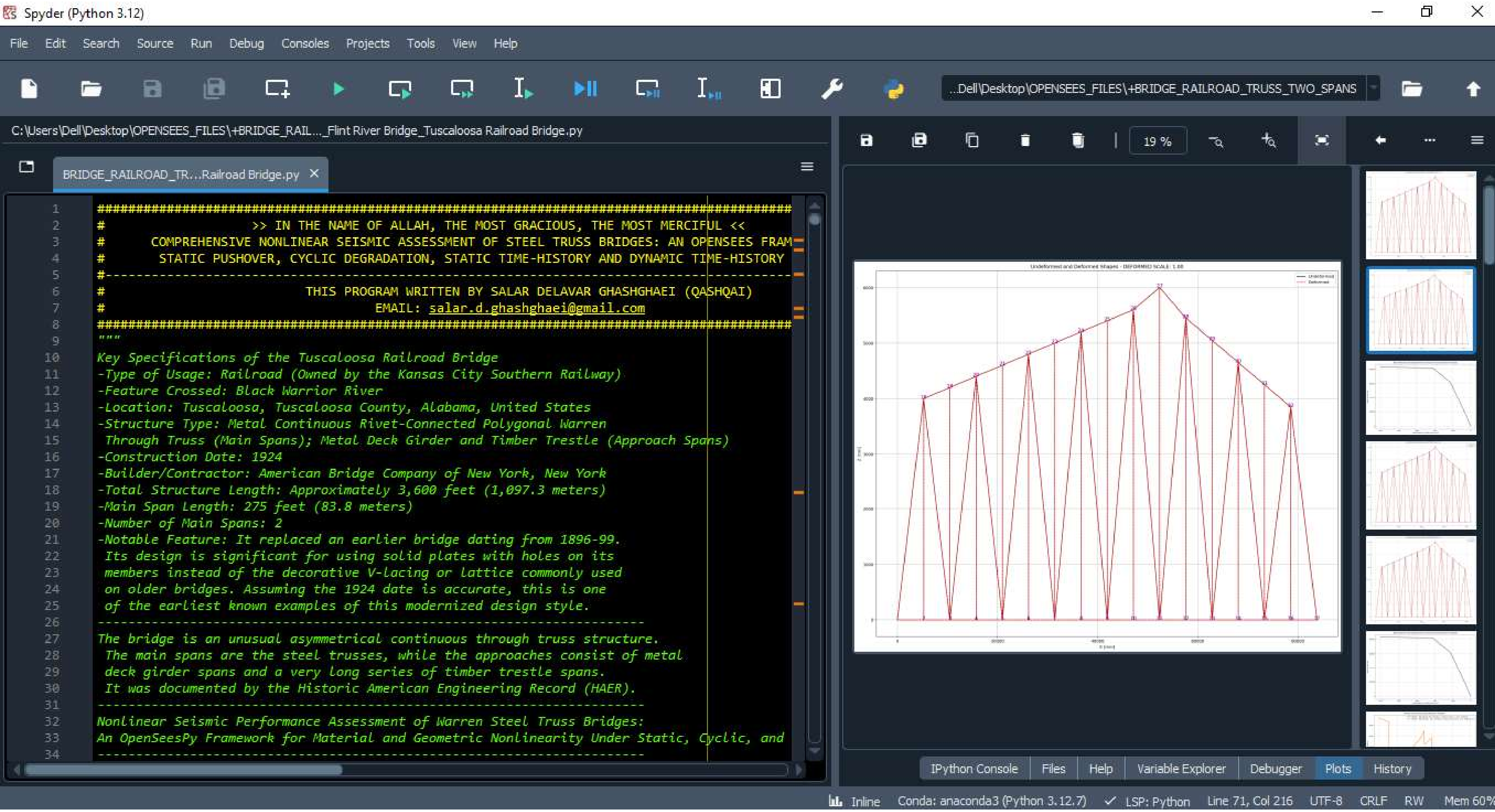
TRUSS ELEMENT ELASTIC MATERIAL



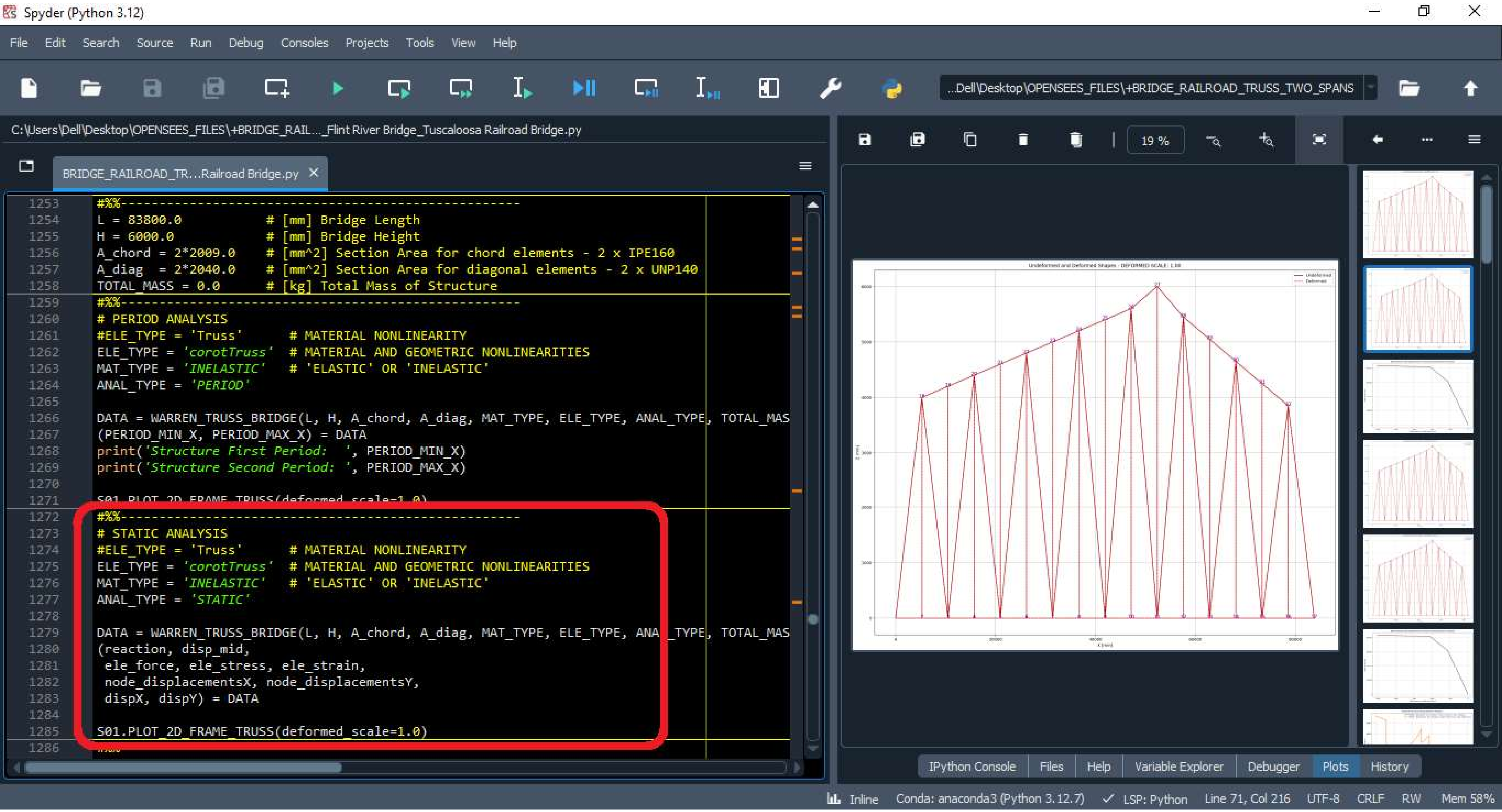
TRUSS ELEMENT INELASTIC MATERIAL

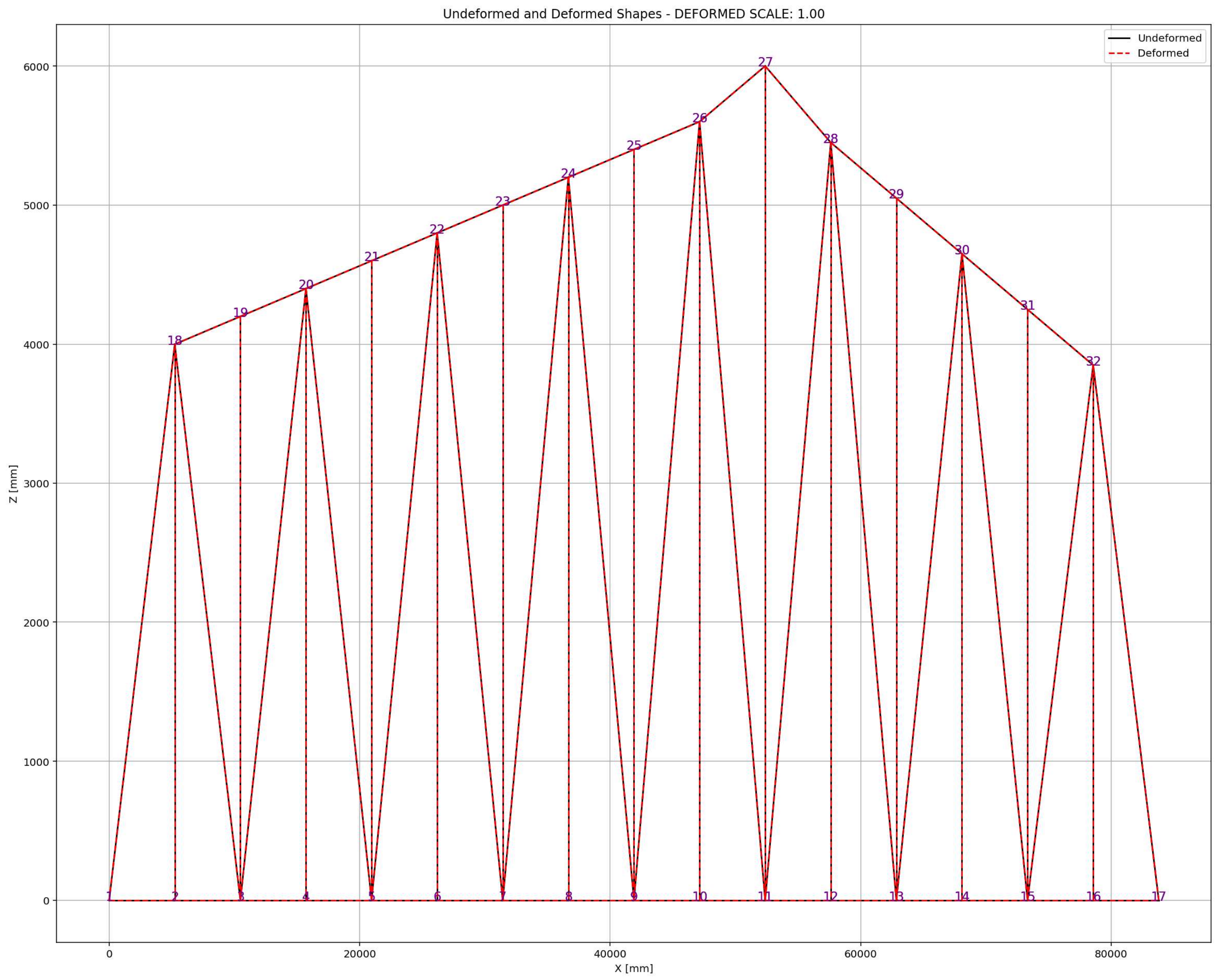
$$Axial\ Ductility\ Damage\ Index = \frac{\varepsilon_d - \varepsilon_y}{\varepsilon_u - \varepsilon_y}$$

$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAILROAD_TRUSS_TWO_SPANS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAIL..._Flint River Bridge_Tuscaloosa Railroad Bridge.py
BRIDGE_RAILROAD_TR...Railroad Bridge.py
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL TRUSS BRIDGES: AN OPENSEES FRAM
STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY
THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
Key Specifications of the Tuscaloosa Railroad Bridge
-Type of Usage: Railroad (Owned by the Kansas City Southern Railway)
-Feature Crossed: Black Warrior River
-Location: Tuscaloosa, Tuscaloosa County, Alabama, United States
-Structure Type: Metal Continuous Rivet-Connected Polygonal Warren
 Through Truss (Main Spans); Metal Deck Girder and Timber Trestle (Approach Spans)
-Construction Date: 1924
-Builder/Contractor: American Bridge Company of New York, New York
-Total Structure Length: Approximately 3,600 feet (1,097.3 meters)
-Main Span Length: 275 feet (83.8 meters)
-Number of Main Spans: 2
-Notable Feature: It replaced an earlier bridge dating from 1896-99.
 Its design is significant for using solid plates with holes on its
 members instead of the decorative V-lacing or lattice commonly used
 on older bridges. Assuming the 1924 date is accurate, this is one
 of the earliest known examples of this modernized design style.
The bridge is an unusual asymmetrical continuous through truss structure.
 The main spans are the steel trusses, while the approaches consist of metal
 deck girder spans and a very long series of timber trestle spans.
 It was documented by the Historic American Engineering Record (HAER).
Nonlinear Seismic Performance Assessment of Warren Steel Truss Bridges:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 71, Col 216 UTF-8 CRLF RW Mem 60%

STATIC ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAILROAD_TRUSS_TWO_SPANS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAIL..._Flint River Bridge_Tuscaloosa Railroad Bridge.py
BRIDGE_RAILROAD_TR...Railroad Bridge.py
#%%---
L = 83800.0   # [mm] Bridge Length
H = 6000.0   # [mm] Bridge Height
A_chord = 2*2009.0   # [mm^2] Section Area for chord elements - 2 x IPE160
A_diag   = 2*2040.0   # [mm^2] Section Area for diagonal elements - 2 x UNP140
TOTAL_MASS = 0.0   # [kg] Total Mass of Structure
#%%---
PERIOD ANALYSIS
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PERIOD'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(PERIOD_MIN_X, PERIOD_MAX_X) = DATA
print('Structure First Period: ', PERIOD_MIN_X)
print('Structure Second Period: ', PERIOD_MAX_X)
S01.PLOT_2D_FRAME_TRUSS(deformed_scale=1.0)
#%%---
STATIC ANALYSIS
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(reaction, disp_mid,
 ele_force, ele_stress, ele_strain,
 node_displacementsX, node_displacementsY,
 dispX, dispY) = DATA
S01.PLOT_2D_FRAME_TRUSS(deformed_scale=1.0)
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 71, Col 216 UTF-8 CRLF RW Mem 58%



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
1000
2000
3000
4000
5000
6000
20000
40000
60000
80000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
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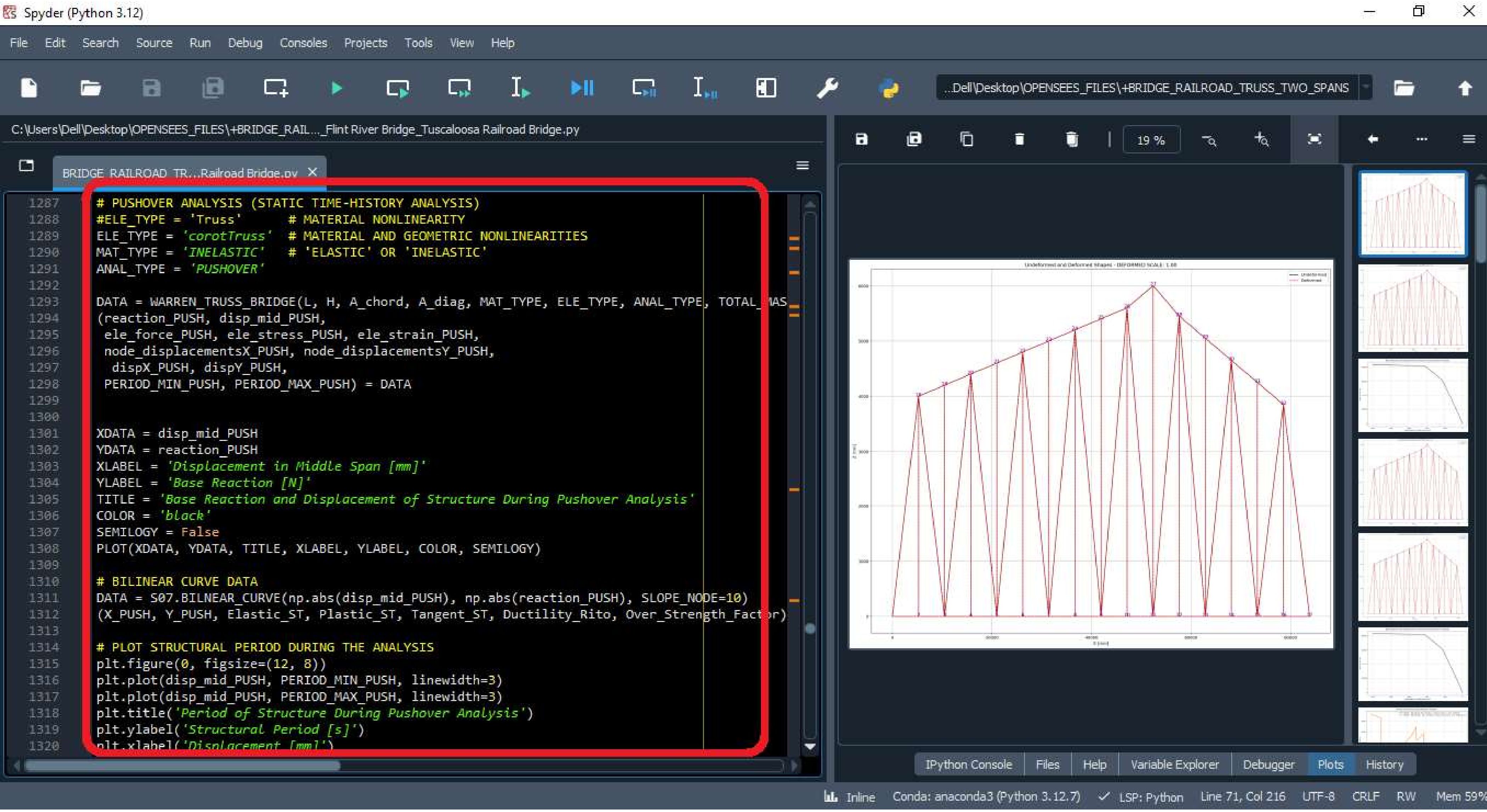
28
29
30
31
32

PUSHOVER ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAILROAD_TRUSS_TWO_SPANS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAIL..._Flint River Bridge_Tuscaloosa Railroad Bridge.py
BRIDGE_RAILROAD_TR...Railroad Bridge.py
PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(reaction_PUSH, disp_mid_PUSH,
 ele_force_PUSH, ele_stress_PUSH, ele_strain_PUSH,
 node_displacementsX_PUSH, node_displacementsY_PUSH,
  dispX_PUSH, dispY_PUSH,
 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
XDATA = disp_mid_PUSH
YDATA = reaction_PUSH
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Pushover Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# BILINEAR CURVE DATA
DATA = S07.BILNEAR_CURVE(np.abs(disp_mid_PUSH), np.abs(reaction_PUSH), SLOPE_NODE=10)
(X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_Strength_Factor)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_PUSH, PERIOD_MIN_PUSH, linewidth=3)
plt.plot(disp_mid_PUSH, PERIOD_MAX_PUSH, linewidth=3)
plt.title('Period of Structure During Pushover Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
19 %
IPython Console
Files
Help
Variable Explorer
Debugger
Plots
History
Inline
Conda: anaconda3 (Python 3.12.7)
LSP: Python
Line 71, Col 216
UTF-8
CRLF
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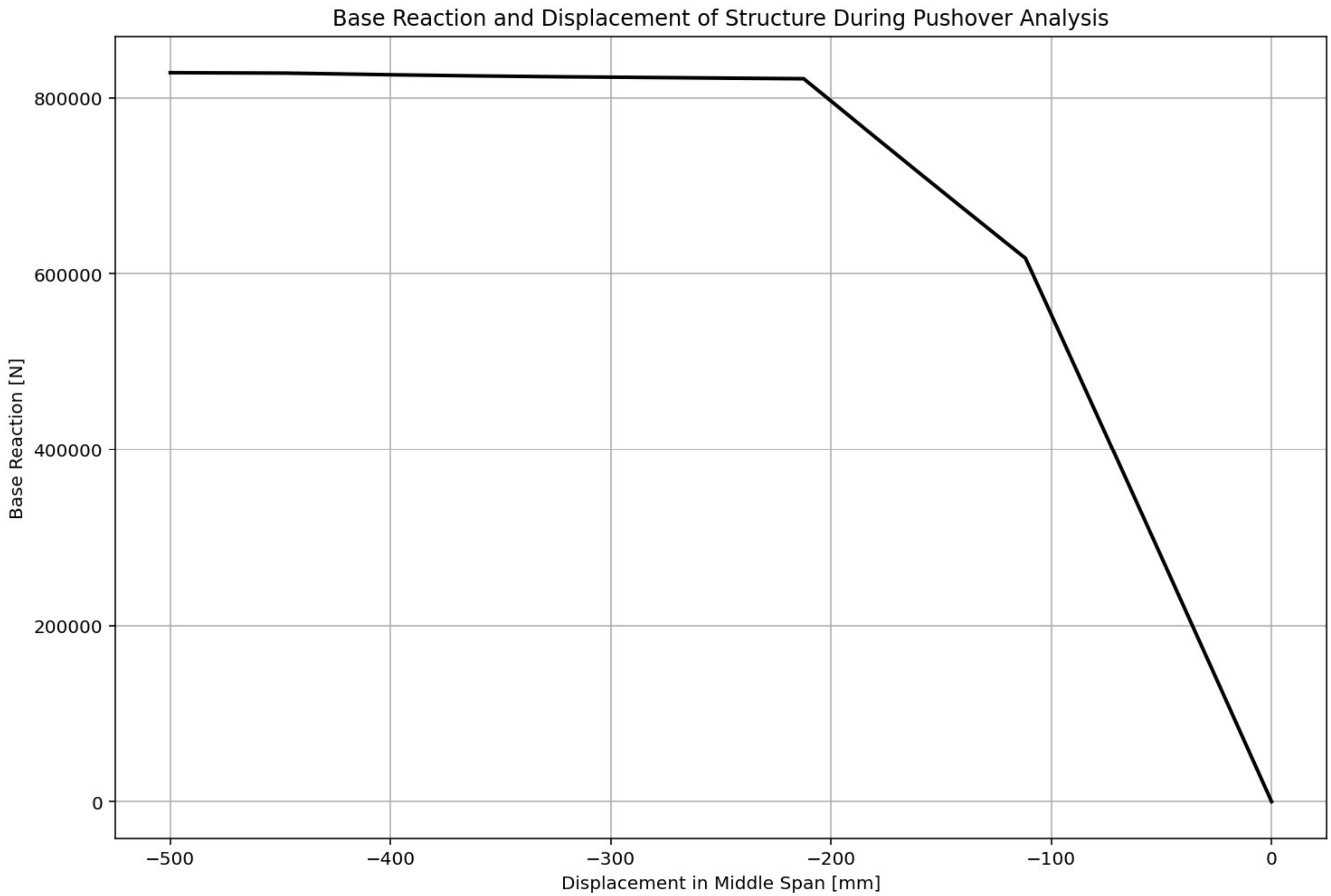
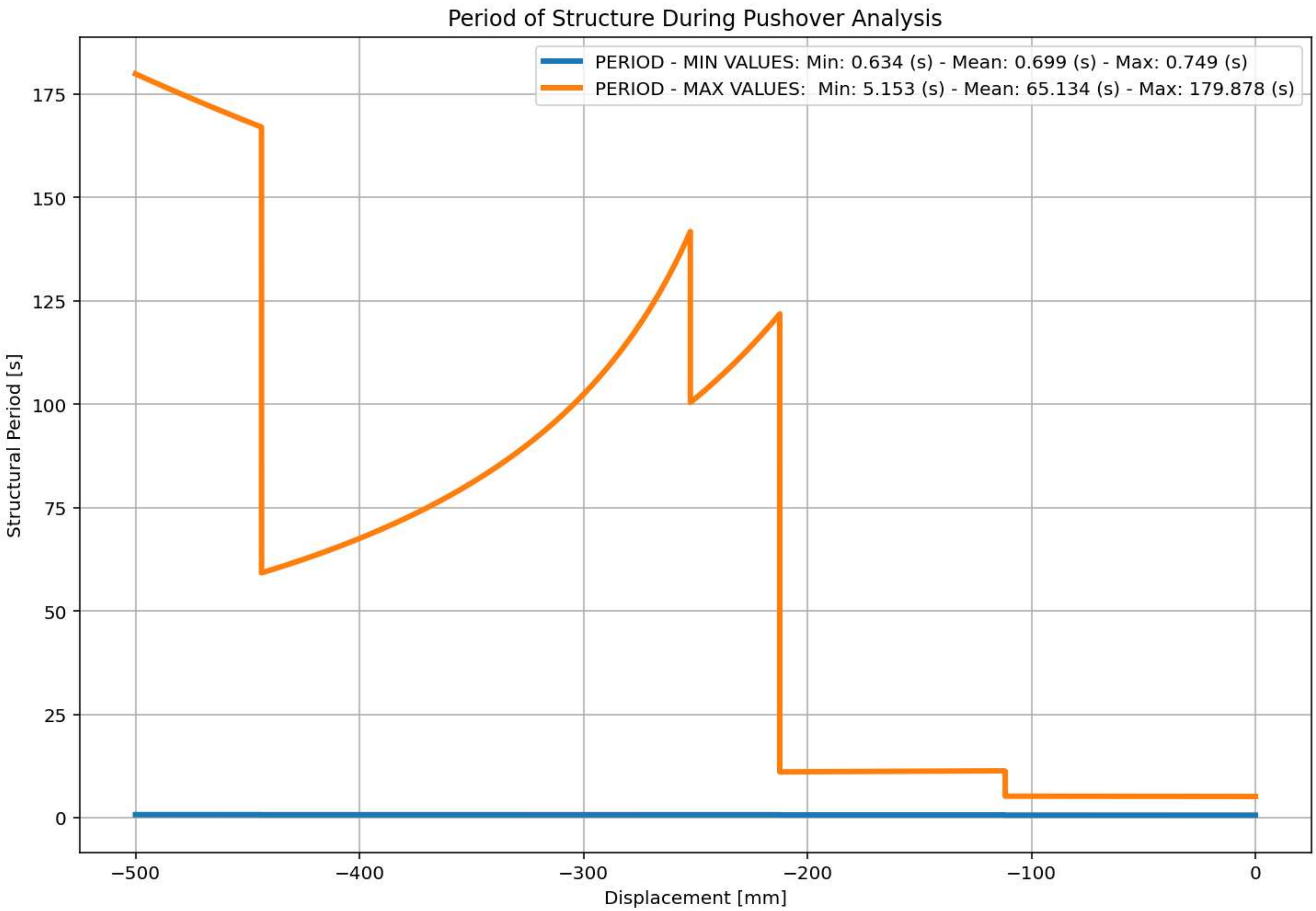
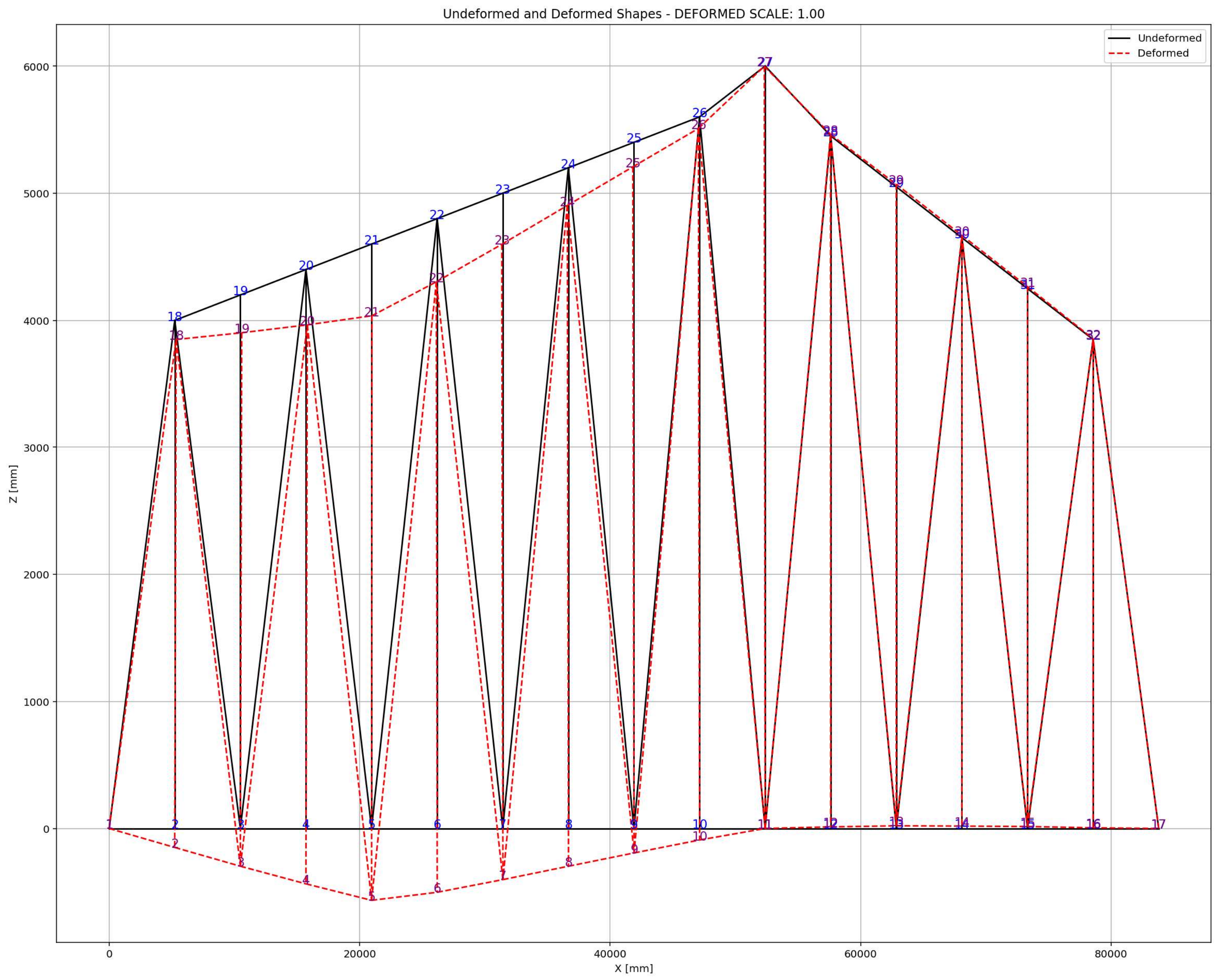
Mem 59%


Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
800000
600000
400000
200000
0
−500
−400
−300
−200
−100
0
Displacement in Middle Span [mm]

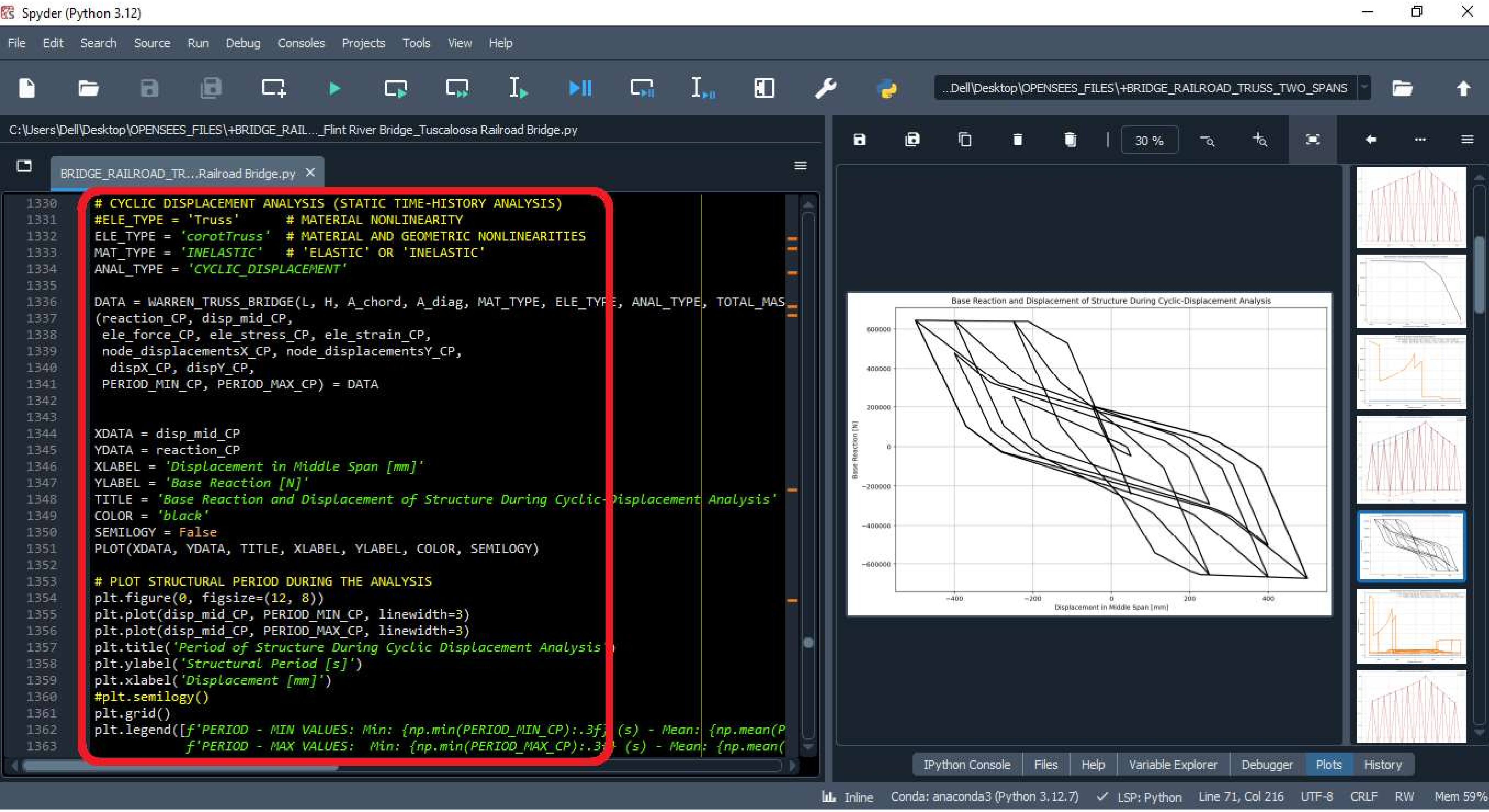


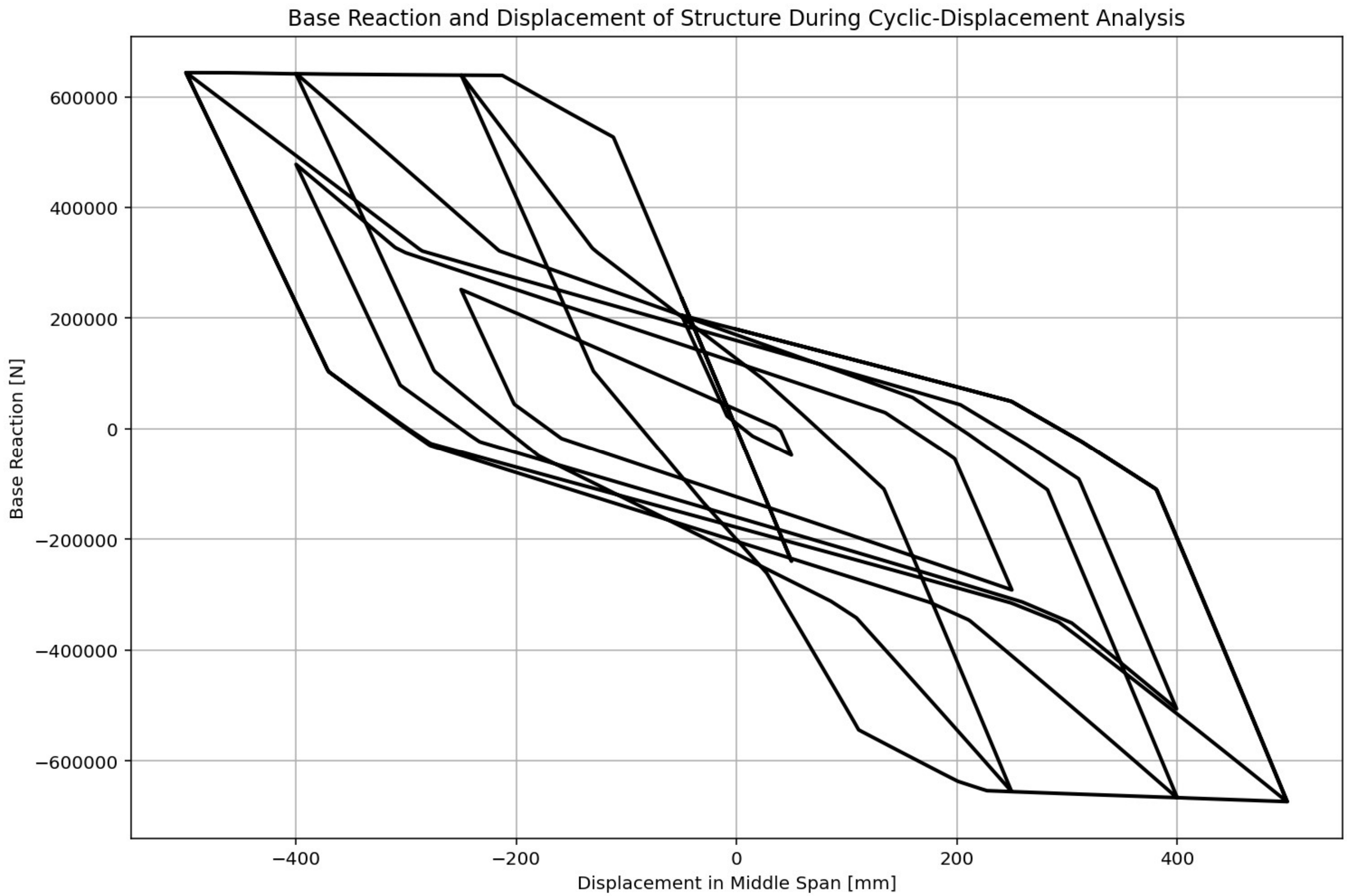
Period of Structure During Pushover Analysis
PERIOD - MIN VALUES: Min: 0.634 (s) - Mean: 0.699 (s) - Max: 0.749 (s)
PERIOD - MAX VALUES: Min: 5.153 (s) - Mean: 65.134 (s) - Max: 179.878 (s)
Structural Period [s]
Displacement [mm]
175
150
125
100
75
50
25
0
−500
−400
−300
−200
−100
0

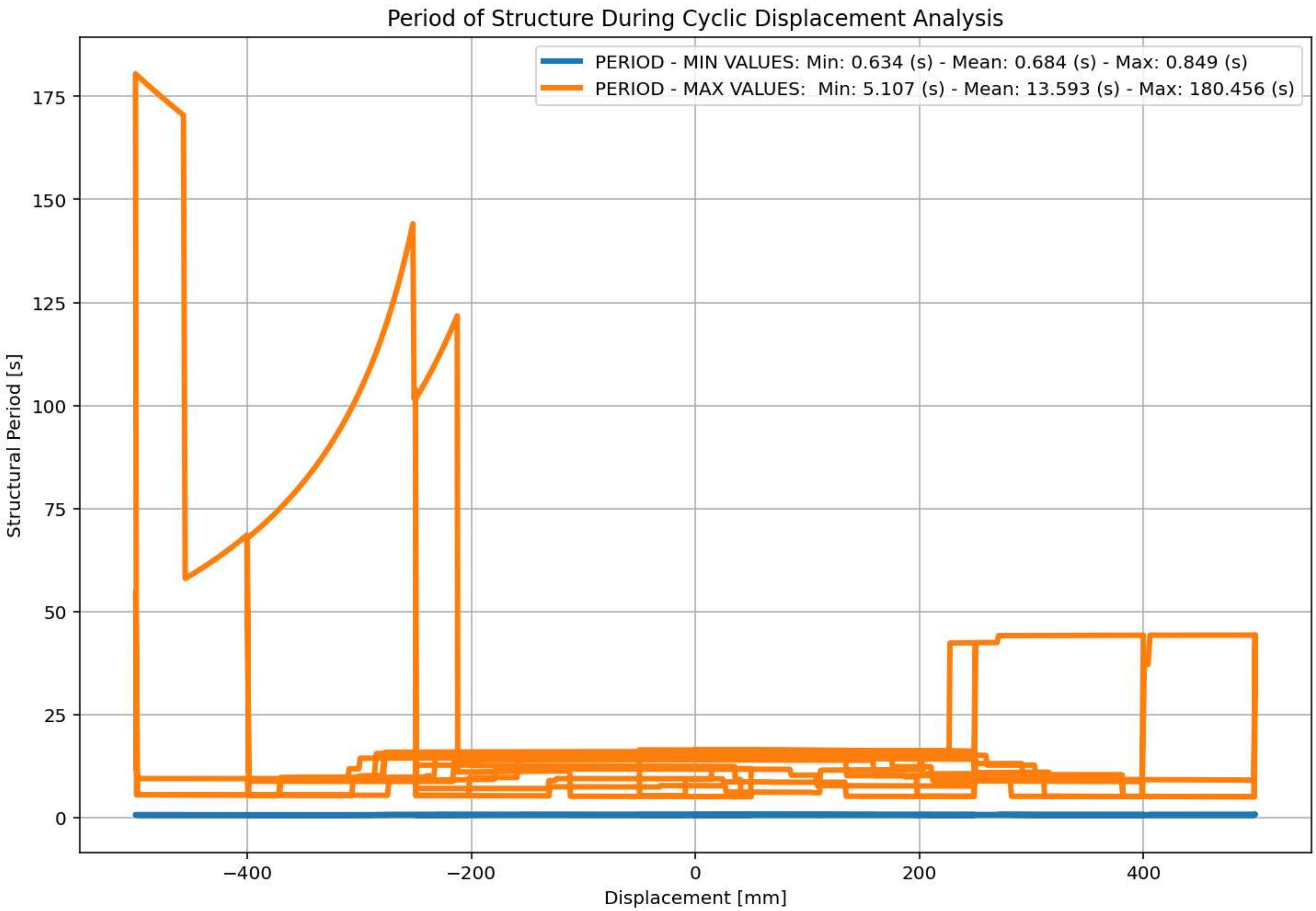


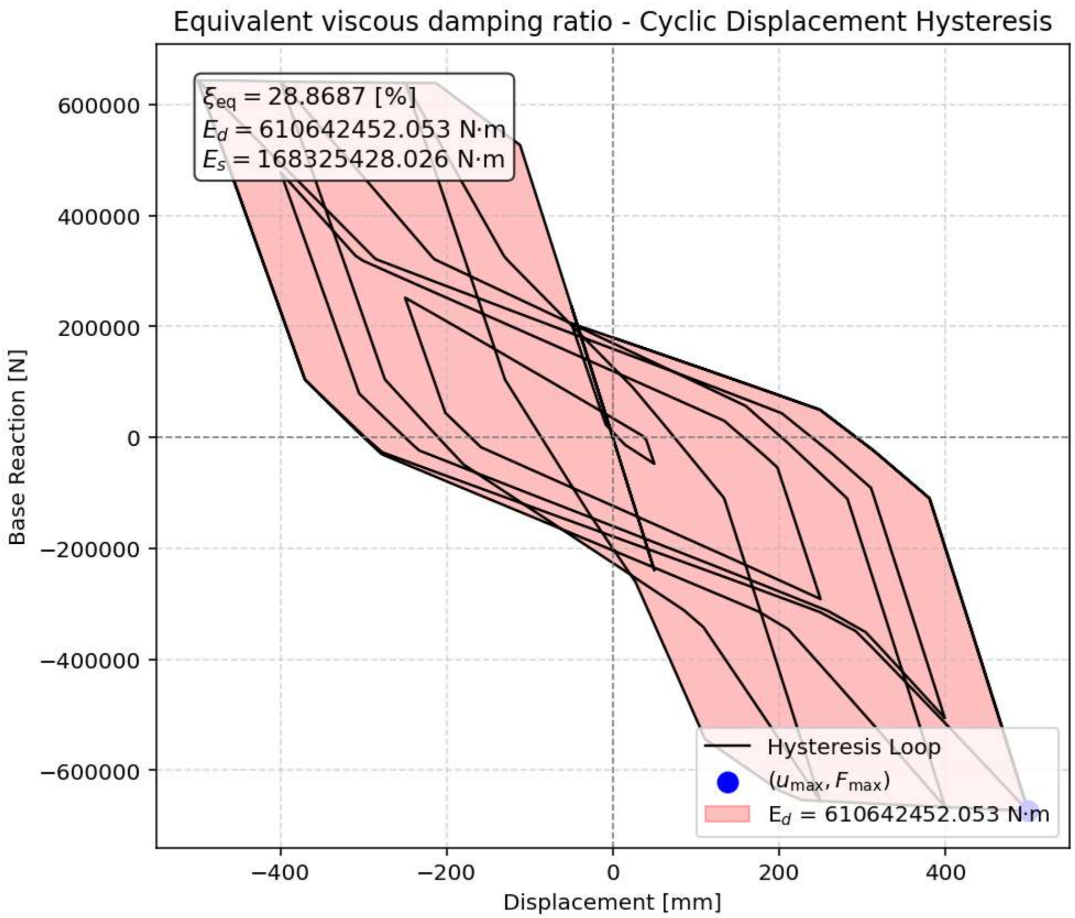
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
20000
40000
60000
80000
1000
2000
3000
4000
5000
6000

CYCLIC DISPLACEMENT ANALYSIS

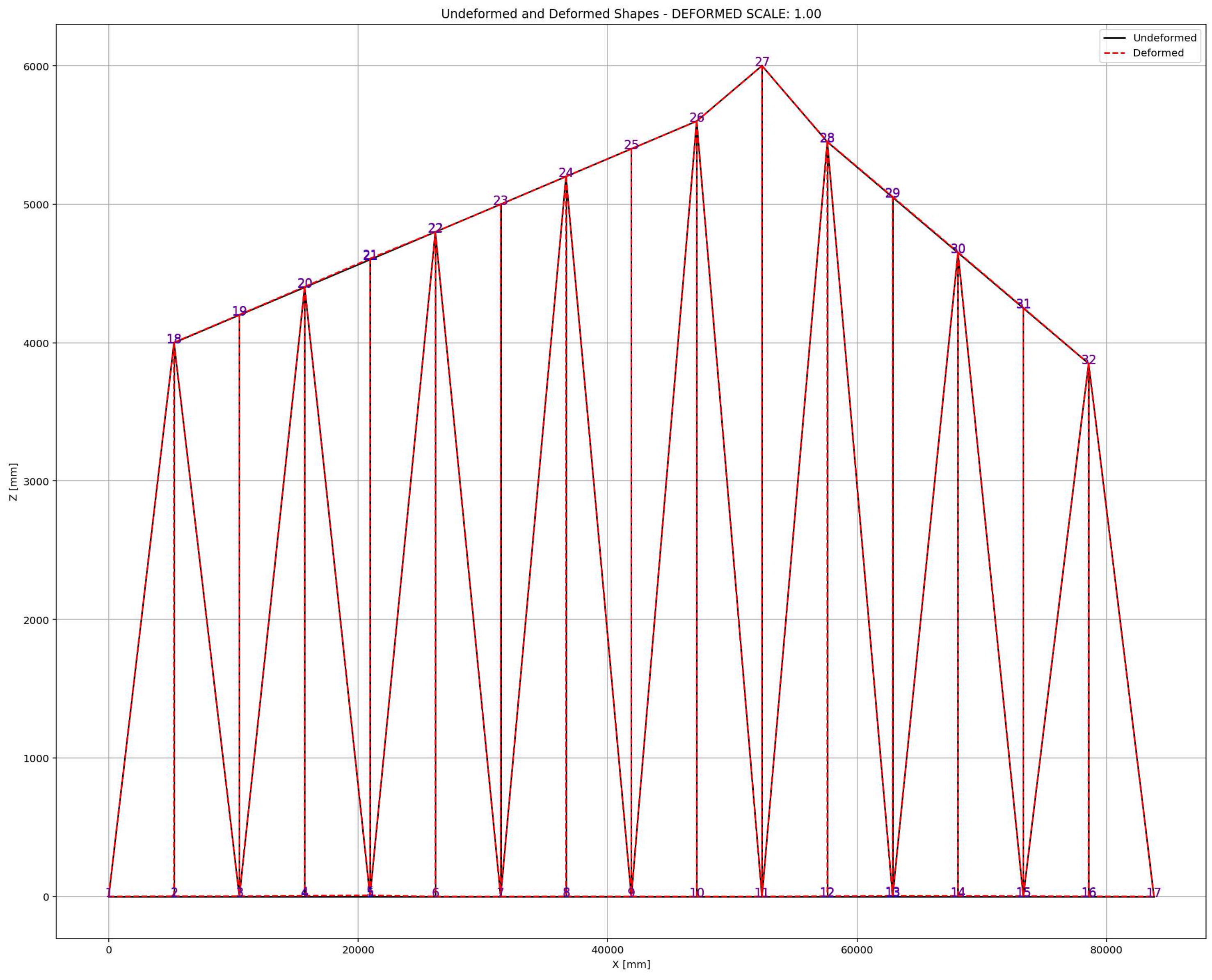

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAILROAD_TRUSS_TWO_SPANS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAIL..._Flint River Bridge_Tuscaloosa Railroad Bridge.py
BRIDGE_RAILROAD_TR...Railroad Bridge.py
CYCLIC DISPLACEMENT ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'CYCLIC_DISPLACEMENT'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(reaction_CP, disp_mid_CP,
ele_force_CP, ele_stress_CP, ele_strain_CP,
node_displacementsX_CP, node_displacementsY_CP,
dispX_CP, dispY_CP,
PERIOD_MIN_CP, PERIOD_MAX_CP) = DATA
XDATA = disp_mid_CP
YDATA = reaction_CP
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_CP, PERIOD_MIN_CP, linewidth=3)
plt.plot(disp_mid_CP, PERIOD_MAX_CP, linewidth=3)
plt.title('Period of Structure During Cyclic Displacement Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_CP):.3f} (s) - Mean: {np.mean(P
f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_CP):.3f} (s) - Mean: {np.mean(
30 %
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 71, Col 216 UTF-8 CRLF RW Mem 59%


Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
600000
400000
200000
0
−200000
−400000
−600000
Base Reaction [N]
−400
−200
0
200
400
Displacement in Middle Span [mm]


Period of Structure During Cyclic Displacement Analysis
PERIOD - MIN VALUES: Min: 0.634 (s) - Mean: 0.684 (s) - Max: 0.849 (s)
PERIOD - MAX VALUES: Min: 5.107 (s) - Mean: 13.593 (s) - Max: 180.456 (s)
Structural Period [s]
Displacement [mm]
175
150
125
100
75
50
25
0
−400
−200
0
200
400



Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 28.8687$ [%]
$E_d = 610642452.053$ N·m
$E_s = 168325428.026$ N·m
600000
400000
200000
0
−200000
−400000
−600000
Base Reaction [N]
−400
−200
0
200
400
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 610642452.053 N·m



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
1000
2000
3000
4000
5000
6000
0
20000
40000
60000
80000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
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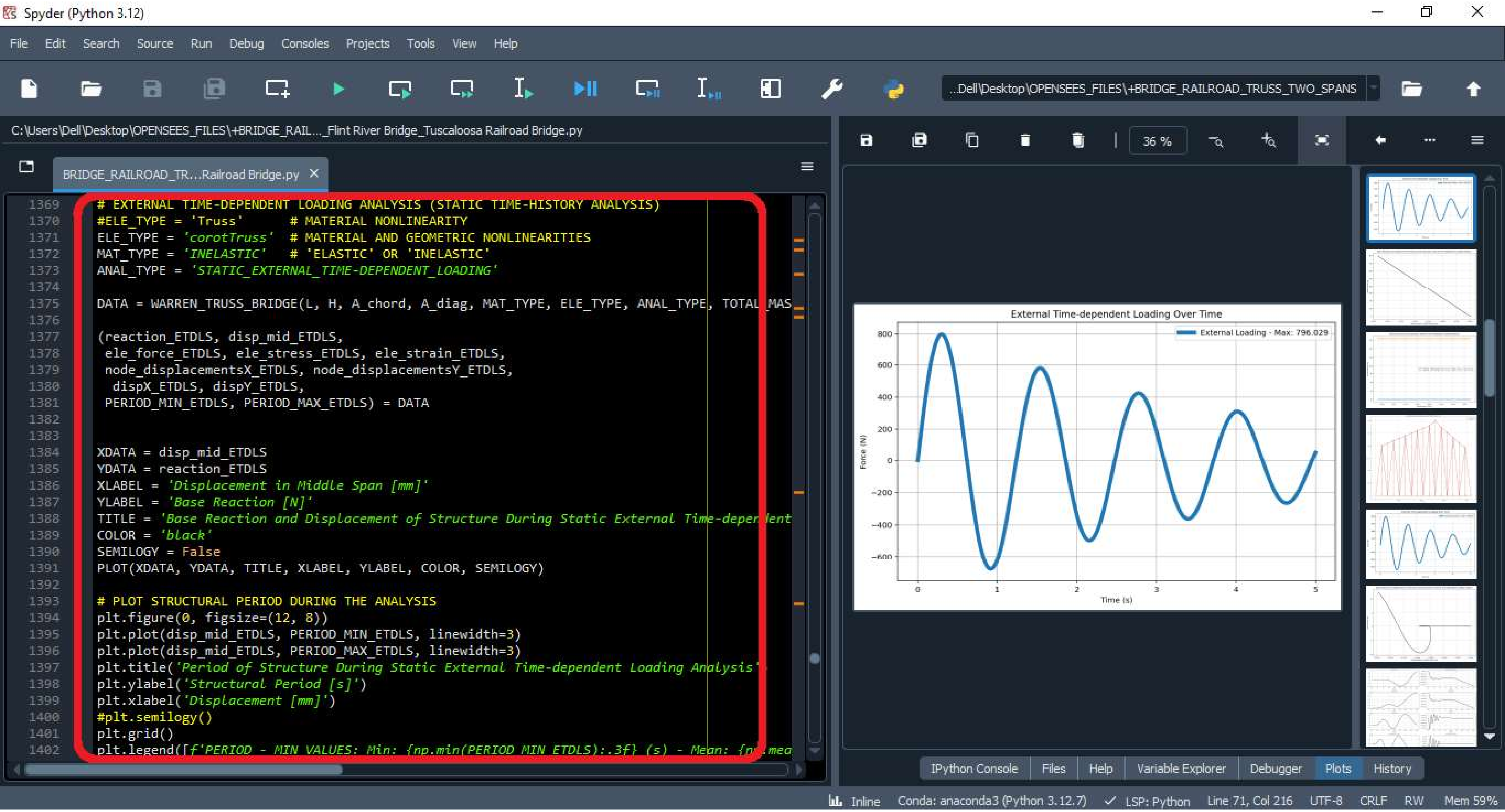
31
32

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS


Spyder (Python 3.12)
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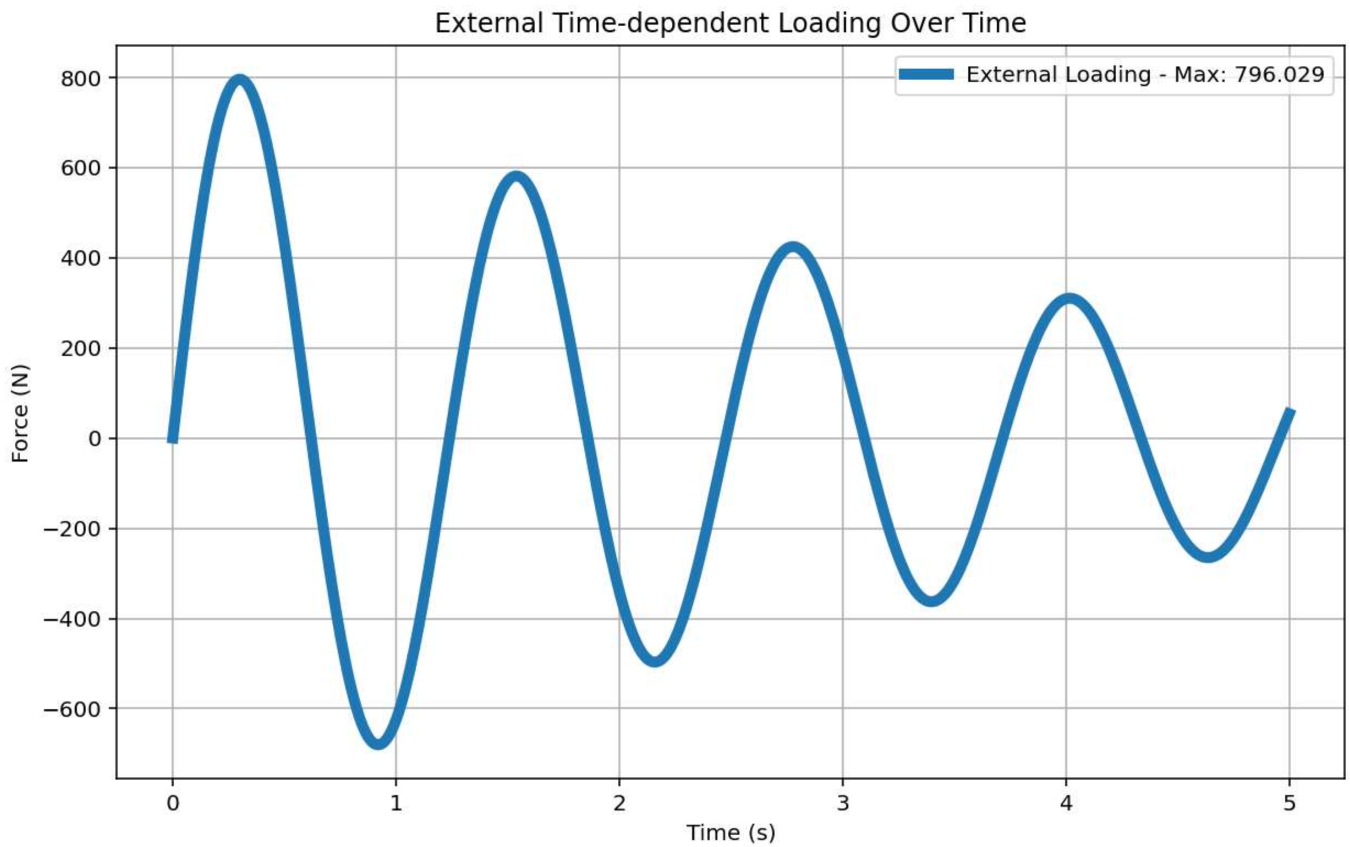
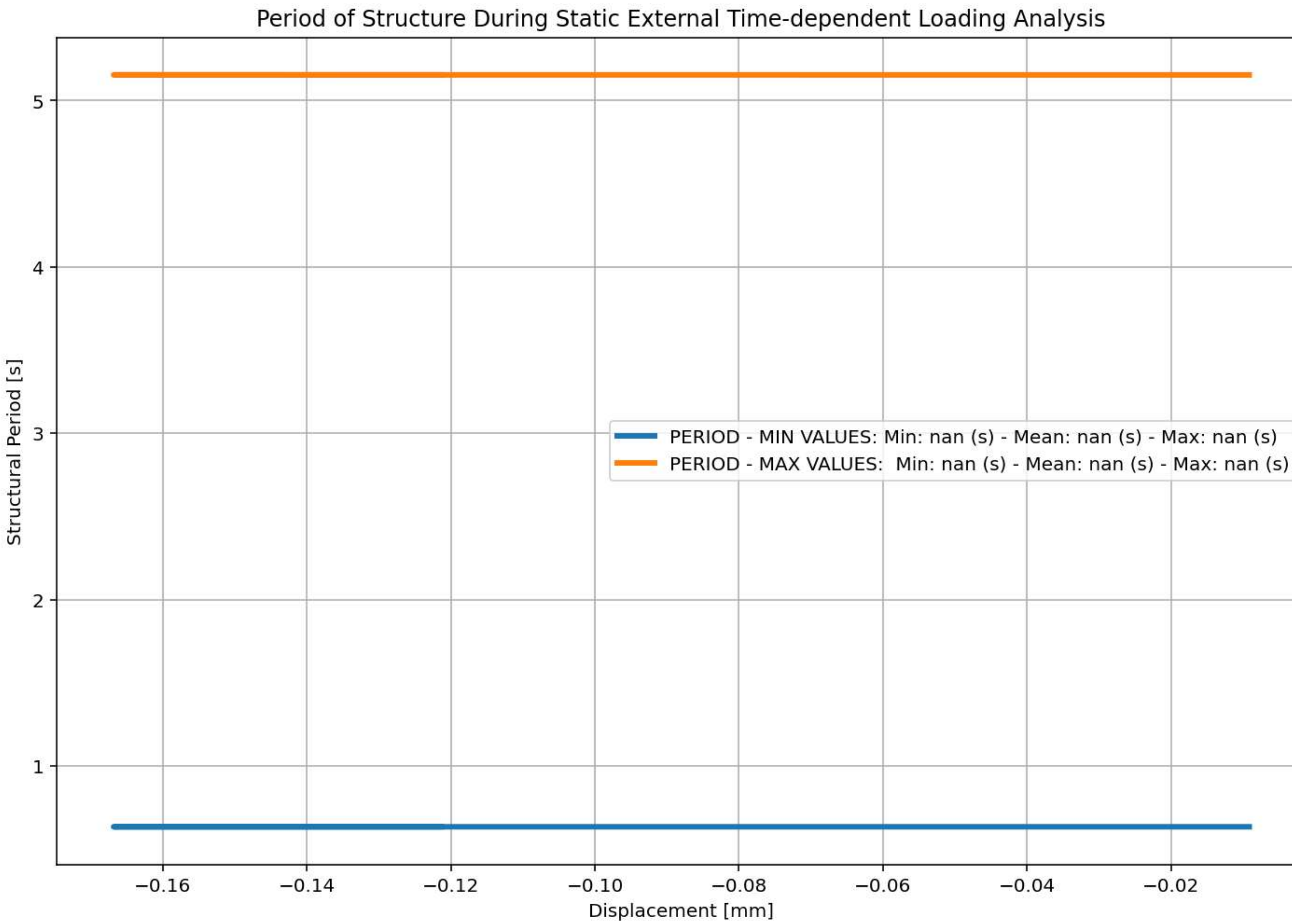
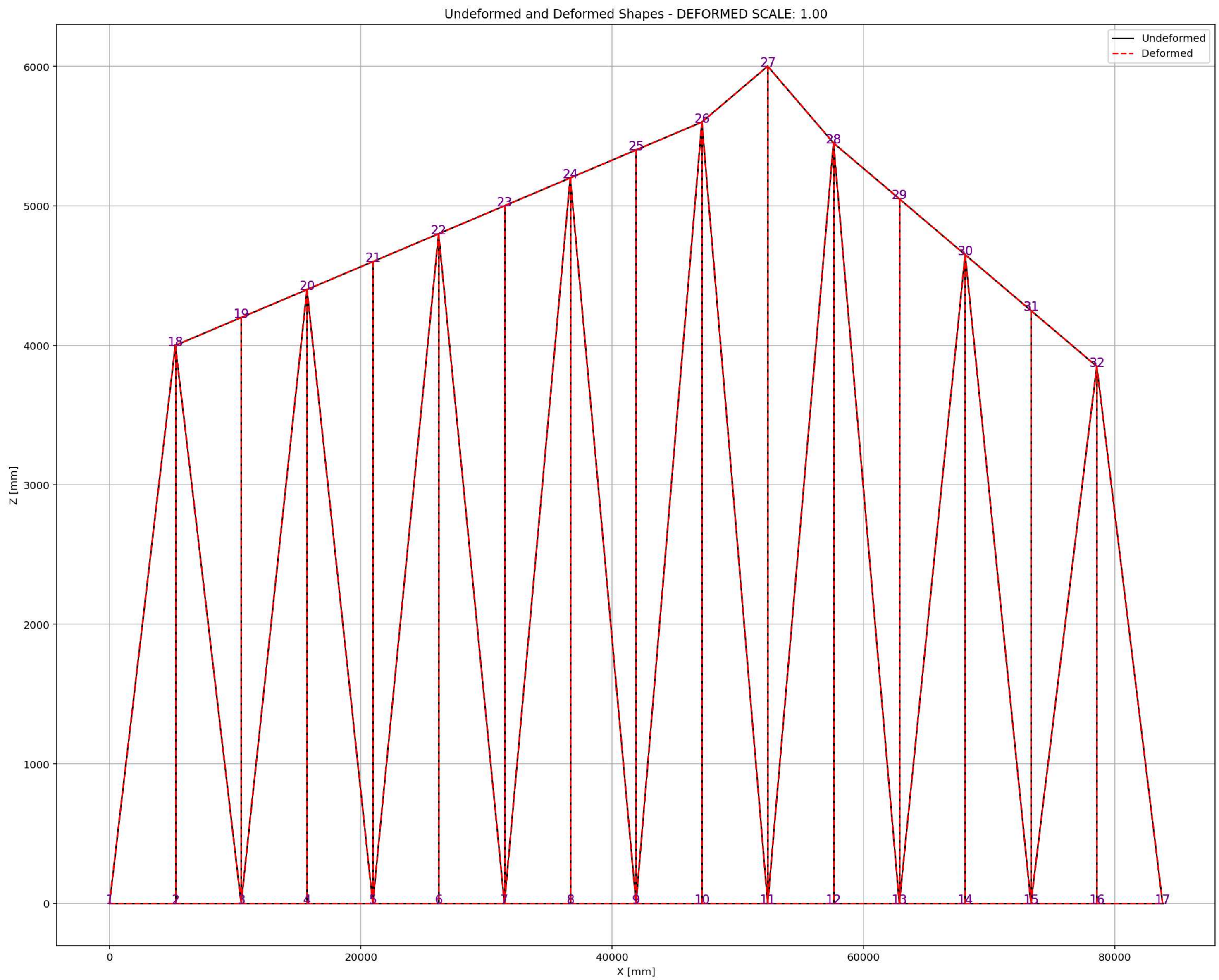
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAIL..._Flint River Bridge_Tuscaloosa Railroad Bridge.py
BRIDGE_RAILROAD_TR...Railroad Bridge.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(reaction_ETDLS, disp_mid_ETDLS,
 ele_force_ETDLS, ele_stress_ETDLS, ele_strain_ETDLS,
 node_displacementsX_ETDLS, node_displacementsY_ETDLS,
  dispX_ETDLS, dispY_ETDLS,
 PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
XDATA = disp_mid_ETDLS
YDATA = reaction_ETDLS
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mea
36 %
External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
Time (s)
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 71, Col 216 UTF-8 CRLF RW Mem 59%


External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
Time (s)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
800
700
600
500
400
300
200
100
0
−0.175
−0.150
−0.125
−0.100
−0.075
−0.050
−0.025
0.000



Period of Structure During Static External Time-dependent Loading Analysis
Structural Period [s]
Displacement [mm]
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
5
4
3
2
1
−0.16
−0.14
−0.12
−0.10
−0.08
−0.06
−0.04
−0.02



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
1000
2000
3000
4000
5000
6000
0
20000
40000
60000
80000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
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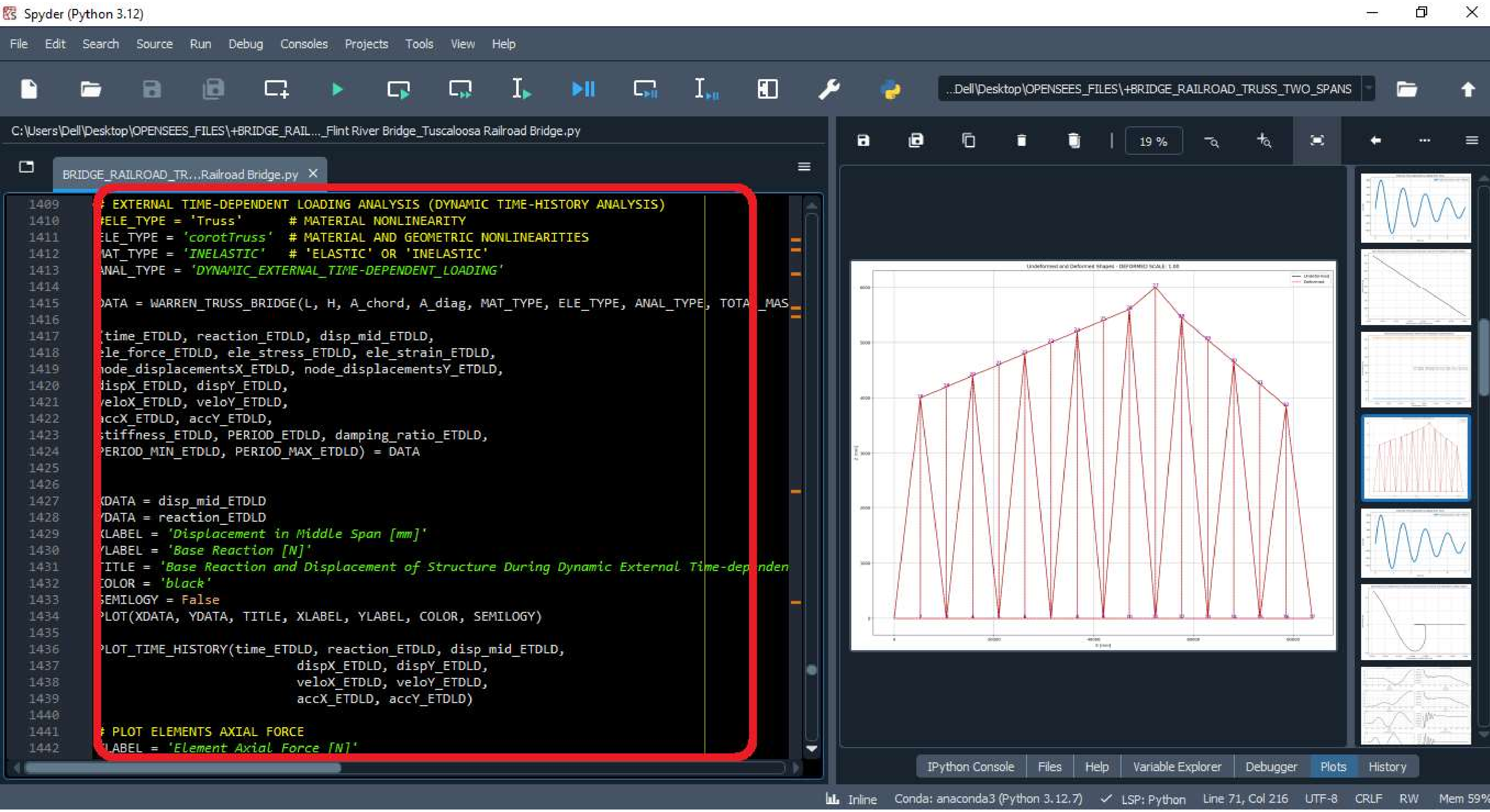
30
31
32

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS


Spyder (Python 3.12)
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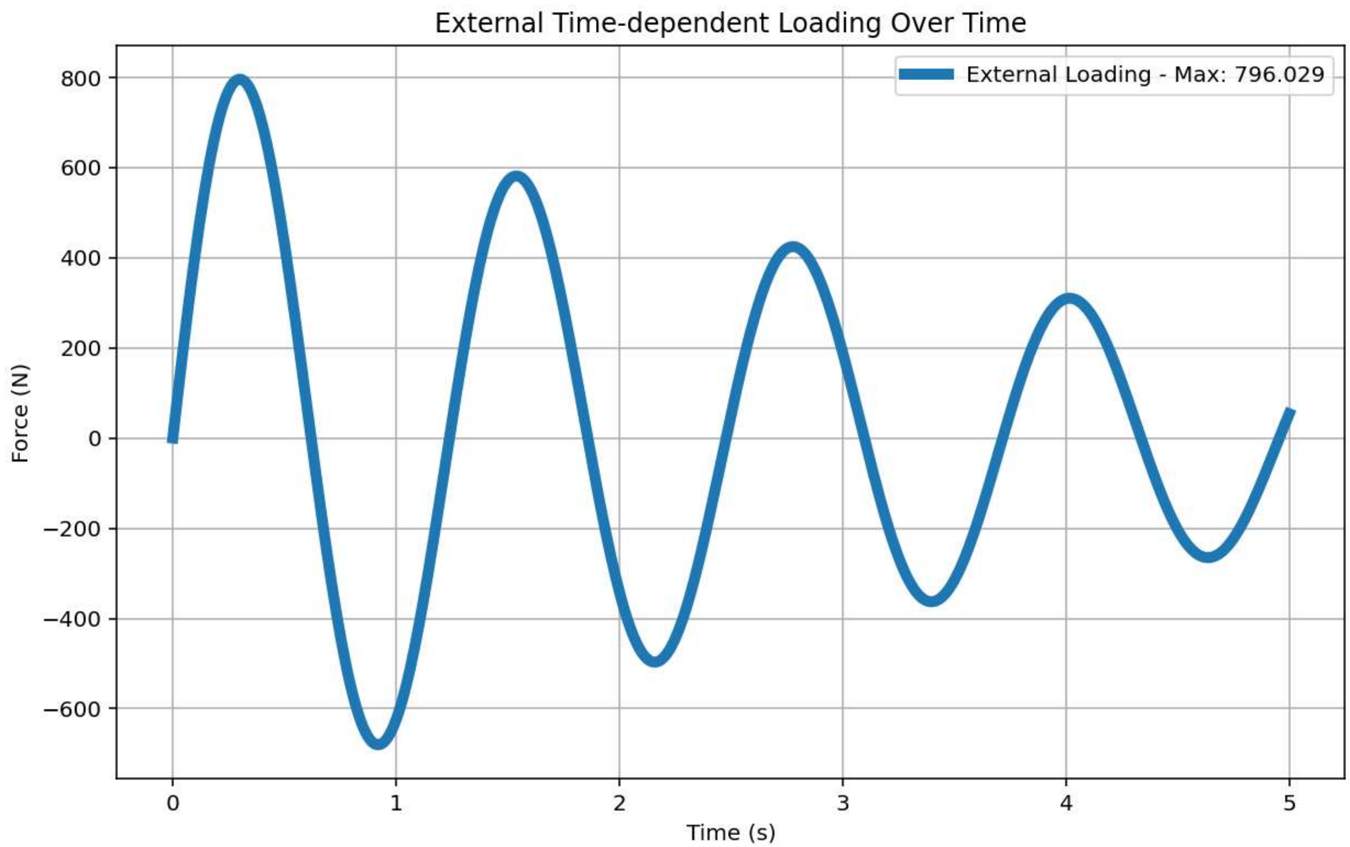
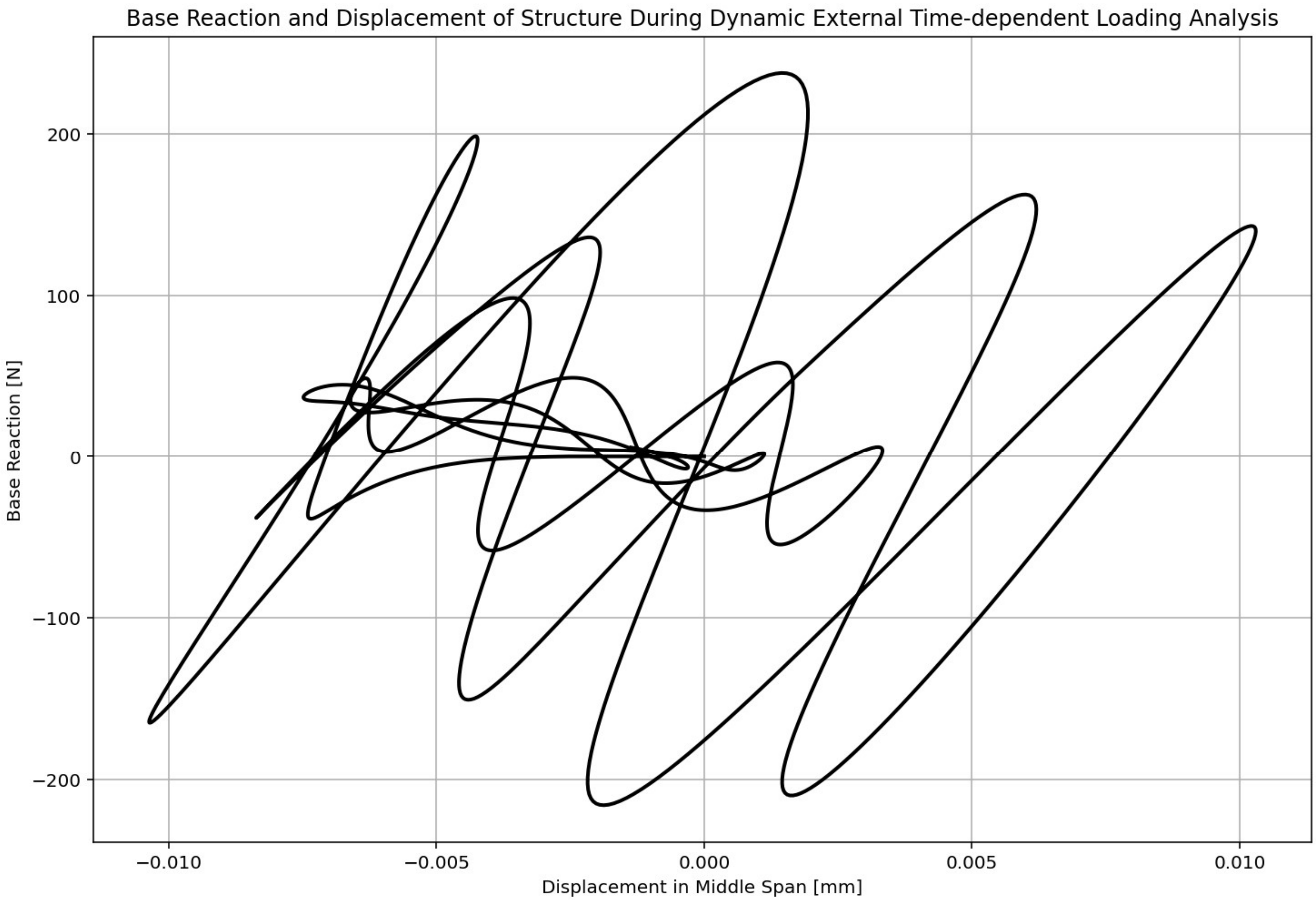
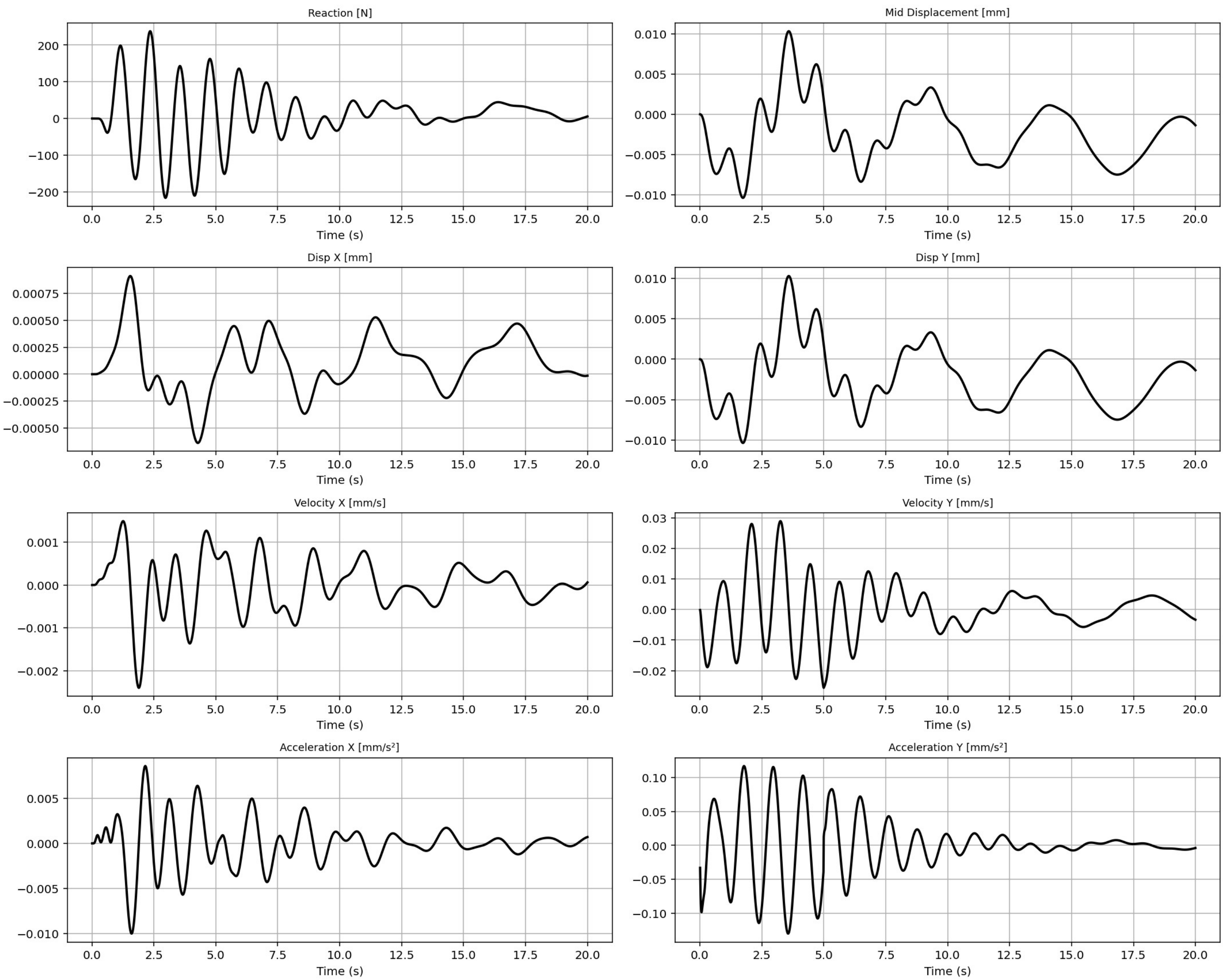
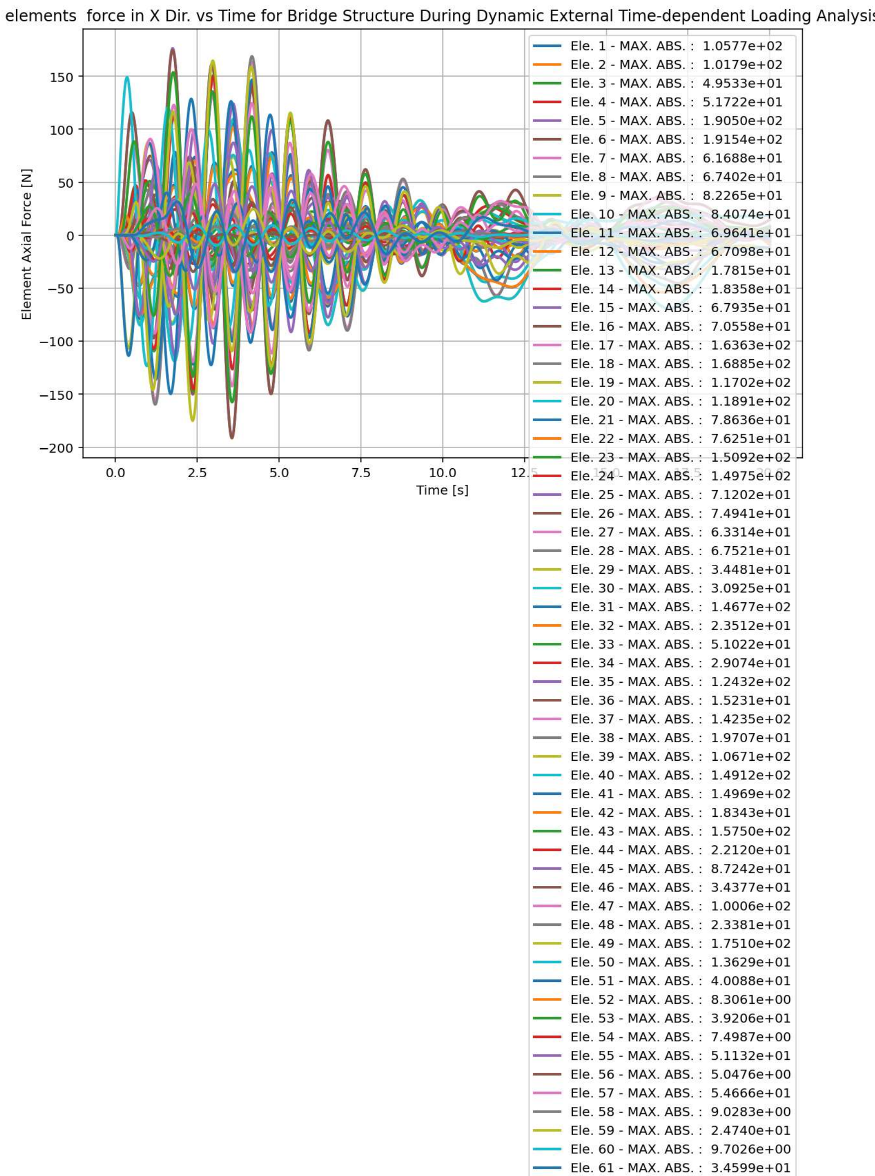
..Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAILROAD_TRUSS_TWO_SPANS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAIL..._Flint River Bridge_Tuscaloosa Railroad Bridge.py
BRIDGE_RAILROAD_TR...Railroad Bridge.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
ele_force_ETDLD, ele_stress_ETDLD, ele_strain_ETDLD,
node_displacementsX_ETDLD, node_displacementsY_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD,
stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
XDATA = disp_mid_ETDLD
YDATA = reaction_ETDLD
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependen
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 71, Col 216 UTF-8 CRLF RW Mem 59%

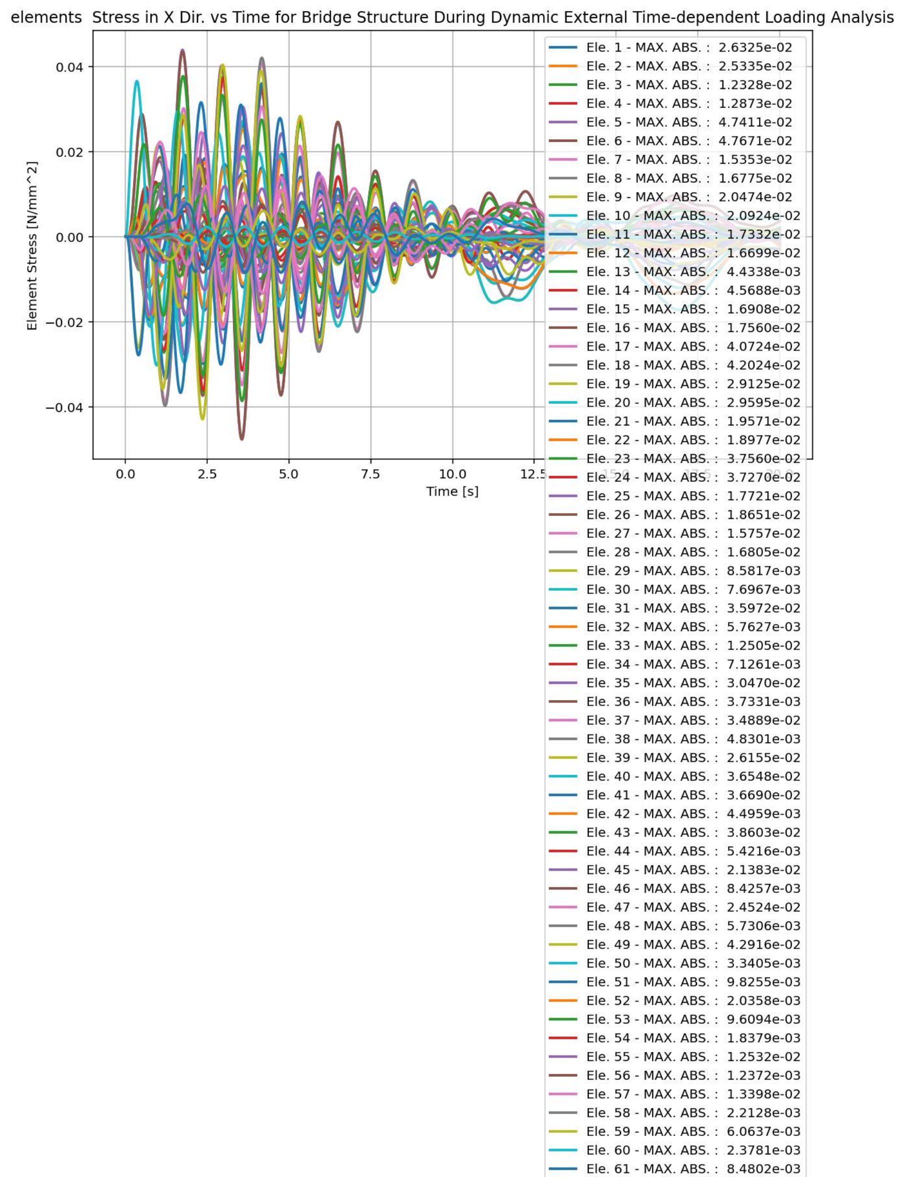

External Time-dependent Loading Over Time
External Loading - Max: 796.029
800
600
400
200
0
−200
−400
−600
Force (N)
0
1
2
3
4
5
Time (s)

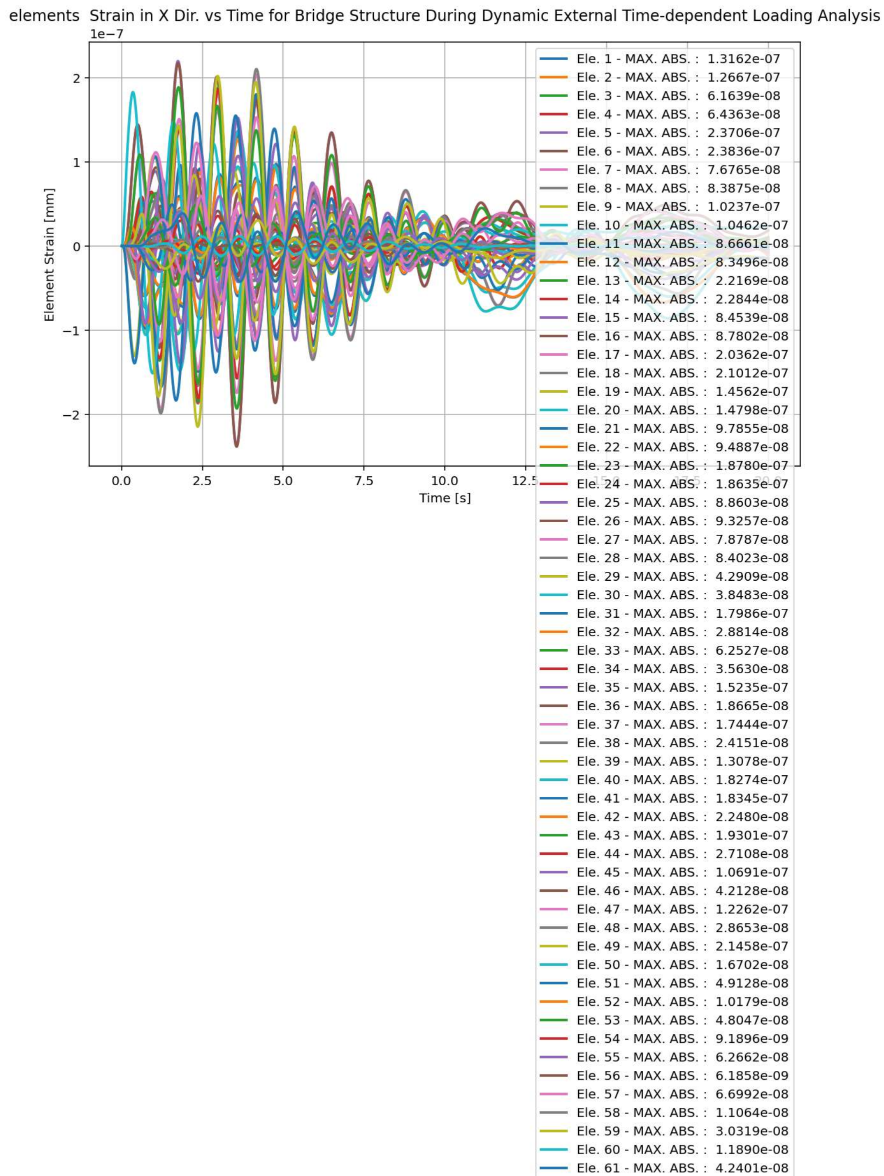


Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
200
100
0
−100
−200
Base Reaction [N]
−0.010
−0.005
0.000
0.005
0.010
Displacement in Middle Span [mm]

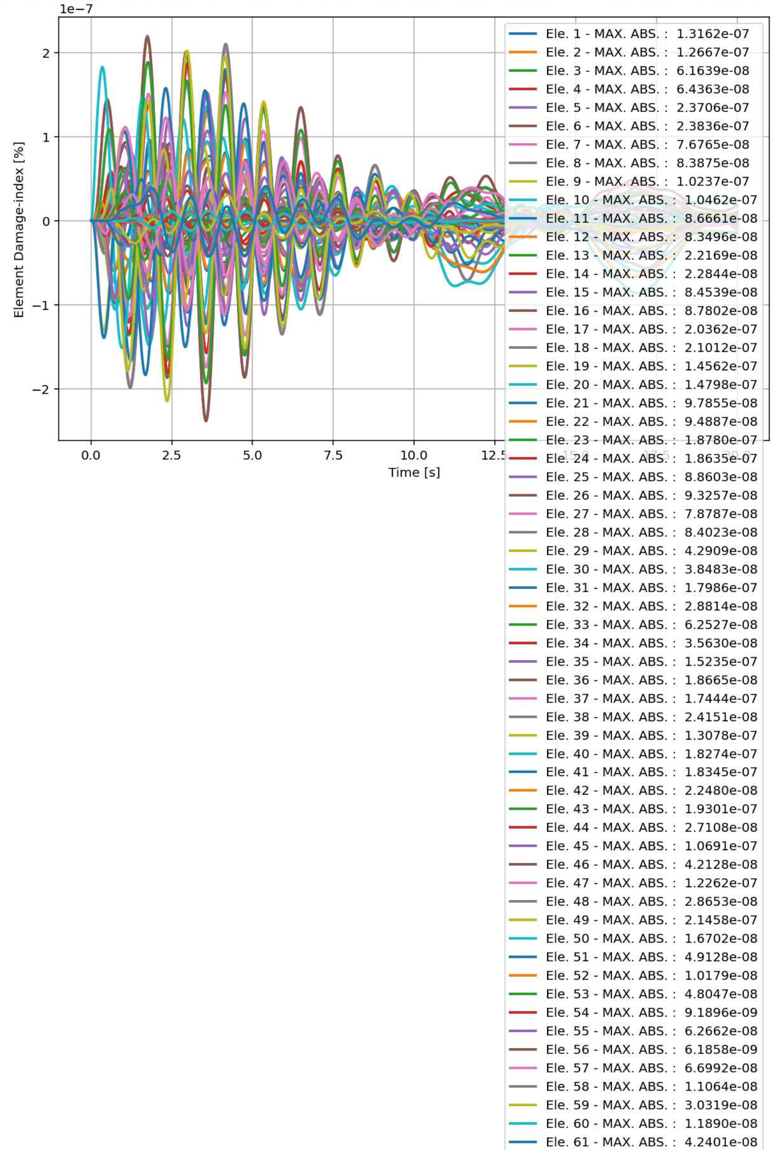

Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

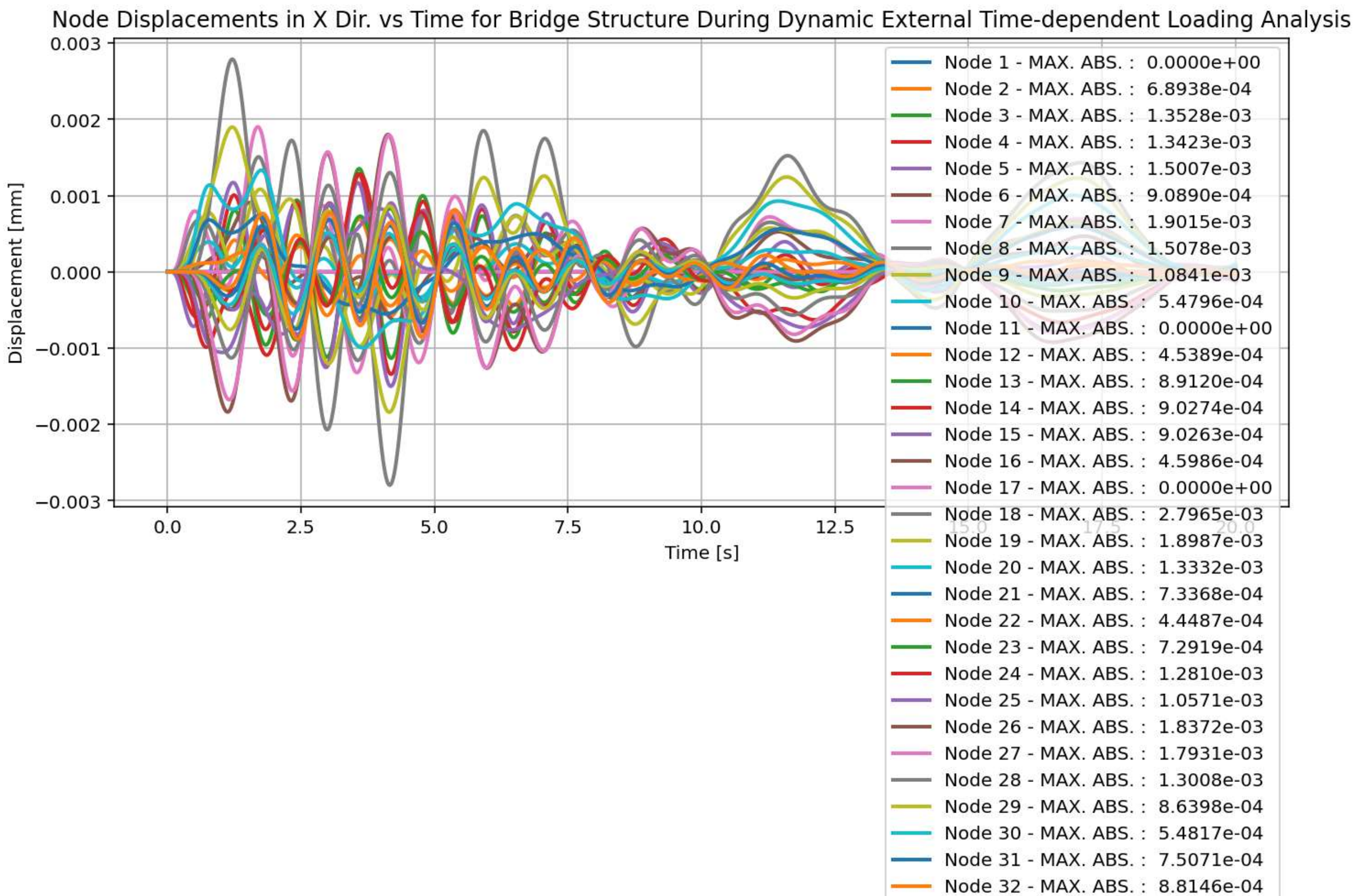

elements force in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Element Axial Force [N]
Time [s]
150
100
50
0
−50
−100
−150
−200
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 1.0577e+02
Ele. 2 - MAX. ABS. : 1.0179e+02
Ele. 3 - MAX. ABS. : 4.9533e+01
Ele. 4 - MAX. ABS. : 5.1722e+01
Ele. 5 - MAX. ABS. : 1.9050e+02
Ele. 6 - MAX. ABS. : 1.9154e+02
Ele. 7 - MAX. ABS. : 6.1688e+01
Ele. 8 - MAX. ABS. : 6.7402e+01
Ele. 9 - MAX. ABS. : 8.2265e+01
Ele. 10 - MAX. ABS. : 8.4074e+01
Ele. 11 - MAX. ABS. : 6.9641e+01
Ele. 12 - MAX. ABS. : 6.7098e+01
Ele. 13 - MAX. ABS. : 1.7815e+01
Ele. 14 - MAX. ABS. : 1.8358e+01
Ele. 15 - MAX. ABS. : 6.7935e+01
Ele. 16 - MAX. ABS. : 7.0558e+01
Ele. 17 - MAX. ABS. : 1.6363e+02
Ele. 18 - MAX. ABS. : 1.6885e+02
Ele. 19 - MAX. ABS. : 1.1702e+02
Ele. 20 - MAX. ABS. : 1.1891e+02
Ele. 21 - MAX. ABS. : 7.8636e+01
Ele. 22 - MAX. ABS. : 7.6251e+01
Ele. 23 - MAX. ABS. : 1.5092e+02
Ele. 24 - MAX. ABS. : 1.4975e+02
Ele. 25 - MAX. ABS. : 7.1202e+01
Ele. 26 - MAX. ABS. : 7.4941e+01
Ele. 27 - MAX. ABS. : 6.3314e+01
Ele. 28 - MAX. ABS. : 6.7521e+01
Ele. 29 - MAX. ABS. : 3.4481e+01
Ele. 30 - MAX. ABS. : 3.0925e+01
Ele. 31 - MAX. ABS. : 1.4677e+02
Ele. 32 - MAX. ABS. : 2.3512e+01
Ele. 33 - MAX. ABS. : 5.1022e+01
Ele. 34 - MAX. ABS. : 2.9074e+01
Ele. 35 - MAX. ABS. : 1.2432e+02
Ele. 36 - MAX. ABS. : 1.5231e+01
Ele. 37 - MAX. ABS. : 1.4235e+02
Ele. 38 - MAX. ABS. : 1.9707e+01
Ele. 39 - MAX. ABS. : 1.0671e+02
Ele. 40 - MAX. ABS. : 1.4912e+02
Ele. 41 - MAX. ABS. : 1.4969e+02
Ele. 42 - MAX. ABS. : 1.8343e+01
Ele. 43 - MAX. ABS. : 1.5750e+02
Ele. 44 - MAX. ABS. : 2.2120e+01
Ele. 45 - MAX. ABS. : 8.7242e+01
Ele. 46 - MAX. ABS. : 3.4377e+01
Ele. 47 - MAX. ABS. : 1.0006e+02
Ele. 48 - MAX. ABS. : 2.3381e+01
Ele. 49 - MAX. ABS. : 1.7510e+02
Ele. 50 - MAX. ABS. : 1.3629e+01
Ele. 51 - MAX. ABS. : 4.0088e+01
Ele. 52 - MAX. ABS. : 8.3061e+00
Ele. 53 - MAX. ABS. : 3.9206e+01
Ele. 54 - MAX. ABS. : 7.4987e+00
Ele. 55 - MAX. ABS. : 5.1132e+01
Ele. 56 - MAX. ABS. : 5.0476e+00
Ele. 57 - MAX. ABS. : 5.4666e+01
Ele. 58 - MAX. ABS. : 9.0283e+00
Ele. 59 - MAX. ABS. : 2.4740e+01
Ele. 60 - MAX. ABS. : 9.7026e+00
Ele. 61 - MAX. ABS. : 3.4599e+01

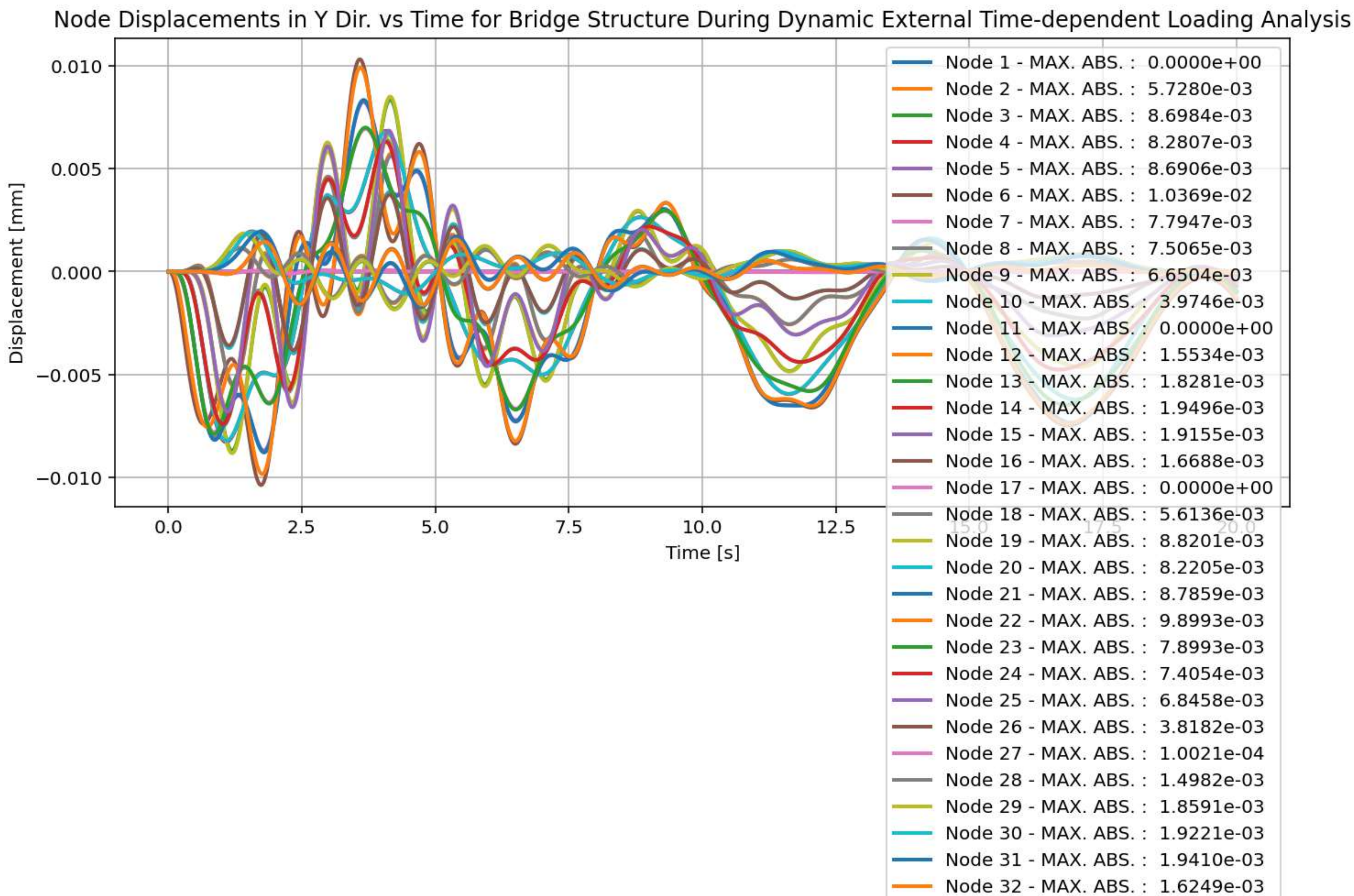

elements Stress in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Element Stress [N/mm^2]
0.04
0.02
0.00
−0.02
−0.04
Time [s]
0.0
2.5
5.0
7.5
10.0
12.5
Ele. 1 - MAX. ABS. : 2.6325e-02
Ele. 2 - MAX. ABS. : 2.5335e-02
Ele. 3 - MAX. ABS. : 1.2328e-02
Ele. 4 - MAX. ABS. : 1.2873e-02
Ele. 5 - MAX. ABS. : 4.7411e-02
Ele. 6 - MAX. ABS. : 4.7671e-02
Ele. 7 - MAX. ABS. : 1.5353e-02
Ele. 8 - MAX. ABS. : 1.6775e-02
Ele. 9 - MAX. ABS. : 2.0474e-02
Ele. 10 - MAX. ABS. : 2.0924e-02
Ele. 11 - MAX. ABS. : 1.7332e-02
Ele. 12 - MAX. ABS. : 1.6699e-02
Ele. 13 - MAX. ABS. : 4.4338e-03
Ele. 14 - MAX. ABS. : 4.5688e-03
Ele. 15 - MAX. ABS. : 1.6908e-02
Ele. 16 - MAX. ABS. : 1.7560e-02
Ele. 17 - MAX. ABS. : 4.0724e-02
Ele. 18 - MAX. ABS. : 4.2024e-02
Ele. 19 - MAX. ABS. : 2.9125e-02
Ele. 20 - MAX. ABS. : 2.9595e-02
Ele. 21 - MAX. ABS. : 1.9571e-02
Ele. 22 - MAX. ABS. : 1.8977e-02
Ele. 23 - MAX. ABS. : 3.7560e-02
Ele. 24 - MAX. ABS. : 3.7270e-02
Ele. 25 - MAX. ABS. : 1.7721e-02
Ele. 26 - MAX. ABS. : 1.8651e-02
Ele. 27 - MAX. ABS. : 1.5757e-02
Ele. 28 - MAX. ABS. : 1.6805e-02
Ele. 29 - MAX. ABS. : 8.5817e-03
Ele. 30 - MAX. ABS. : 7.6967e-03
Ele. 31 - MAX. ABS. : 3.5972e-02
Ele. 32 - MAX. ABS. : 5.7627e-03
Ele. 33 - MAX. ABS. : 1.2505e-02
Ele. 34 - MAX. ABS. : 7.1261e-03
Ele. 35 - MAX. ABS. : 3.0470e-02
Ele. 36 - MAX. ABS. : 3.7331e-03
Ele. 37 - MAX. ABS. : 3.4889e-02
Ele. 38 - MAX. ABS. : 4.8301e-03
Ele. 39 - MAX. ABS. : 2.6155e-02
Ele. 40 - MAX. ABS. : 3.6548e-02
Ele. 41 - MAX. ABS. : 3.6690e-02
Ele. 42 - MAX. ABS. : 4.4959e-03
Ele. 43 - MAX. ABS. : 3.8603e-02
Ele. 44 - MAX. ABS. : 5.4216e-03
Ele. 45 - MAX. ABS. : 2.1383e-02
Ele. 46 - MAX. ABS. : 8.4257e-03
Ele. 47 - MAX. ABS. : 2.4524e-02
Ele. 48 - MAX. ABS. : 5.7306e-03
Ele. 49 - MAX. ABS. : 4.2916e-02
Ele. 50 - MAX. ABS. : 3.3405e-03
Ele. 51 - MAX. ABS. : 9.8255e-03
Ele. 52 - MAX. ABS. : 2.0358e-03
Ele. 53 - MAX. ABS. : 9.6094e-03
Ele. 54 - MAX. ABS. : 1.8379e-03
Ele. 55 - MAX. ABS. : 1.2532e-02
Ele. 56 - MAX. ABS. : 1.2372e-03
Ele. 57 - MAX. ABS. : 1.3398e-02
Ele. 58 - MAX. ABS. : 2.2128e-03
Ele. 59 - MAX. ABS. : 6.0637e-03
Ele. 60 - MAX. ABS. : 2.3781e-03
Ele. 61 - MAX. ABS. : 8.4802e-03

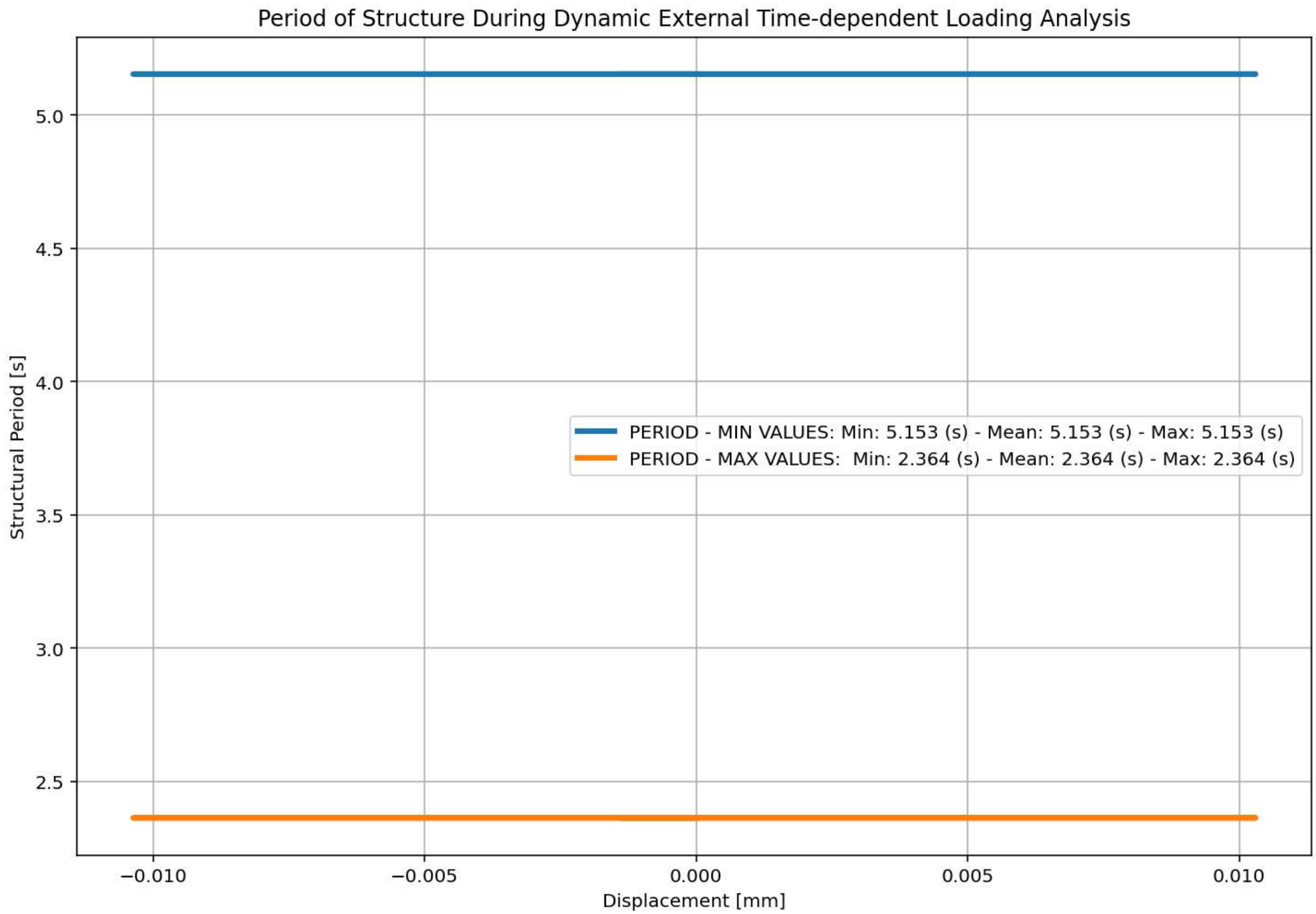

elements Strain in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
1e−7
Element Strain [mm]
Time [s]
Ele. 1 - MAX. ABS. : 1.3162e-07
Ele. 2 - MAX. ABS. : 1.2667e-07
Ele. 3 - MAX. ABS. : 6.1639e-08
Ele. 4 - MAX. ABS. : 6.4363e-08
Ele. 5 - MAX. ABS. : 2.3706e-07
Ele. 6 - MAX. ABS. : 2.3836e-07
Ele. 7 - MAX. ABS. : 7.6765e-08
Ele. 8 - MAX. ABS. : 8.3875e-08
Ele. 9 - MAX. ABS. : 1.0237e-07
Ele. 10 - MAX. ABS. : 1.0462e-07
Ele. 11 - MAX. ABS. : 8.6661e-08
Ele. 12 - MAX. ABS. : 8.3496e-08
Ele. 13 - MAX. ABS. : 2.2169e-08
Ele. 14 - MAX. ABS. : 2.2844e-08
Ele. 15 - MAX. ABS. : 8.4539e-08
Ele. 16 - MAX. ABS. : 8.7802e-08
Ele. 17 - MAX. ABS. : 2.0362e-07
Ele. 18 - MAX. ABS. : 2.1012e-07
Ele. 19 - MAX. ABS. : 1.4562e-07
Ele. 20 - MAX. ABS. : 1.4798e-07
Ele. 21 - MAX. ABS. : 9.7855e-08
Ele. 22 - MAX. ABS. : 9.4887e-08
Ele. 23 - MAX. ABS. : 1.8780e-07
Ele. 24 - MAX. ABS. : 1.8635e-07
Ele. 25 - MAX. ABS. : 8.8603e-08
Ele. 26 - MAX. ABS. : 9.3257e-08
Ele. 27 - MAX. ABS. : 7.8787e-08
Ele. 28 - MAX. ABS. : 8.4023e-08
Ele. 29 - MAX. ABS. : 4.2909e-08
Ele. 30 - MAX. ABS. : 3.8483e-08
Ele. 31 - MAX. ABS. : 1.7986e-07
Ele. 32 - MAX. ABS. : 2.8814e-08
Ele. 33 - MAX. ABS. : 6.2527e-08
Ele. 34 - MAX. ABS. : 3.5630e-08
Ele. 35 - MAX. ABS. : 1.5235e-07
Ele. 36 - MAX. ABS. : 1.8665e-08
Ele. 37 - MAX. ABS. : 1.7444e-07
Ele. 38 - MAX. ABS. : 2.4151e-08
Ele. 39 - MAX. ABS. : 1.3078e-07
Ele. 40 - MAX. ABS. : 1.8274e-07
Ele. 41 - MAX. ABS. : 1.8345e-07
Ele. 42 - MAX. ABS. : 2.2480e-08
Ele. 43 - MAX. ABS. : 1.9301e-07
Ele. 44 - MAX. ABS. : 2.7108e-08
Ele. 45 - MAX. ABS. : 1.0691e-07
Ele. 46 - MAX. ABS. : 4.2128e-08
Ele. 47 - MAX. ABS. : 1.2262e-07
Ele. 48 - MAX. ABS. : 2.8653e-08
Ele. 49 - MAX. ABS. : 2.1458e-07
Ele. 50 - MAX. ABS. : 1.6702e-08
Ele. 51 - MAX. ABS. : 4.9128e-08
Ele. 52 - MAX. ABS. : 1.0179e-08
Ele. 53 - MAX. ABS. : 4.8047e-08
Ele. 54 - MAX. ABS. : 9.1896e-09
Ele. 55 - MAX. ABS. : 6.2662e-08
Ele. 56 - MAX. ABS. : 6.1858e-09
Ele. 57 - MAX. ABS. : 6.6992e-08
Ele. 58 - MAX. ABS. : 1.1064e-08
Ele. 59 - MAX. ABS. : 3.0319e-08
Ele. 60 - MAX. ABS. : 1.1890e-08
Ele. 61 - MAX. ABS. : 4.2401e-08

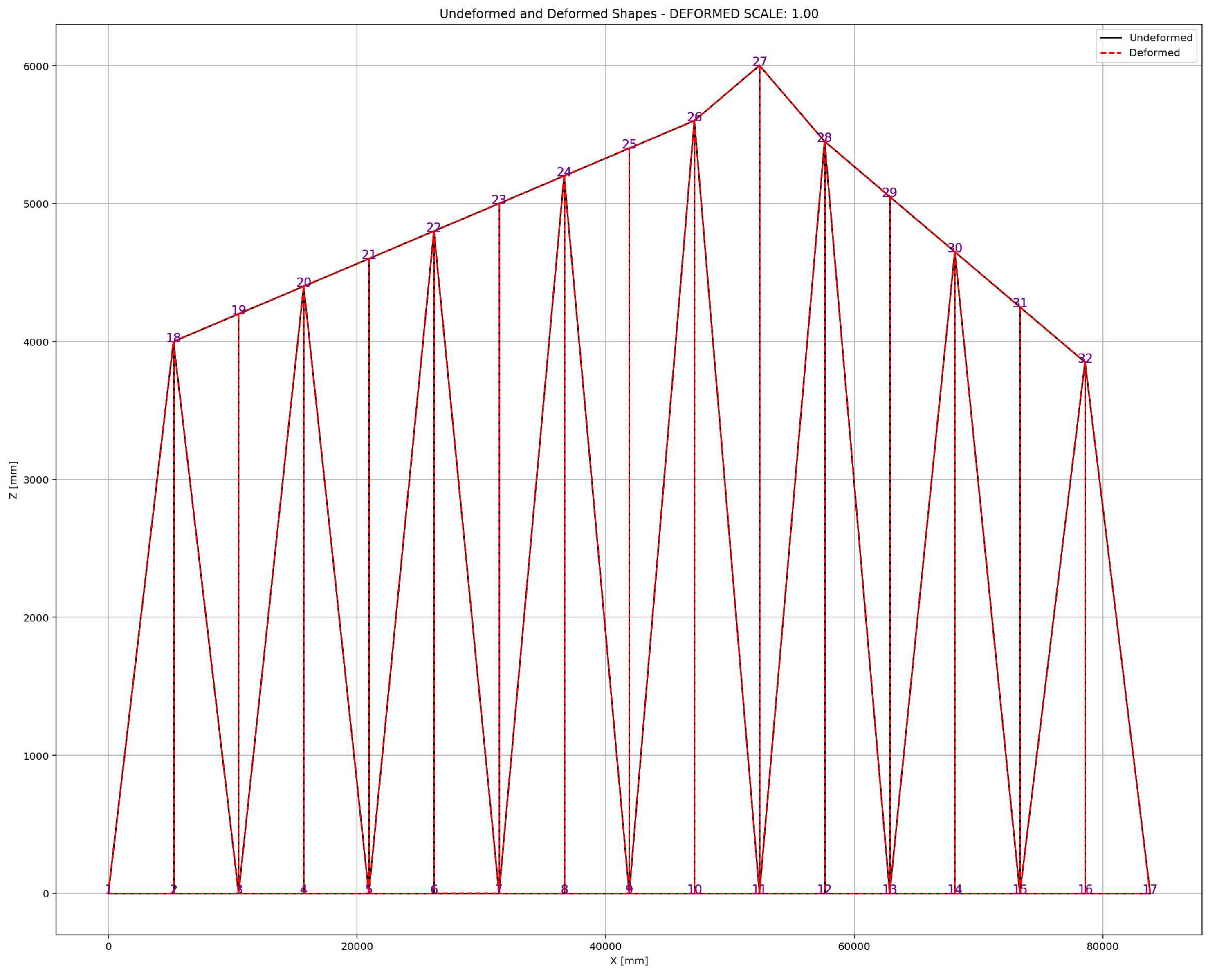
elements Damage-index [%] in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis




Node Displacements in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
0.003
0.002
0.001
0.000
−0.001
−0.002
−0.003
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 6.8938e-04
Node 3 - MAX. ABS. : 1.3528e-03
Node 4 - MAX. ABS. : 1.3423e-03
Node 5 - MAX. ABS. : 1.5007e-03
Node 6 - MAX. ABS. : 9.0890e-04
Node 7 - MAX. ABS. : 1.9015e-03
Node 8 - MAX. ABS. : 1.5078e-03
Node 9 - MAX. ABS. : 1.0841e-03
Node 10 - MAX. ABS. : 5.4796e-04
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 4.5389e-04
Node 13 - MAX. ABS. : 8.9120e-04
Node 14 - MAX. ABS. : 9.0274e-04
Node 15 - MAX. ABS. : 9.0263e-04
Node 16 - MAX. ABS. : 4.5986e-04
Node 17 - MAX. ABS. : 0.0000e+00
Node 18 - MAX. ABS. : 2.7965e-03
Node 19 - MAX. ABS. : 1.8987e-03
Node 20 - MAX. ABS. : 1.3332e-03
Node 21 - MAX. ABS. : 7.3368e-04
Node 22 - MAX. ABS. : 4.4487e-04
Node 23 - MAX. ABS. : 7.2919e-04
Node 24 - MAX. ABS. : 1.2810e-03
Node 25 - MAX. ABS. : 1.0571e-03
Node 26 - MAX. ABS. : 1.8372e-03
Node 27 - MAX. ABS. : 1.7931e-03
Node 28 - MAX. ABS. : 1.3008e-03
Node 29 - MAX. ABS. : 8.6398e-04
Node 30 - MAX. ABS. : 5.4817e-04
Node 31 - MAX. ABS. : 7.5071e-04
Node 32 - MAX. ABS. : 8.8146e-04


Node Displacements in Y Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Displacement [mm]
Time [s]
0.010
0.005
0.000
−0.005
−0.010
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 5.7280e-03
Node 3 - MAX. ABS. : 8.6984e-03
Node 4 - MAX. ABS. : 8.2807e-03
Node 5 - MAX. ABS. : 8.6906e-03
Node 6 - MAX. ABS. : 1.0369e-02
Node 7 - MAX. ABS. : 7.7947e-03
Node 8 - MAX. ABS. : 7.5065e-03
Node 9 - MAX. ABS. : 6.6504e-03
Node 10 - MAX. ABS. : 3.9746e-03
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 1.5534e-03
Node 13 - MAX. ABS. : 1.8281e-03
Node 14 - MAX. ABS. : 1.9496e-03
Node 15 - MAX. ABS. : 1.9155e-03
Node 16 - MAX. ABS. : 1.6688e-03
Node 17 - MAX. ABS. : 0.0000e+00
Node 18 - MAX. ABS. : 5.6136e-03
Node 19 - MAX. ABS. : 8.8201e-03
Node 20 - MAX. ABS. : 8.2205e-03
Node 21 - MAX. ABS. : 8.7859e-03
Node 22 - MAX. ABS. : 9.8993e-03
Node 23 - MAX. ABS. : 7.8993e-03
Node 24 - MAX. ABS. : 7.4054e-03
Node 25 - MAX. ABS. : 6.8458e-03
Node 26 - MAX. ABS. : 3.8182e-03
Node 27 - MAX. ABS. : 1.0021e-04
Node 28 - MAX. ABS. : 1.4982e-03
Node 29 - MAX. ABS. : 1.8591e-03
Node 30 - MAX. ABS. : 1.9221e-03
Node 31 - MAX. ABS. : 1.9410e-03
Node 32 - MAX. ABS. : 1.6249e-03


Period of Structure During Dynamic External Time-dependent Loading Analysis
Structural Period [s]
Displacement [mm]
5.0
4.5
4.0
3.5
3.0
2.5
−0.010
−0.005
0.000
0.005
0.010
PERIOD - MIN VALUES: Min: 5.153 (s) - Mean: 5.153 (s) - Max: 5.153 (s)
PERIOD - MAX VALUES: Min: 2.364 (s) - Mean: 2.364 (s) - Max: 2.364 (s)



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
20000
40000
60000
80000
1000
2000
3000
4000
5000
6000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
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27
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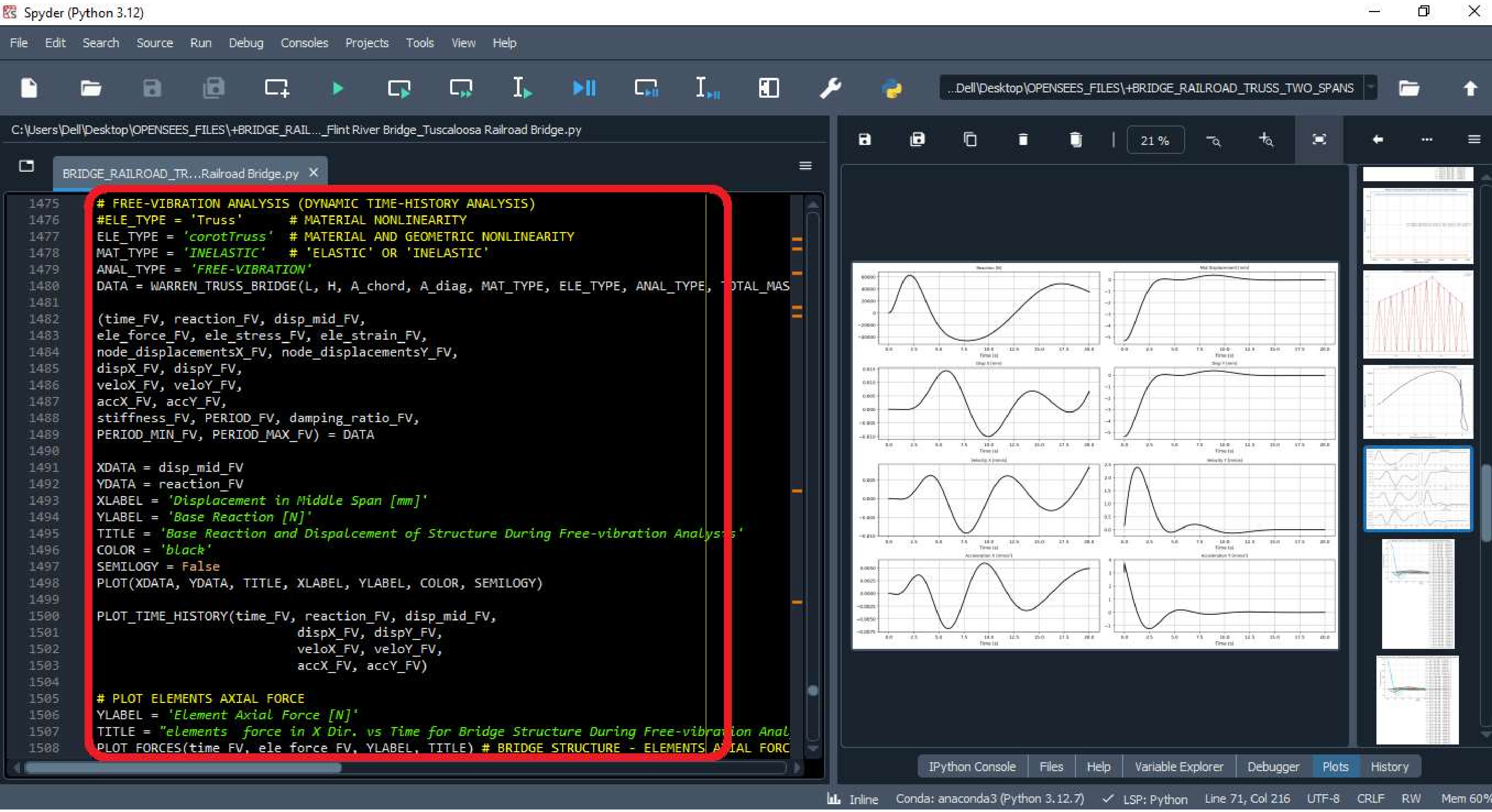
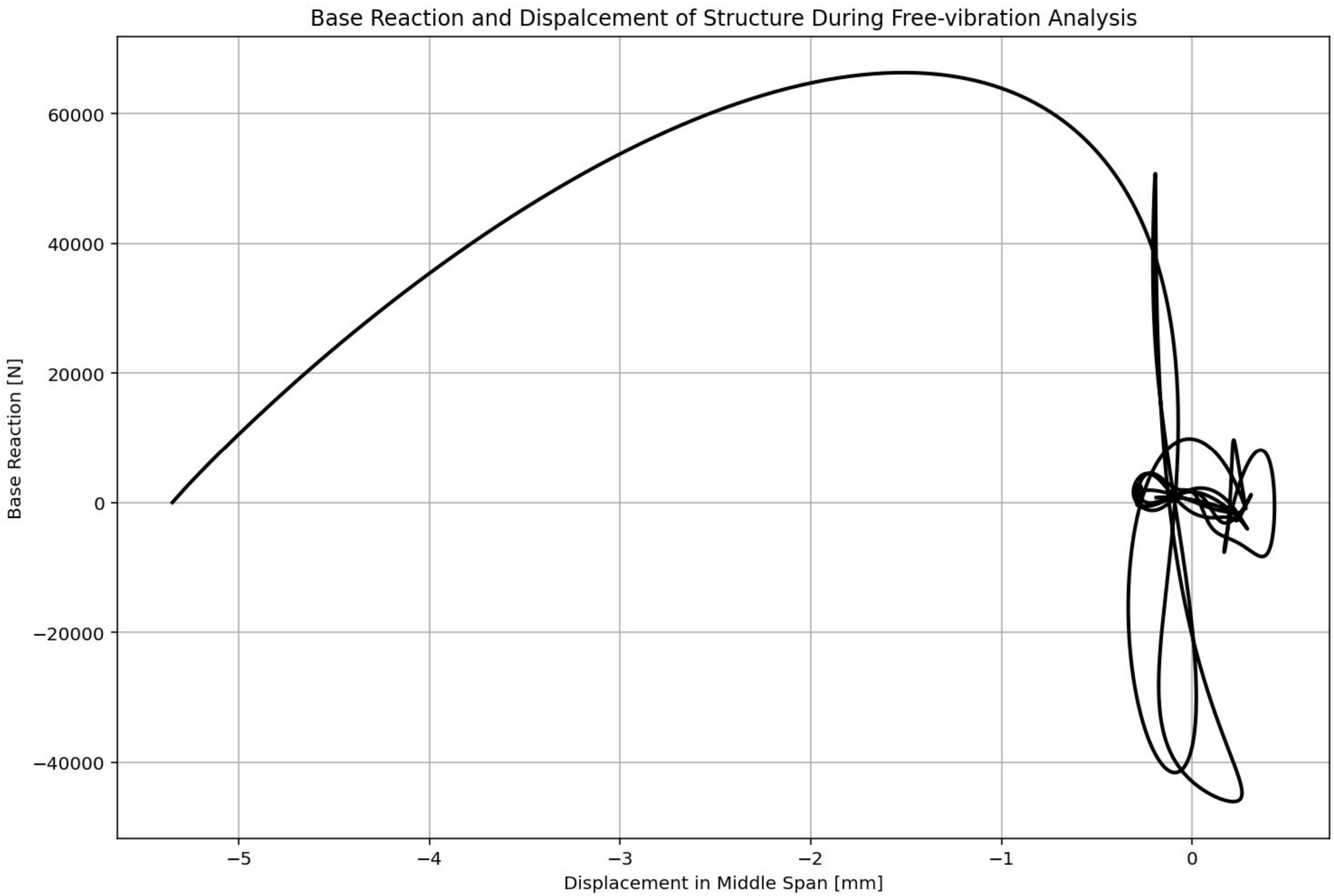
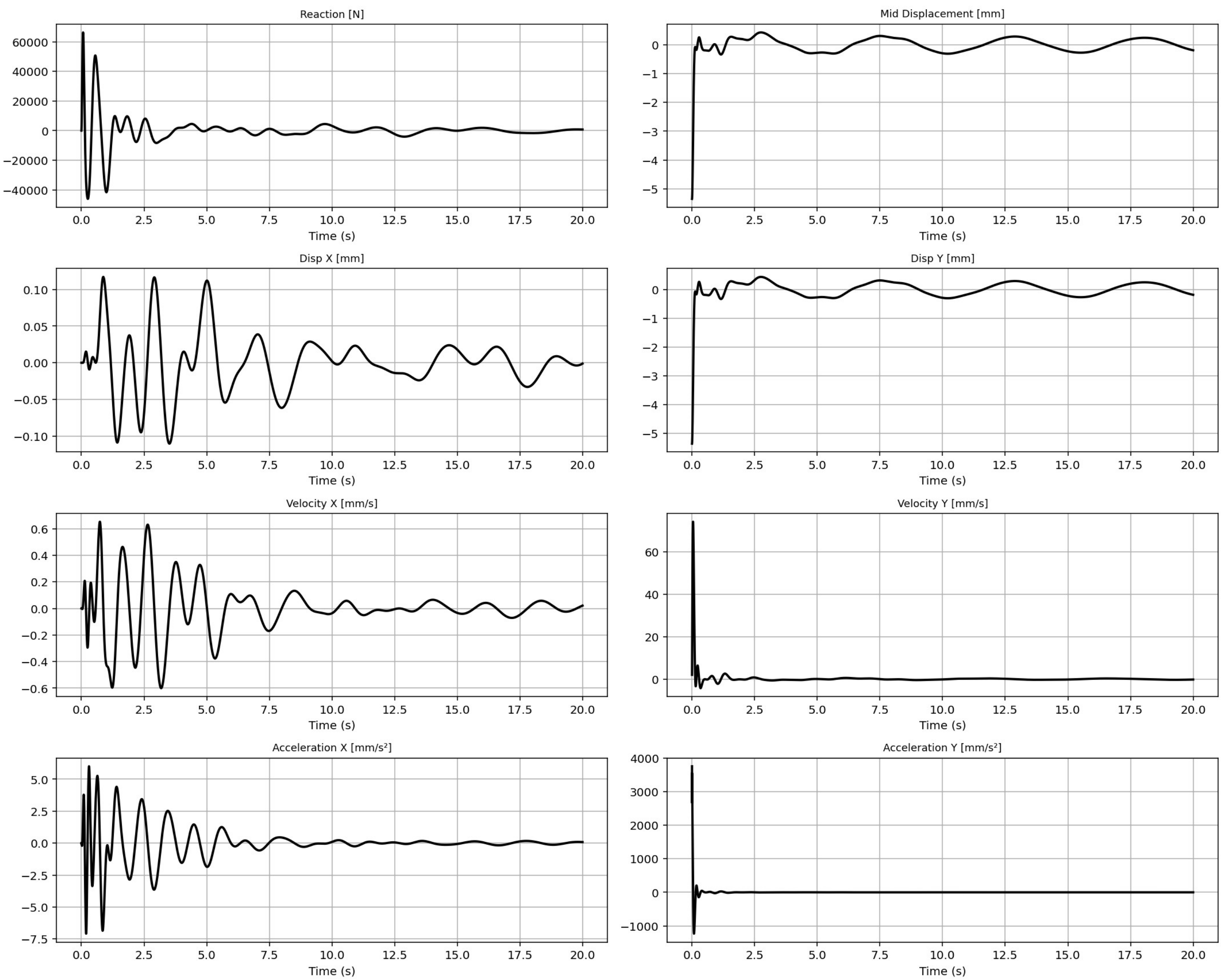
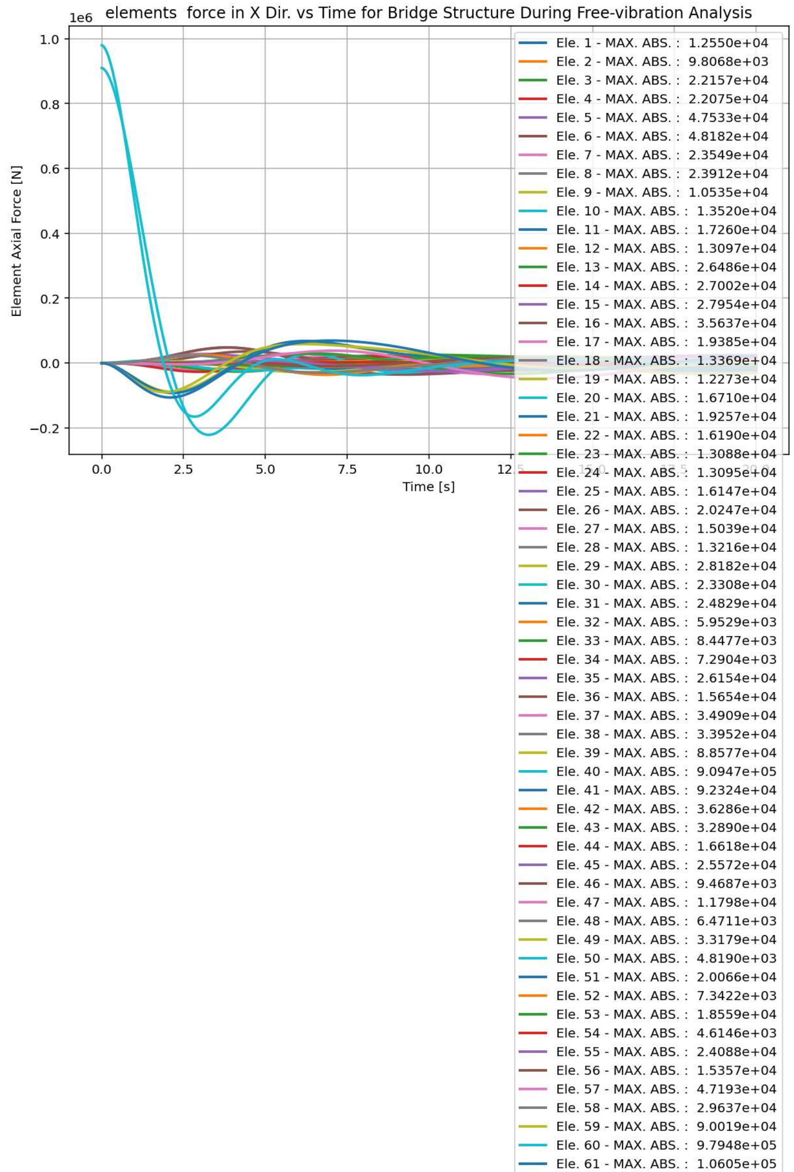
29
30
31
32

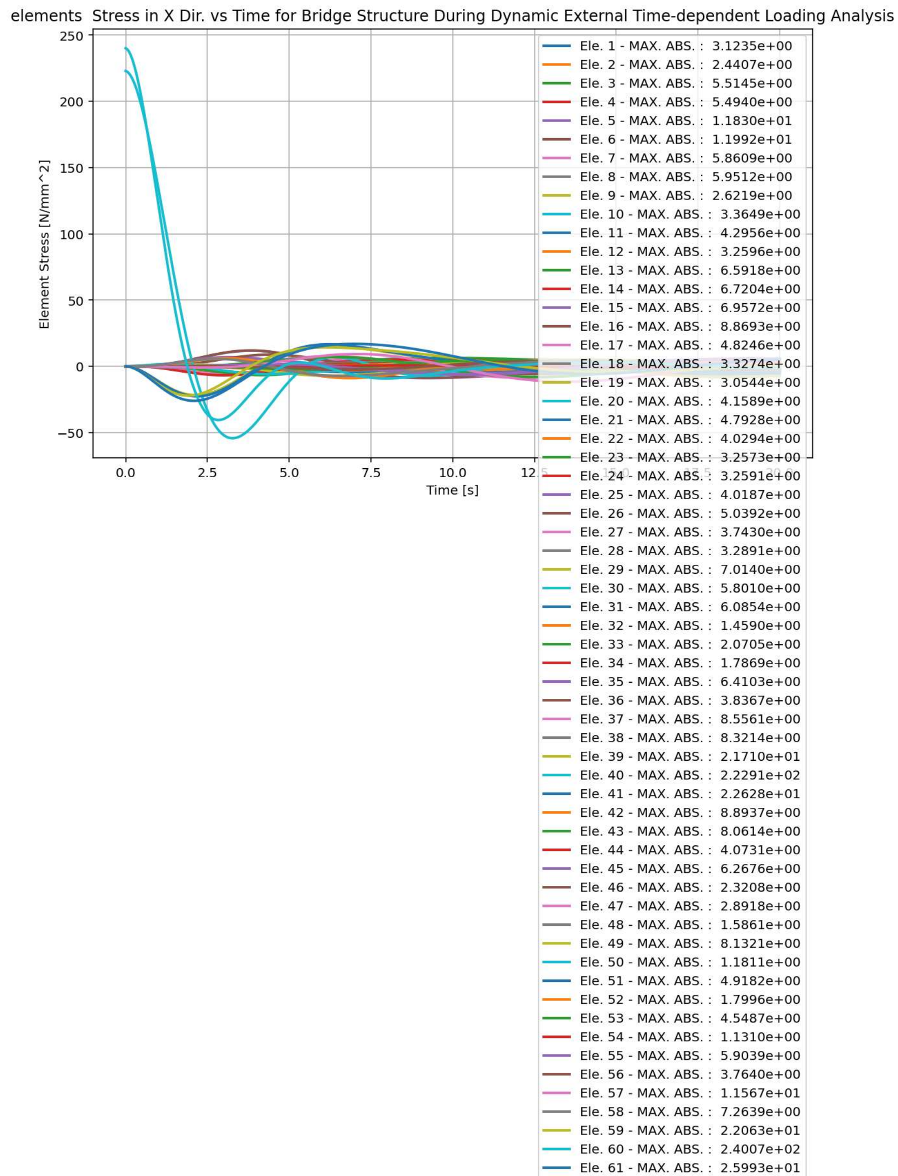
FREE-VIBRATION ANALYSIS

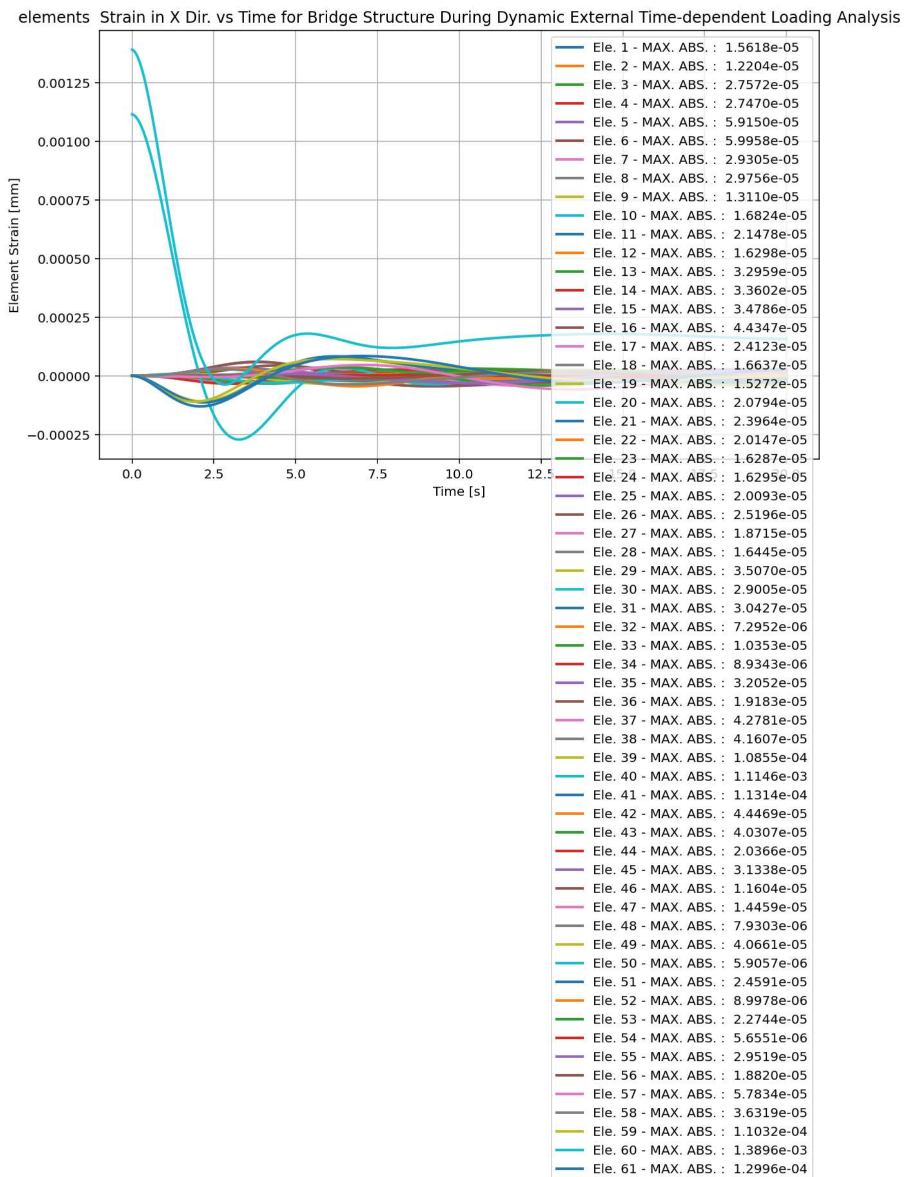

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
..Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAILROAD_TRUSS_TWO_SPANS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAIL..._Flint River Bridge_Tuscaloosa Railroad Bridge.py
BRIDGE_RAILROAD_TR...Railroad Bridge.py
FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITY
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'FREE-VIBRATION'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOTAL_MAS
(time_FV, reaction_FV, disp_mid_FV,
ele_force_FV, ele_stress_FV, ele_strain_FV,
node_displacementsX_FV, node_displacementsY_FV,
dispX_FV, dispY_FV,
veloX_FV, veloY_FV,
accX_FV, accY_FV,
stiffness_FV, PERIOD_FV, damping_ratio_FV,
PERIOD_MIN_FV, PERIOD_MAX_FV) = DATA
XDATA = disp_mid_FV
YDATA = reaction_FV
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_mid_FV,
dispX_FV, dispY_FV,
veloX_FV, veloY_FV,
accX_FV, accY_FV)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
TITLE = "elements force in X Dir. vs Time for Bridge Structure During Free-vibration Anal
PLOT_FORCES(time_FV, ele_force_FV, YLABEL, TITLE) # BRIDGE STRUCTURE - ELEMENTS AXIAL FORC
21 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 71, Col 216 UTF-8 CRLF RW Mem 60%

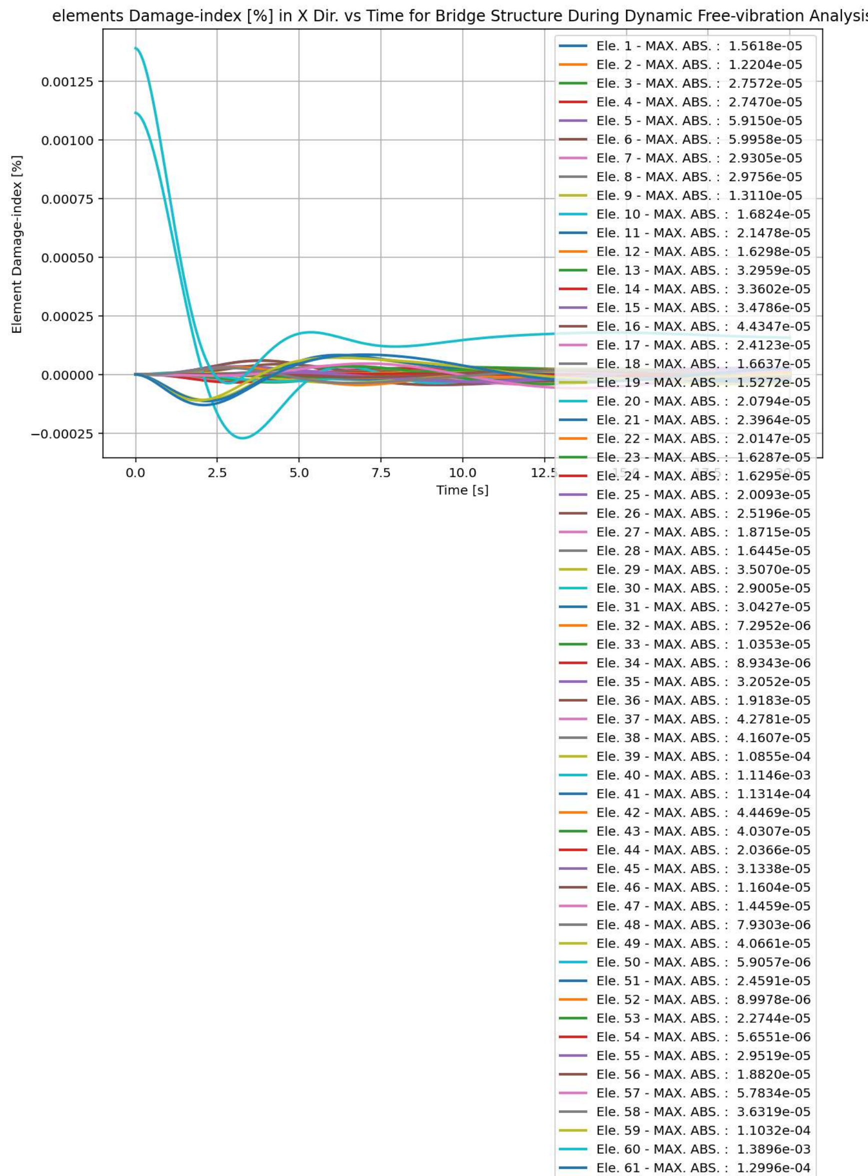

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
60000
40000
20000
0
−20000
−40000
Base Reaction [N]
−5
−4
−3
−2
−1
0
Displacement in Middle Span [mm]

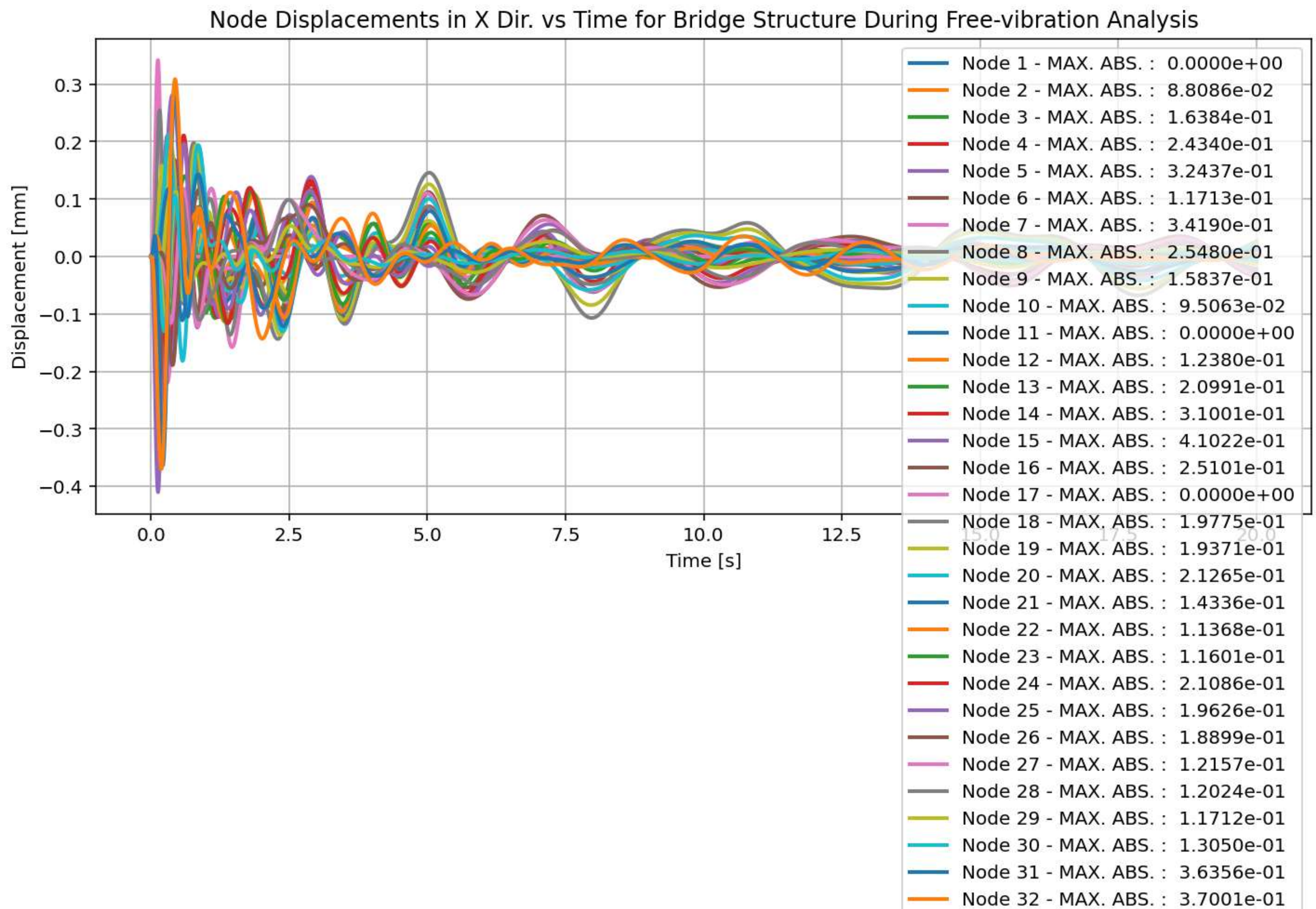

Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

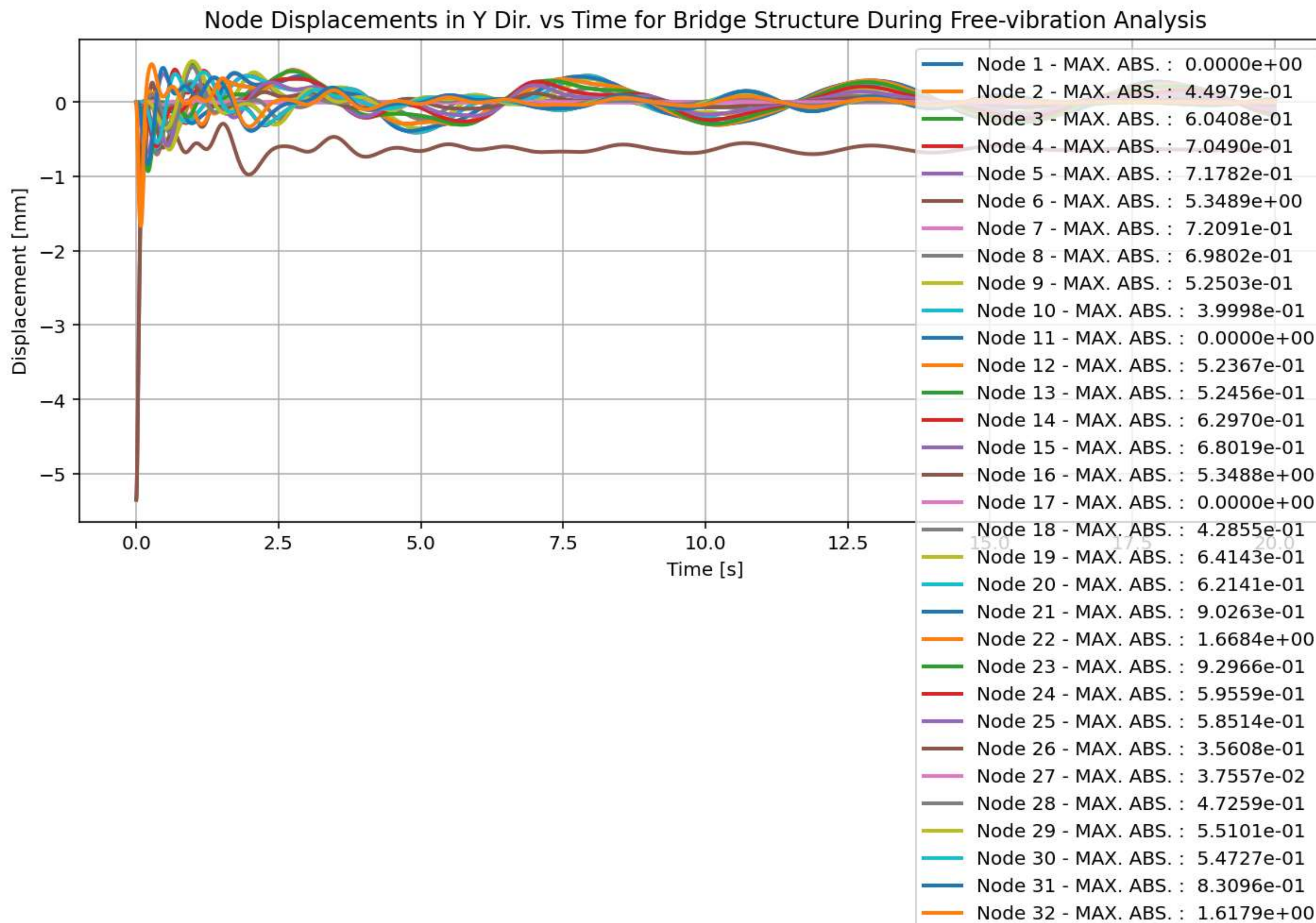

elements force in X Dir. vs Time for Bridge Structure During Free-vibration Analysis
1e6
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
Element Axial Force [N]
0.0
2.5
5.0
7.5
10.0
12.5
Time [s]
Ele. 1 - MAX. ABS. : 1.2550e+04
Ele. 2 - MAX. ABS. : 9.8068e+03
Ele. 3 - MAX. ABS. : 2.2157e+04
Ele. 4 - MAX. ABS. : 2.2075e+04
Ele. 5 - MAX. ABS. : 4.7533e+04
Ele. 6 - MAX. ABS. : 4.8182e+04
Ele. 7 - MAX. ABS. : 2.3549e+04
Ele. 8 - MAX. ABS. : 2.3912e+04
Ele. 9 - MAX. ABS. : 1.0535e+04
Ele. 10 - MAX. ABS. : 1.3520e+04
Ele. 11 - MAX. ABS. : 1.7260e+04
Ele. 12 - MAX. ABS. : 1.3097e+04
Ele. 13 - MAX. ABS. : 2.6486e+04
Ele. 14 - MAX. ABS. : 2.7002e+04
Ele. 15 - MAX. ABS. : 2.7954e+04
Ele. 16 - MAX. ABS. : 3.5637e+04
Ele. 17 - MAX. ABS. : 1.9385e+04
Ele. 18 - MAX. ABS. : 1.3369e+04
Ele. 19 - MAX. ABS. : 1.2273e+04
Ele. 20 - MAX. ABS. : 1.6710e+04
Ele. 21 - MAX. ABS. : 1.9257e+04
Ele. 22 - MAX. ABS. : 1.6190e+04
Ele. 23 - MAX. ABS. : 1.3088e+04
Ele. 24 - MAX. ABS. : 1.3095e+04
Ele. 25 - MAX. ABS. : 1.6147e+04
Ele. 26 - MAX. ABS. : 2.0247e+04
Ele. 27 - MAX. ABS. : 1.5039e+04
Ele. 28 - MAX. ABS. : 1.3216e+04
Ele. 29 - MAX. ABS. : 2.8182e+04
Ele. 30 - MAX. ABS. : 2.3308e+04
Ele. 31 - MAX. ABS. : 2.4829e+04
Ele. 32 - MAX. ABS. : 5.9529e+03
Ele. 33 - MAX. ABS. : 8.4477e+03
Ele. 34 - MAX. ABS. : 7.2904e+03
Ele. 35 - MAX. ABS. : 2.6154e+04
Ele. 36 - MAX. ABS. : 1.5654e+04
Ele. 37 - MAX. ABS. : 3.4909e+04
Ele. 38 - MAX. ABS. : 3.3952e+04
Ele. 39 - MAX. ABS. : 8.8577e+04
Ele. 40 - MAX. ABS. : 9.0947e+05
Ele. 41 - MAX. ABS. : 9.2324e+04
Ele. 42 - MAX. ABS. : 3.6286e+04
Ele. 43 - MAX. ABS. : 3.2890e+04
Ele. 44 - MAX. ABS. : 1.6618e+04
Ele. 45 - MAX. ABS. : 2.5572e+04
Ele. 46 - MAX. ABS. : 9.4687e+03
Ele. 47 - MAX. ABS. : 1.1798e+04
Ele. 48 - MAX. ABS. : 6.4711e+03
Ele. 49 - MAX. ABS. : 3.3179e+04
Ele. 50 - MAX. ABS. : 4.8190e+03
Ele. 51 - MAX. ABS. : 2.0066e+04
Ele. 52 - MAX. ABS. : 7.3422e+03
Ele. 53 - MAX. ABS. : 1.8559e+04
Ele. 54 - MAX. ABS. : 4.6146e+03
Ele. 55 - MAX. ABS. : 2.4088e+04
Ele. 56 - MAX. ABS. : 1.5357e+04
Ele. 57 - MAX. ABS. : 4.7193e+04
Ele. 58 - MAX. ABS. : 2.9637e+04
Ele. 59 - MAX. ABS. : 9.0019e+04
Ele. 60 - MAX. ABS. : 9.7948e+05
Ele. 61 - MAX. ABS. : 1.0605e+05

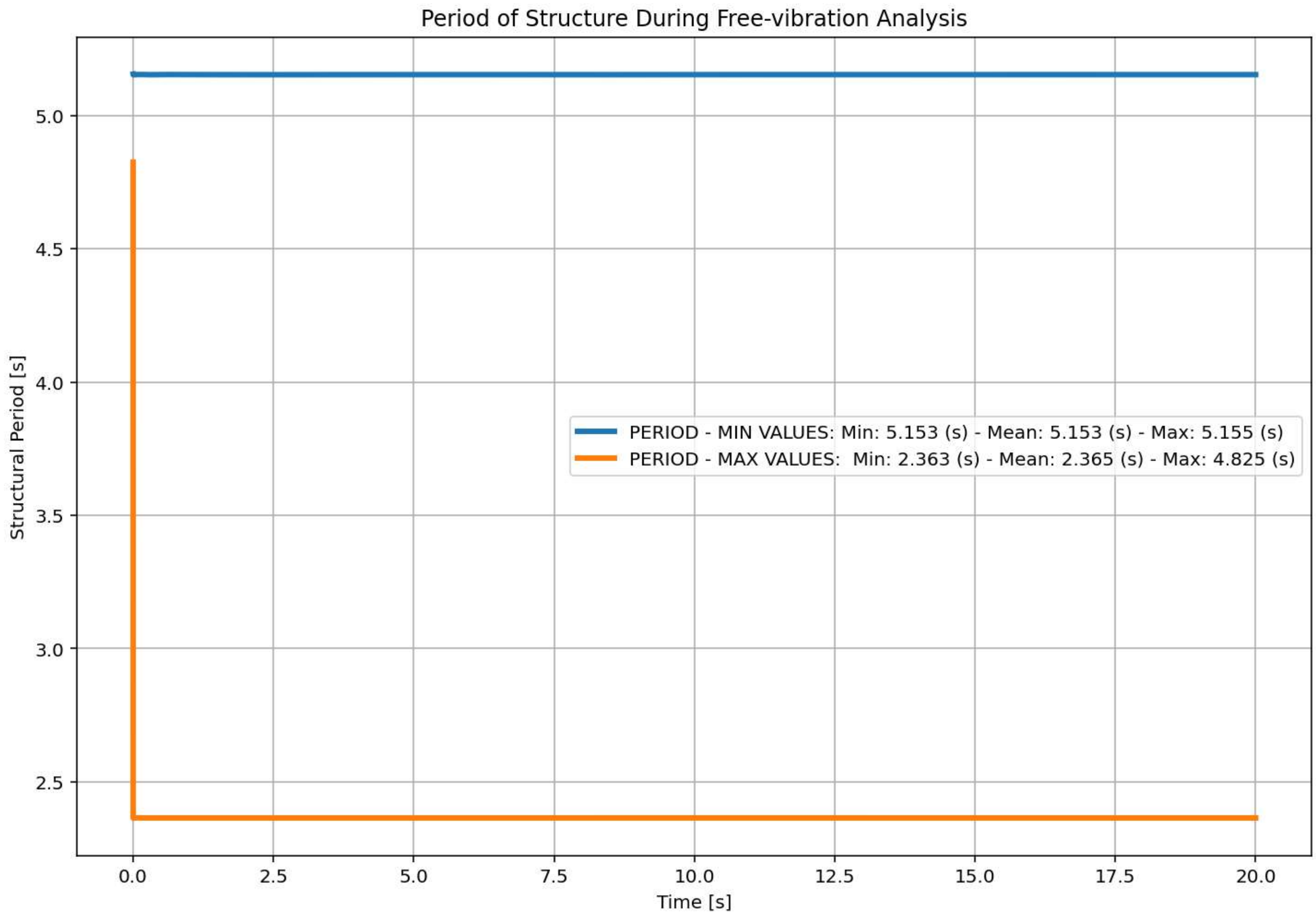

elements Stress in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Element Stress [N/mm^2]
Time [s]
Ele. 1 - MAX. ABS. : 3.1235e+00
Ele. 2 - MAX. ABS. : 2.4407e+00
Ele. 3 - MAX. ABS. : 5.5145e+00
Ele. 4 - MAX. ABS. : 5.4940e+00
Ele. 5 - MAX. ABS. : 1.1830e+01
Ele. 6 - MAX. ABS. : 1.1992e+01
Ele. 7 - MAX. ABS. : 5.8609e+00
Ele. 8 - MAX. ABS. : 5.9512e+00
Ele. 9 - MAX. ABS. : 2.6219e+00
Ele. 10 - MAX. ABS. : 3.3649e+00
Ele. 11 - MAX. ABS. : 4.2956e+00
Ele. 12 - MAX. ABS. : 3.2596e+00
Ele. 13 - MAX. ABS. : 6.5918e+00
Ele. 14 - MAX. ABS. : 6.7204e+00
Ele. 15 - MAX. ABS. : 6.9572e+00
Ele. 16 - MAX. ABS. : 8.8693e+00
Ele. 17 - MAX. ABS. : 4.8246e+00
Ele. 18 - MAX. ABS. : 3.3274e+00
Ele. 19 - MAX. ABS. : 3.0544e+00
Ele. 20 - MAX. ABS. : 4.1589e+00
Ele. 21 - MAX. ABS. : 4.7928e+00
Ele. 22 - MAX. ABS. : 4.0294e+00
Ele. 23 - MAX. ABS. : 3.2573e+00
Ele. 24 - MAX. ABS. : 3.2591e+00
Ele. 25 - MAX. ABS. : 4.0187e+00
Ele. 26 - MAX. ABS. : 5.0392e+00
Ele. 27 - MAX. ABS. : 3.7430e+00
Ele. 28 - MAX. ABS. : 3.2891e+00
Ele. 29 - MAX. ABS. : 7.0140e+00
Ele. 30 - MAX. ABS. : 5.8010e+00
Ele. 31 - MAX. ABS. : 6.0854e+00
Ele. 32 - MAX. ABS. : 1.4590e+00
Ele. 33 - MAX. ABS. : 2.0705e+00
Ele. 34 - MAX. ABS. : 1.7869e+00
Ele. 35 - MAX. ABS. : 6.4103e+00
Ele. 36 - MAX. ABS. : 3.8367e+00
Ele. 37 - MAX. ABS. : 8.5561e+00
Ele. 38 - MAX. ABS. : 8.3214e+00
Ele. 39 - MAX. ABS. : 2.1710e+01
Ele. 40 - MAX. ABS. : 2.2291e+02
Ele. 41 - MAX. ABS. : 2.2628e+01
Ele. 42 - MAX. ABS. : 8.8937e+00
Ele. 43 - MAX. ABS. : 8.0614e+00
Ele. 44 - MAX. ABS. : 4.0731e+00
Ele. 45 - MAX. ABS. : 6.2676e+00
Ele. 46 - MAX. ABS. : 2.3208e+00
Ele. 47 - MAX. ABS. : 2.8918e+00
Ele. 48 - MAX. ABS. : 1.5861e+00
Ele. 49 - MAX. ABS. : 8.1321e+00
Ele. 50 - MAX. ABS. : 1.1811e+00
Ele. 51 - MAX. ABS. : 4.9182e+00
Ele. 52 - MAX. ABS. : 1.7996e+00
Ele. 53 - MAX. ABS. : 4.5487e+00
Ele. 54 - MAX. ABS. : 1.1310e+00
Ele. 55 - MAX. ABS. : 5.9039e+00
Ele. 56 - MAX. ABS. : 3.7640e+00
Ele. 57 - MAX. ABS. : 1.1567e+01
Ele. 58 - MAX. ABS. : 7.2639e+00
Ele. 59 - MAX. ABS. : 2.2063e+01
Ele. 60 - MAX. ABS. : 2.4007e+02
Ele. 61 - MAX. ABS. : 2.5993e+01

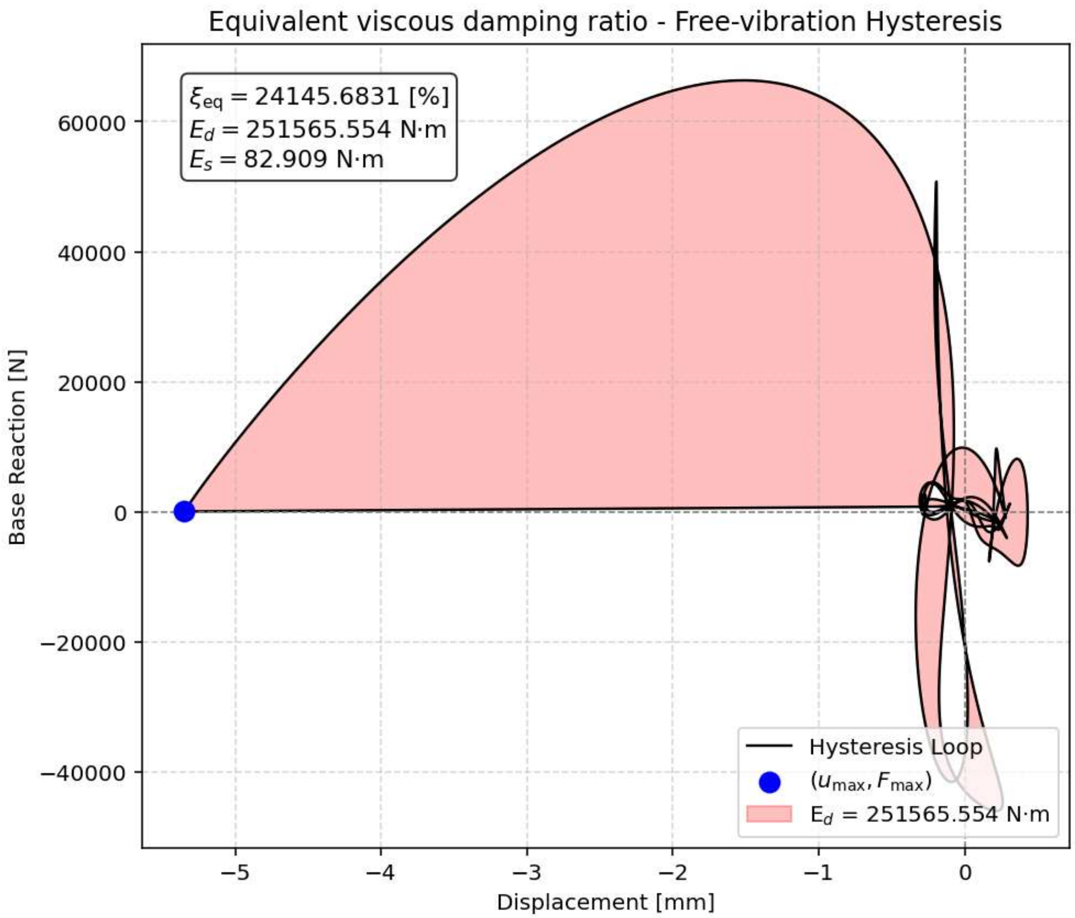

elements Strain in X Dir. vs Time for Bridge Structure During Dynamic External Time-dependent Loading Analysis
Element Strain [mm]
Time [s]
0.00125
0.00100
0.00075
0.00050
0.00025
0.00000
−0.00025
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 1.5618e-05
Ele. 2 - MAX. ABS. : 1.2204e-05
Ele. 3 - MAX. ABS. : 2.7572e-05
Ele. 4 - MAX. ABS. : 2.7470e-05
Ele. 5 - MAX. ABS. : 5.9150e-05
Ele. 6 - MAX. ABS. : 5.9958e-05
Ele. 7 - MAX. ABS. : 2.9305e-05
Ele. 8 - MAX. ABS. : 2.9756e-05
Ele. 9 - MAX. ABS. : 1.3110e-05
Ele. 10 - MAX. ABS. : 1.6824e-05
Ele. 11 - MAX. ABS. : 2.1478e-05
Ele. 12 - MAX. ABS. : 1.6298e-05
Ele. 13 - MAX. ABS. : 3.2959e-05
Ele. 14 - MAX. ABS. : 3.3602e-05
Ele. 15 - MAX. ABS. : 3.4786e-05
Ele. 16 - MAX. ABS. : 4.4347e-05
Ele. 17 - MAX. ABS. : 2.4123e-05
Ele. 18 - MAX. ABS. : 1.6637e-05
Ele. 19 - MAX. ABS. : 1.5272e-05
Ele. 20 - MAX. ABS. : 2.0794e-05
Ele. 21 - MAX. ABS. : 2.3964e-05
Ele. 22 - MAX. ABS. : 2.0147e-05
Ele. 23 - MAX. ABS. : 1.6287e-05
Ele. 24 - MAX. ABS. : 1.6295e-05
Ele. 25 - MAX. ABS. : 2.0093e-05
Ele. 26 - MAX. ABS. : 2.5196e-05
Ele. 27 - MAX. ABS. : 1.8715e-05
Ele. 28 - MAX. ABS. : 1.6445e-05
Ele. 29 - MAX. ABS. : 3.5070e-05
Ele. 30 - MAX. ABS. : 2.9005e-05
Ele. 31 - MAX. ABS. : 3.0427e-05
Ele. 32 - MAX. ABS. : 7.2952e-06
Ele. 33 - MAX. ABS. : 1.0353e-05
Ele. 34 - MAX. ABS. : 8.9343e-06
Ele. 35 - MAX. ABS. : 3.2052e-05
Ele. 36 - MAX. ABS. : 1.9183e-05
Ele. 37 - MAX. ABS. : 4.2781e-05
Ele. 38 - MAX. ABS. : 4.1607e-05
Ele. 39 - MAX. ABS. : 1.0855e-04
Ele. 40 - MAX. ABS. : 1.1146e-03
Ele. 41 - MAX. ABS. : 1.1314e-04
Ele. 42 - MAX. ABS. : 4.4469e-05
Ele. 43 - MAX. ABS. : 4.0307e-05
Ele. 44 - MAX. ABS. : 2.0366e-05
Ele. 45 - MAX. ABS. : 3.1338e-05
Ele. 46 - MAX. ABS. : 1.1604e-05
Ele. 47 - MAX. ABS. : 1.4459e-05
Ele. 48 - MAX. ABS. : 7.9303e-06
Ele. 49 - MAX. ABS. : 4.0661e-05
Ele. 50 - MAX. ABS. : 5.9057e-06
Ele. 51 - MAX. ABS. : 2.4591e-05
Ele. 52 - MAX. ABS. : 8.9978e-06
Ele. 53 - MAX. ABS. : 2.2744e-05
Ele. 54 - MAX. ABS. : 5.6551e-06
Ele. 55 - MAX. ABS. : 2.9519e-05
Ele. 56 - MAX. ABS. : 1.8820e-05
Ele. 57 - MAX. ABS. : 5.7834e-05
Ele. 58 - MAX. ABS. : 3.6319e-05
Ele. 59 - MAX. ABS. : 1.1032e-04
Ele. 60 - MAX. ABS. : 1.3896e-03
Ele. 61 - MAX. ABS. : 1.2996e-04


elements Damage-index [%] in X Dir. vs Time for Bridge Structure During Dynamic Free-vibration Analysis
Element Damage-index [%]
Time [s]
0.00125
0.00100
0.00075
0.00050
0.00025
0.00000
−0.00025
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 1.5618e-05
Ele. 2 - MAX. ABS. : 1.2204e-05
Ele. 3 - MAX. ABS. : 2.7572e-05
Ele. 4 - MAX. ABS. : 2.7470e-05
Ele. 5 - MAX. ABS. : 5.9150e-05
Ele. 6 - MAX. ABS. : 5.9958e-05
Ele. 7 - MAX. ABS. : 2.9305e-05
Ele. 8 - MAX. ABS. : 2.9756e-05
Ele. 9 - MAX. ABS. : 1.3110e-05
Ele. 10 - MAX. ABS. : 1.6824e-05
Ele. 11 - MAX. ABS. : 2.1478e-05
Ele. 12 - MAX. ABS. : 1.6298e-05
Ele. 13 - MAX. ABS. : 3.2959e-05
Ele. 14 - MAX. ABS. : 3.3602e-05
Ele. 15 - MAX. ABS. : 3.4786e-05
Ele. 16 - MAX. ABS. : 4.4347e-05
Ele. 17 - MAX. ABS. : 2.4123e-05
Ele. 18 - MAX. ABS. : 1.6637e-05
Ele. 19 - MAX. ABS. : 1.5272e-05
Ele. 20 - MAX. ABS. : 2.0794e-05
Ele. 21 - MAX. ABS. : 2.3964e-05
Ele. 22 - MAX. ABS. : 2.0147e-05
Ele. 23 - MAX. ABS. : 1.6287e-05
Ele. 24 - MAX. ABS. : 1.6295e-05
Ele. 25 - MAX. ABS. : 2.0093e-05
Ele. 26 - MAX. ABS. : 2.5196e-05
Ele. 27 - MAX. ABS. : 1.8715e-05
Ele. 28 - MAX. ABS. : 1.6445e-05
Ele. 29 - MAX. ABS. : 3.5070e-05
Ele. 30 - MAX. ABS. : 2.9005e-05
Ele. 31 - MAX. ABS. : 3.0427e-05
Ele. 32 - MAX. ABS. : 7.2952e-06
Ele. 33 - MAX. ABS. : 1.0353e-05
Ele. 34 - MAX. ABS. : 8.9343e-06
Ele. 35 - MAX. ABS. : 3.2052e-05
Ele. 36 - MAX. ABS. : 1.9183e-05
Ele. 37 - MAX. ABS. : 4.2781e-05
Ele. 38 - MAX. ABS. : 4.1607e-05
Ele. 39 - MAX. ABS. : 1.0855e-04
Ele. 40 - MAX. ABS. : 1.1146e-03
Ele. 41 - MAX. ABS. : 1.1314e-04
Ele. 42 - MAX. ABS. : 4.4469e-05
Ele. 43 - MAX. ABS. : 4.0307e-05
Ele. 44 - MAX. ABS. : 2.0366e-05
Ele. 45 - MAX. ABS. : 3.1338e-05
Ele. 46 - MAX. ABS. : 1.1604e-05
Ele. 47 - MAX. ABS. : 1.4459e-05
Ele. 48 - MAX. ABS. : 7.9303e-06
Ele. 49 - MAX. ABS. : 4.0661e-05
Ele. 50 - MAX. ABS. : 5.9057e-06
Ele. 51 - MAX. ABS. : 2.4591e-05
Ele. 52 - MAX. ABS. : 8.9978e-06
Ele. 53 - MAX. ABS. : 2.2744e-05
Ele. 54 - MAX. ABS. : 5.6551e-06
Ele. 55 - MAX. ABS. : 2.9519e-05
Ele. 56 - MAX. ABS. : 1.8820e-05
Ele. 57 - MAX. ABS. : 5.7834e-05
Ele. 58 - MAX. ABS. : 3.6319e-05
Ele. 59 - MAX. ABS. : 1.1032e-04
Ele. 60 - MAX. ABS. : 1.3896e-03
Ele. 61 - MAX. ABS. : 1.2996e-04

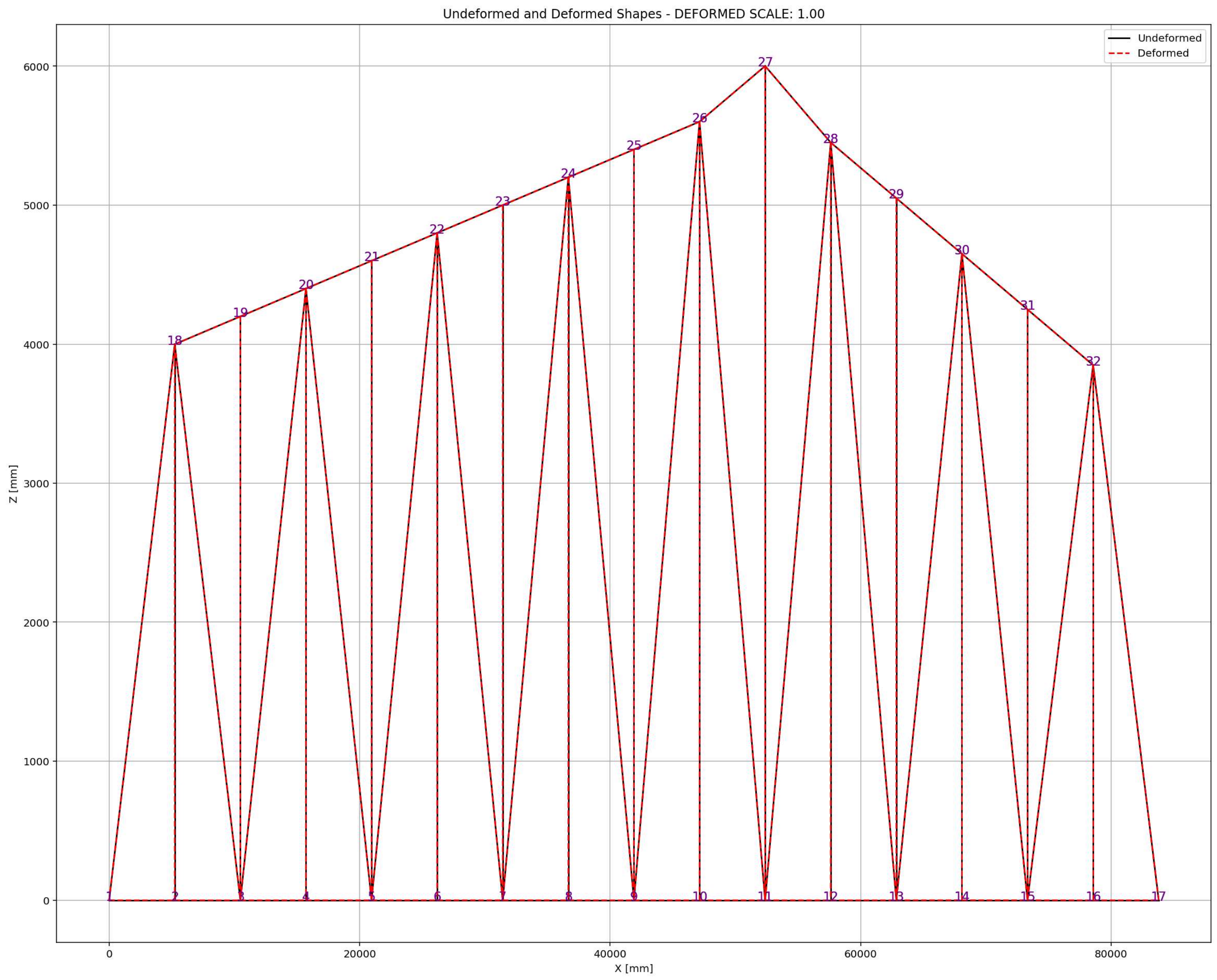

Node Displacements in X Dir. vs Time for Bridge Structure During Free-vibration Analysis
Displacement [mm]
Time [s]
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 8.8086e-02
Node 3 - MAX. ABS. : 1.6384e-01
Node 4 - MAX. ABS. : 2.4340e-01
Node 5 - MAX. ABS. : 3.2437e-01
Node 6 - MAX. ABS. : 1.1713e-01
Node 7 - MAX. ABS. : 3.4190e-01
Node 8 - MAX. ABS. : 2.5480e-01
Node 9 - MAX. ABS. : 1.5837e-01
Node 10 - MAX. ABS. : 9.5063e-02
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 1.2380e-01
Node 13 - MAX. ABS. : 2.0991e-01
Node 14 - MAX. ABS. : 3.1001e-01
Node 15 - MAX. ABS. : 4.1022e-01
Node 16 - MAX. ABS. : 2.5101e-01
Node 17 - MAX. ABS. : 0.0000e+00
Node 18 - MAX. ABS. : 1.9775e-01
Node 19 - MAX. ABS. : 1.9371e-01
Node 20 - MAX. ABS. : 2.1265e-01
Node 21 - MAX. ABS. : 1.4336e-01
Node 22 - MAX. ABS. : 1.1368e-01
Node 23 - MAX. ABS. : 1.1601e-01
Node 24 - MAX. ABS. : 2.1086e-01
Node 25 - MAX. ABS. : 1.9626e-01
Node 26 - MAX. ABS. : 1.8899e-01
Node 27 - MAX. ABS. : 1.2157e-01
Node 28 - MAX. ABS. : 1.2024e-01
Node 29 - MAX. ABS. : 1.1712e-01
Node 30 - MAX. ABS. : 1.3050e-01
Node 31 - MAX. ABS. : 3.6356e-01
Node 32 - MAX. ABS. : 3.7001e-01


Node Displacements in Y Dir. vs Time for Bridge Structure During Free-vibration Analysis
Displacement [mm]
Time [s]
0
−1
−2
−3
−4
−5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 4.4979e-01
Node 3 - MAX. ABS. : 6.0408e-01
Node 4 - MAX. ABS. : 7.0490e-01
Node 5 - MAX. ABS. : 7.1782e-01
Node 6 - MAX. ABS. : 5.3489e+00
Node 7 - MAX. ABS. : 7.2091e-01
Node 8 - MAX. ABS. : 6.9802e-01
Node 9 - MAX. ABS. : 5.2503e-01
Node 10 - MAX. ABS. : 3.9998e-01
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 5.2367e-01
Node 13 - MAX. ABS. : 5.2456e-01
Node 14 - MAX. ABS. : 6.2970e-01
Node 15 - MAX. ABS. : 6.8019e-01
Node 16 - MAX. ABS. : 5.3488e+00
Node 17 - MAX. ABS. : 0.0000e+00
Node 18 - MAX. ABS. : 4.2855e-01
Node 19 - MAX. ABS. : 6.4143e-01
Node 20 - MAX. ABS. : 6.2141e-01
Node 21 - MAX. ABS. : 9.0263e-01
Node 22 - MAX. ABS. : 1.6684e+00
Node 23 - MAX. ABS. : 9.2966e-01
Node 24 - MAX. ABS. : 5.9559e-01
Node 25 - MAX. ABS. : 5.8514e-01
Node 26 - MAX. ABS. : 3.5608e-01
Node 27 - MAX. ABS. : 3.7557e-02
Node 28 - MAX. ABS. : 4.7259e-01
Node 29 - MAX. ABS. : 5.5101e-01
Node 30 - MAX. ABS. : 5.4727e-01
Node 31 - MAX. ABS. : 8.3096e-01
Node 32 - MAX. ABS. : 1.6179e+00

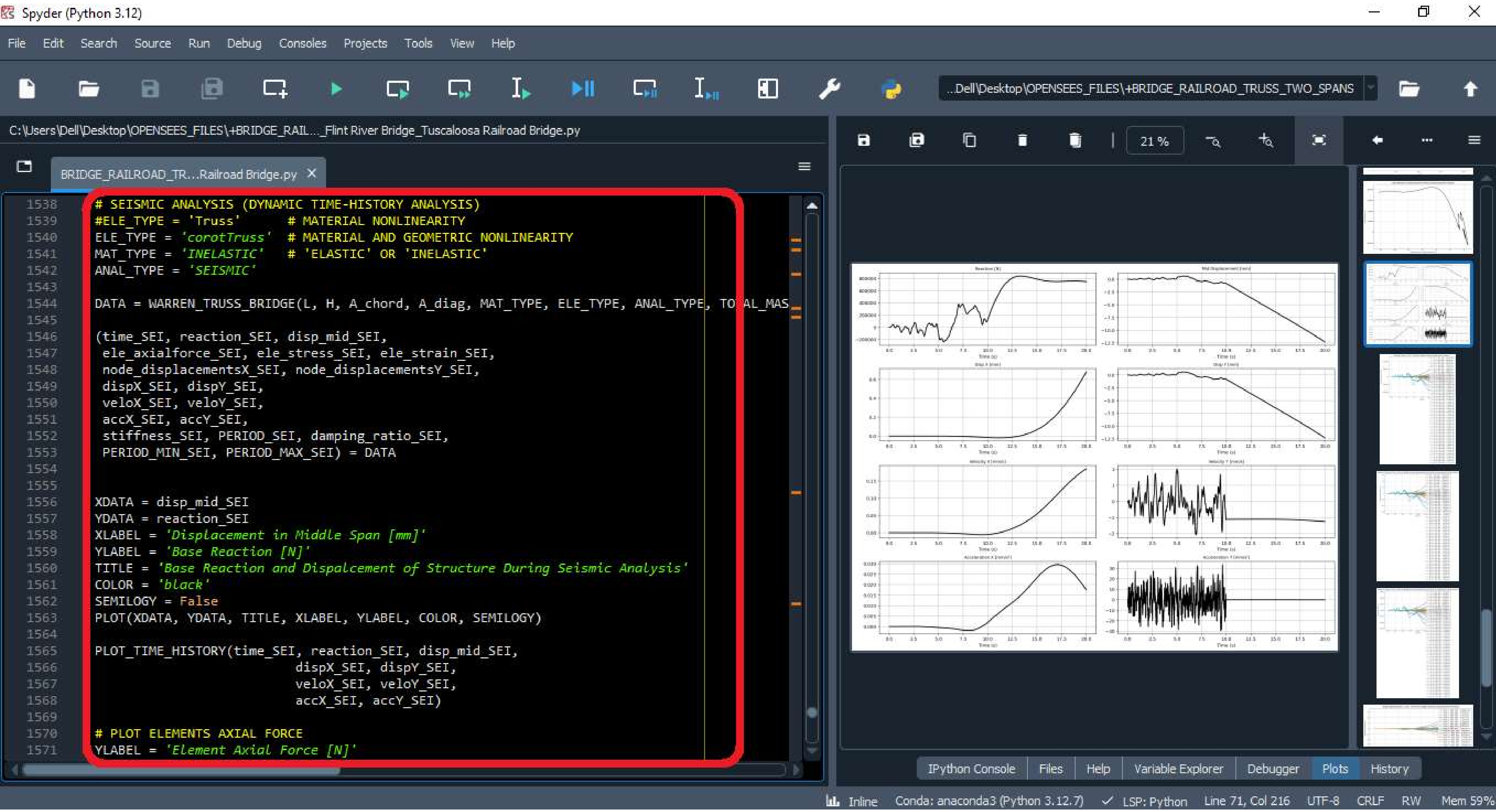

Period of Structure During Free-vibration Analysis
Structural Period [s]
Time [s]
PERIOD - MIN VALUES: Min: 5.153 (s) - Mean: 5.153 (s) - Max: 5.155 (s)
PERIOD - MAX VALUES: Min: 2.363 (s) - Mean: 2.365 (s) - Max: 4.825 (s)
5.0
4.5
4.0
3.5
3.0
2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



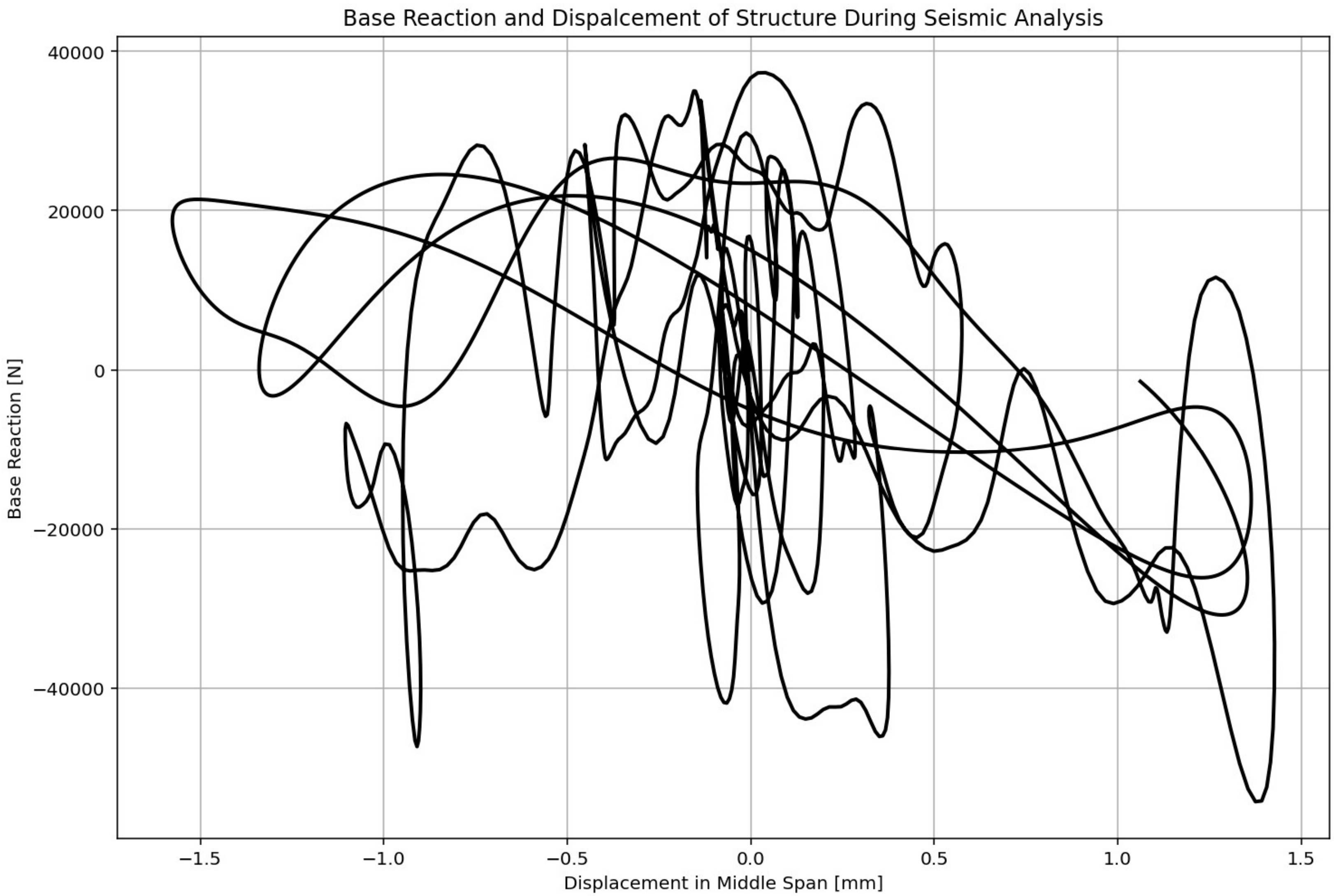
Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 24145.6831$ [%]
$E_d = 251565.554$ N·m
$E_s = 82.909$ N·m
60000
40000
20000
0
−20000
−40000
Base Reaction [N]
−5
−4
−3
−2
−1
0
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E$_d$ = 251565.554 N·m

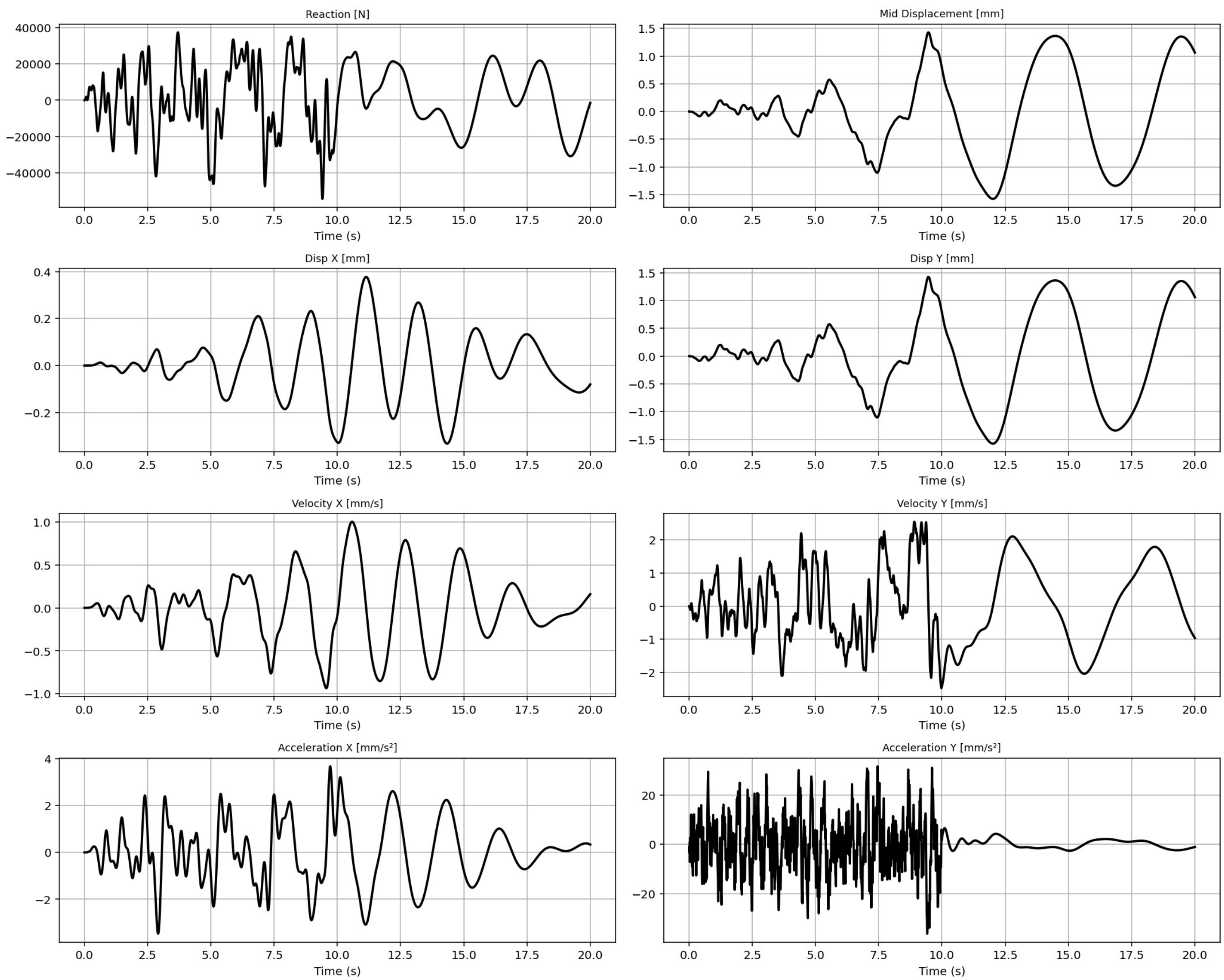

Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
1000
2000
3000
4000
5000
6000
0
20000
40000
60000
80000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32

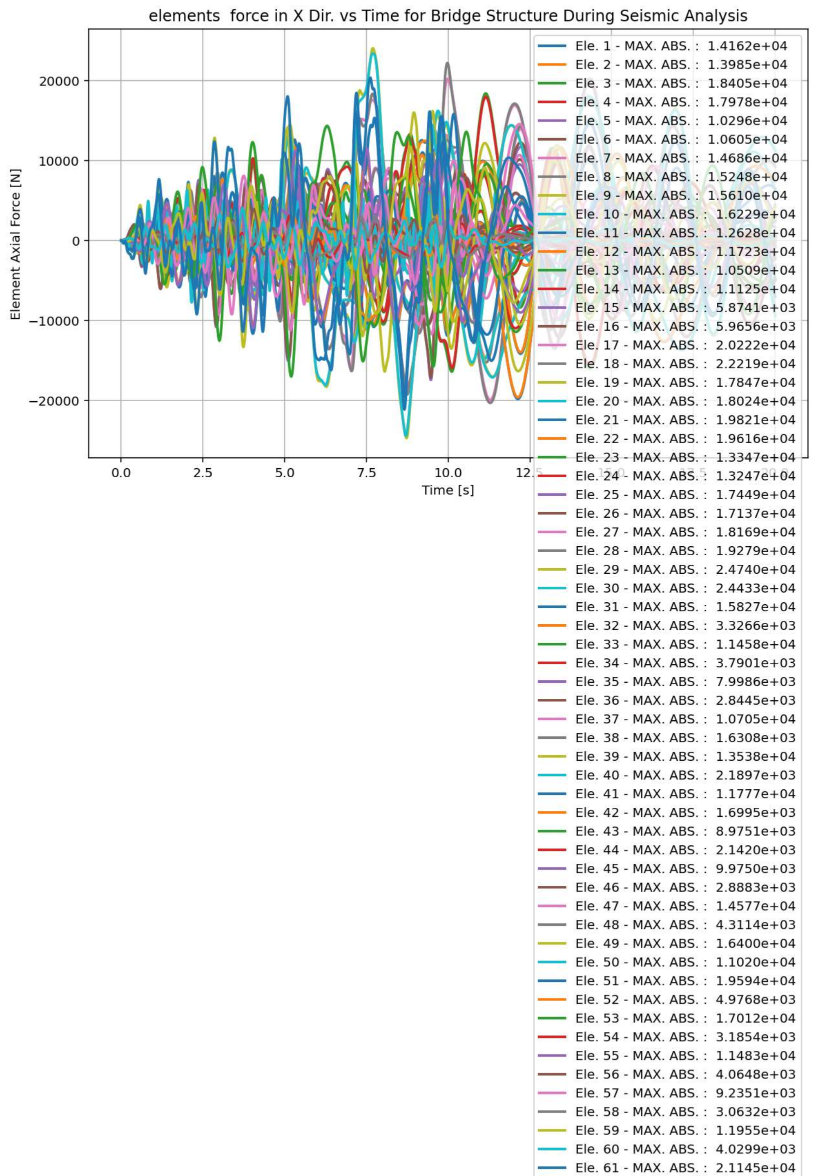
SEISMIC ANALYSIS

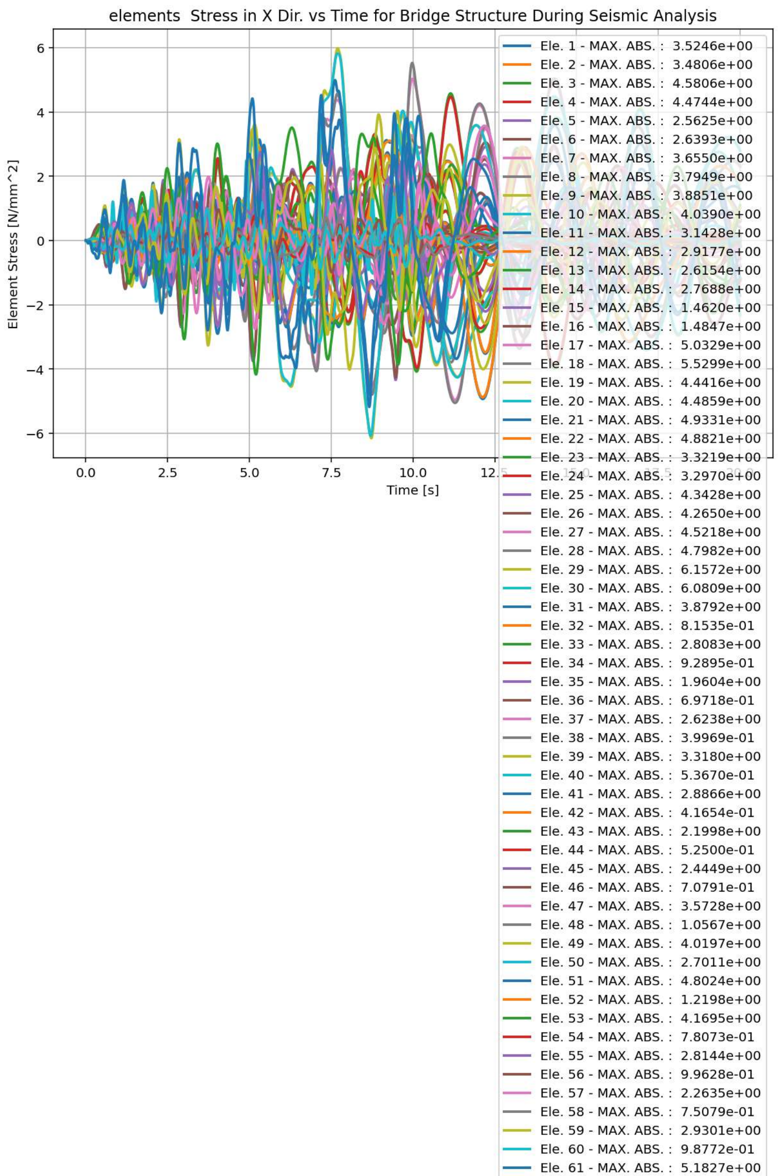


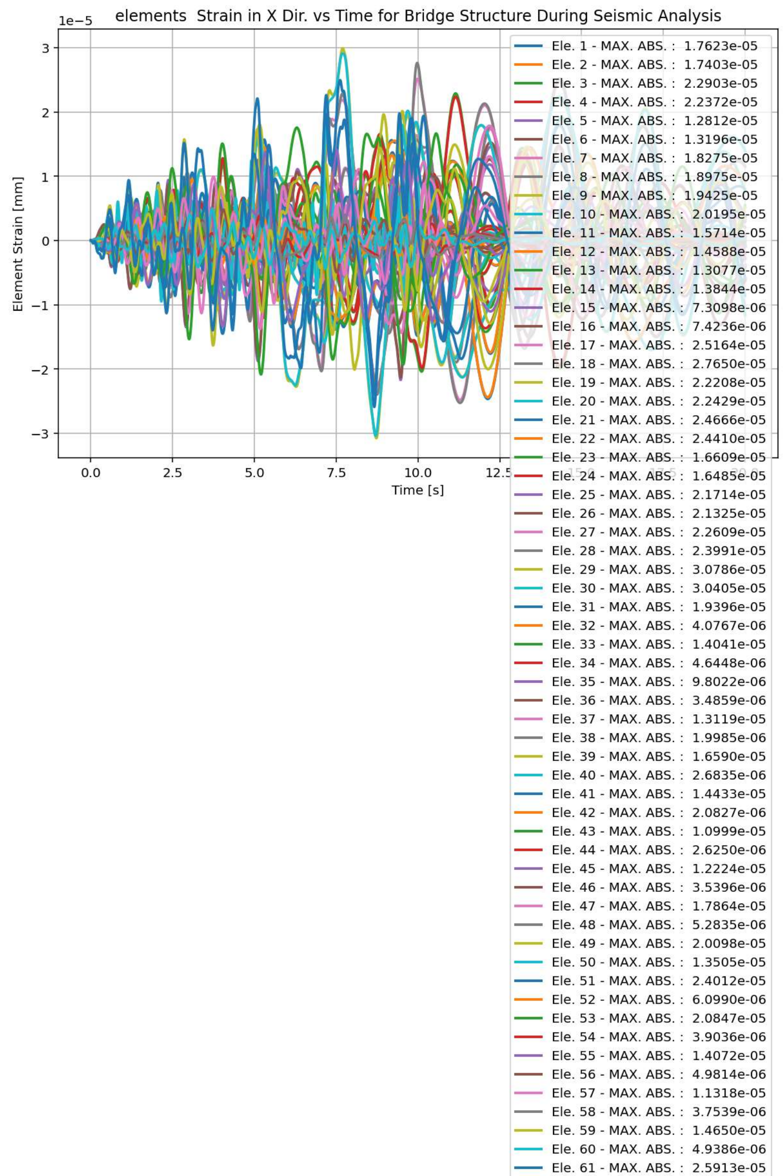
Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAILROAD_TRUSS_TWO_SPANS
C:\Users\Dell\Desktop\OPENSEES_FILES\+BRIDGE_RAIL..._Flint River Bridge_Tuscaloosa Railroad Bridge.py
BRIDGE_RAILROAD_TR...Railroad Bridge.py
SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITY
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'SEISMIC'
DATA = WARREN_TRUSS_BRIDGE(L, H, A_chord, A_diag, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TO AL_MAS
(time_SEI, reaction_SEI, disp_mid_SEI,
 ele_axialforce_SEI, ele_stress_SEI, ele_strain_SEI,
 node_displacementsX_SEI, node_displacementsY_SEI,
 dispX_SEI, dispY_SEI,
 veloX_SEI, veloY_SEI,
 accX_SEI, accY_SEI,
 stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
 PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
XDATA = disp_mid_SEI
YDATA = reaction_SEI
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_mid_SEI,
                  dispX_SEI, dispY_SEI,
                  veloX_SEI, veloY_SEI,
                  accX_SEI, accY_SEI)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
21 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 71, Col 216 UTF-8 CRLF RW Mem 59%

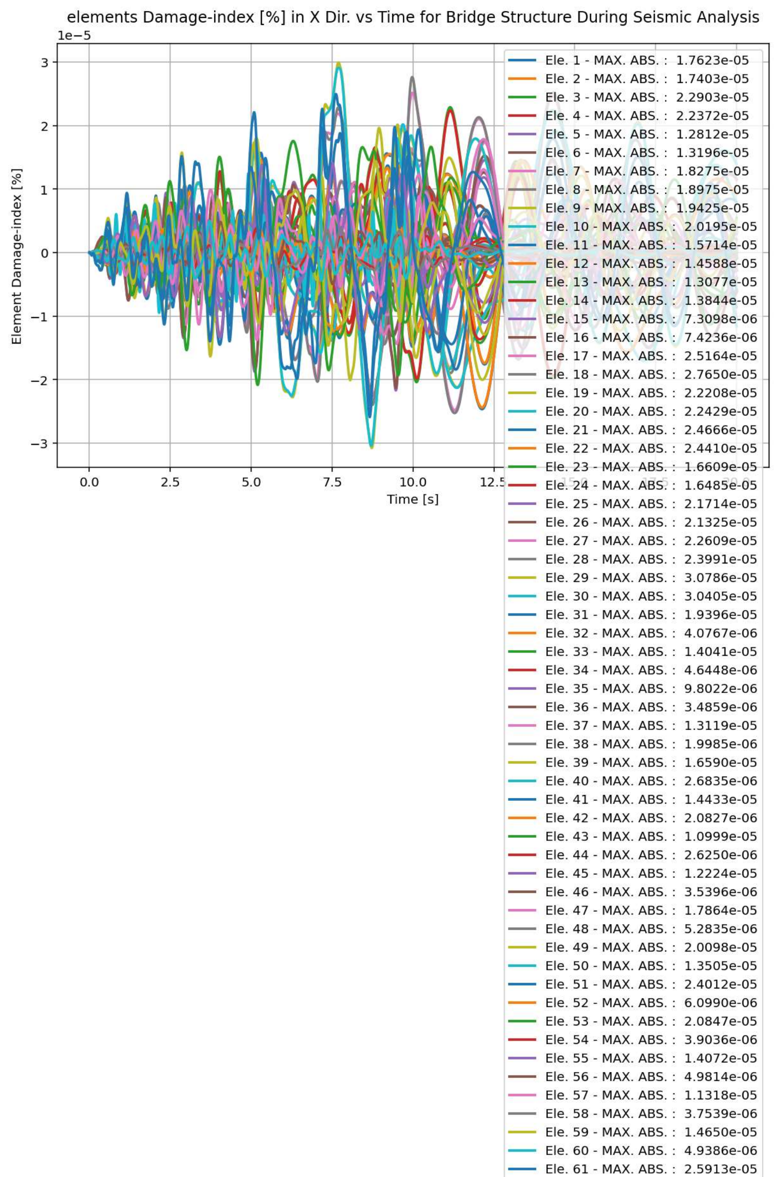

Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
40000
20000
0
−20000
−40000
Displacement in Middle Span [mm]
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5


Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)

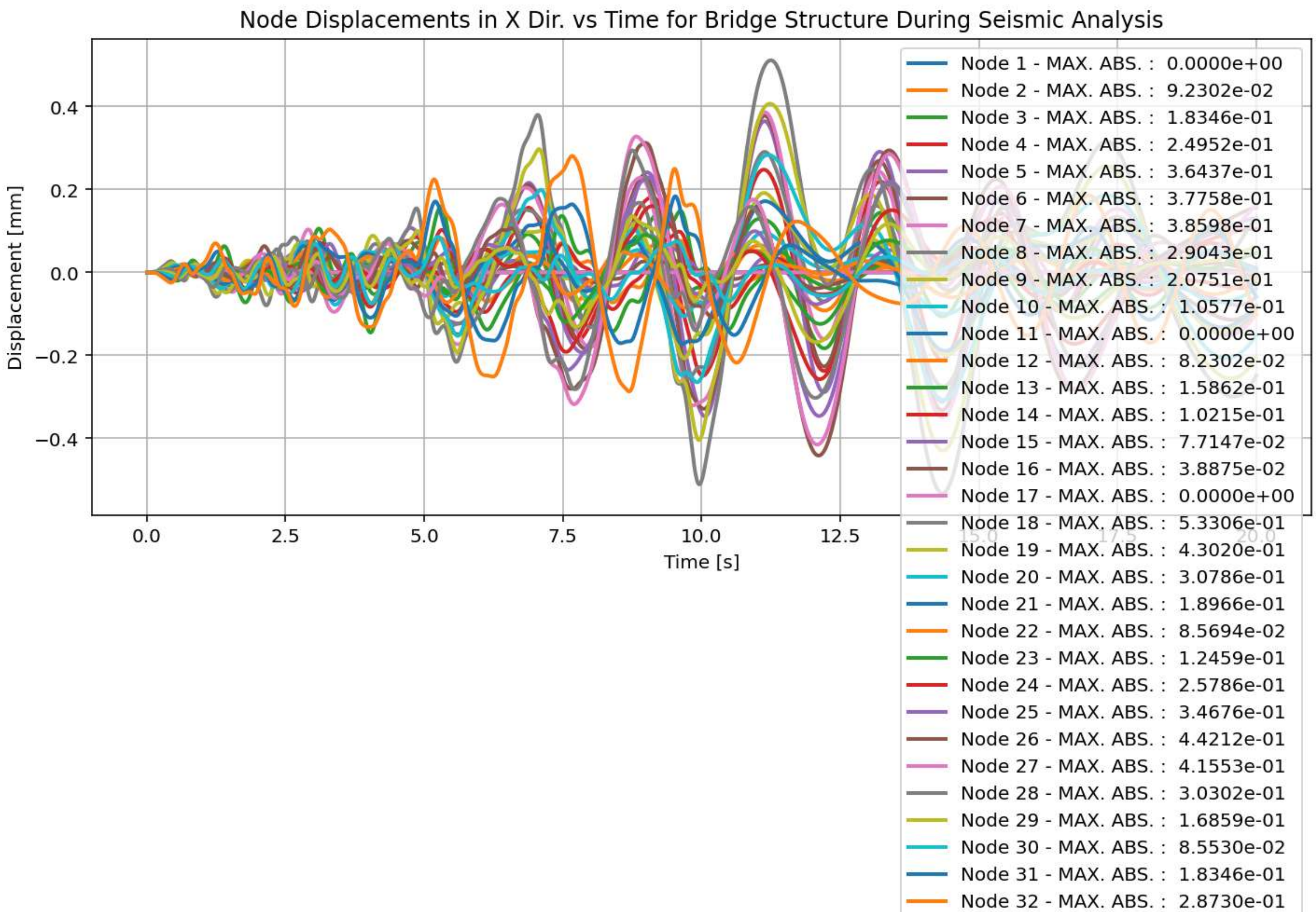

elements force in X Dir. vs Time for Bridge Structure During Seismic Analysis
Element Axial Force [N]
Time [s]
20000
10000
0
−10000
−20000
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 1.4162e+04
Ele. 2 - MAX. ABS. : 1.3985e+04
Ele. 3 - MAX. ABS. : 1.8405e+04
Ele. 4 - MAX. ABS. : 1.7978e+04
Ele. 5 - MAX. ABS. : 1.0296e+04
Ele. 6 - MAX. ABS. : 1.0605e+04
Ele. 7 - MAX. ABS. : 1.4686e+04
Ele. 8 - MAX. ABS. : 1.5248e+04
Ele. 9 - MAX. ABS. : 1.5610e+04
Ele. 10 - MAX. ABS. : 1.6229e+04
Ele. 11 - MAX. ABS. : 1.2628e+04
Ele. 12 - MAX. ABS. : 1.1723e+04
Ele. 13 - MAX. ABS. : 1.0509e+04
Ele. 14 - MAX. ABS. : 1.1125e+04
Ele. 15 - MAX. ABS. : 5.8741e+03
Ele. 16 - MAX. ABS. : 5.9656e+03
Ele. 17 - MAX. ABS. : 2.0222e+04
Ele. 18 - MAX. ABS. : 2.2219e+04
Ele. 19 - MAX. ABS. : 1.7847e+04
Ele. 20 - MAX. ABS. : 1.8024e+04
Ele. 21 - MAX. ABS. : 1.9821e+04
Ele. 22 - MAX. ABS. : 1.9616e+04
Ele. 23 - MAX. ABS. : 1.3347e+04
Ele. 24 - MAX. ABS. : 1.3247e+04
Ele. 25 - MAX. ABS. : 1.7449e+04
Ele. 26 - MAX. ABS. : 1.7137e+04
Ele. 27 - MAX. ABS. : 1.8169e+04
Ele. 28 - MAX. ABS. : 1.9279e+04
Ele. 29 - MAX. ABS. : 2.4740e+04
Ele. 30 - MAX. ABS. : 2.4433e+04
Ele. 31 - MAX. ABS. : 1.5827e+04
Ele. 32 - MAX. ABS. : 3.3266e+03
Ele. 33 - MAX. ABS. : 1.1458e+04
Ele. 34 - MAX. ABS. : 3.7901e+03
Ele. 35 - MAX. ABS. : 7.9986e+03
Ele. 36 - MAX. ABS. : 2.8445e+03
Ele. 37 - MAX. ABS. : 1.0705e+04
Ele. 38 - MAX. ABS. : 1.6308e+03
Ele. 39 - MAX. ABS. : 1.3538e+04
Ele. 40 - MAX. ABS. : 2.1897e+03
Ele. 41 - MAX. ABS. : 1.1777e+04
Ele. 42 - MAX. ABS. : 1.6995e+03
Ele. 43 - MAX. ABS. : 8.9751e+03
Ele. 44 - MAX. ABS. : 2.1420e+03
Ele. 45 - MAX. ABS. : 9.9750e+03
Ele. 46 - MAX. ABS. : 2.8883e+03
Ele. 47 - MAX. ABS. : 1.4577e+04
Ele. 48 - MAX. ABS. : 4.3114e+03
Ele. 49 - MAX. ABS. : 1.6400e+04
Ele. 50 - MAX. ABS. : 1.1020e+04
Ele. 51 - MAX. ABS. : 1.9594e+04
Ele. 52 - MAX. ABS. : 4.9768e+03
Ele. 53 - MAX. ABS. : 1.7012e+04
Ele. 54 - MAX. ABS. : 3.1854e+03
Ele. 55 - MAX. ABS. : 1.1483e+04
Ele. 56 - MAX. ABS. : 4.0648e+03
Ele. 57 - MAX. ABS. : 9.2351e+03
Ele. 58 - MAX. ABS. : 3.0632e+03
Ele. 59 - MAX. ABS. : 1.1955e+04
Ele. 60 - MAX. ABS. : 4.0299e+03
Ele. 61 - MAX. ABS. : 2.1145e+04

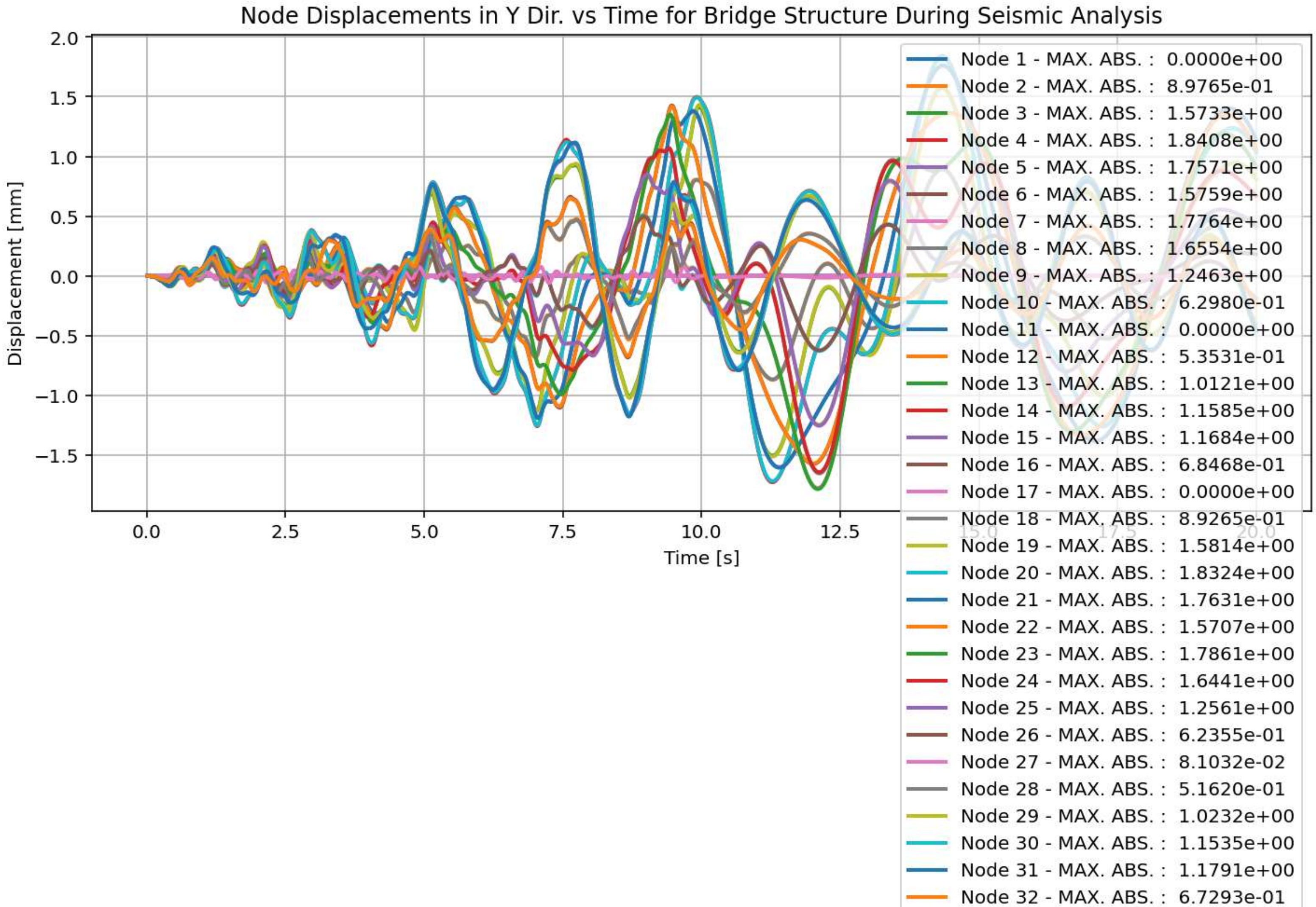

elements Stress in X Dir. vs Time for Bridge Structure During Seismic Analysis
Element Stress [N/mm^2]
Time [s]
6
4
2
0
−2
−4
−6
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 3.5246e+00
Ele. 2 - MAX. ABS. : 3.4806e+00
Ele. 3 - MAX. ABS. : 4.5806e+00
Ele. 4 - MAX. ABS. : 4.4744e+00
Ele. 5 - MAX. ABS. : 2.5625e+00
Ele. 6 - MAX. ABS. : 2.6393e+00
Ele. 7 - MAX. ABS. : 3.6550e+00
Ele. 8 - MAX. ABS. : 3.7949e+00
Ele. 9 - MAX. ABS. : 3.8851e+00
Ele. 10 - MAX. ABS. : 4.0390e+00
Ele. 11 - MAX. ABS. : 3.1428e+00
Ele. 12 - MAX. ABS. : 2.9177e+00
Ele. 13 - MAX. ABS. : 2.6154e+00
Ele. 14 - MAX. ABS. : 2.7688e+00
Ele. 15 - MAX. ABS. : 1.4620e+00
Ele. 16 - MAX. ABS. : 1.4847e+00
Ele. 17 - MAX. ABS. : 5.0329e+00
Ele. 18 - MAX. ABS. : 5.5299e+00
Ele. 19 - MAX. ABS. : 4.4416e+00
Ele. 20 - MAX. ABS. : 4.4859e+00
Ele. 21 - MAX. ABS. : 4.9331e+00
Ele. 22 - MAX. ABS. : 4.8821e+00
Ele. 23 - MAX. ABS. : 3.3219e+00
Ele. 24 - MAX. ABS. : 3.2970e+00
Ele. 25 - MAX. ABS. : 4.3428e+00
Ele. 26 - MAX. ABS. : 4.2650e+00
Ele. 27 - MAX. ABS. : 4.5218e+00
Ele. 28 - MAX. ABS. : 4.7982e+00
Ele. 29 - MAX. ABS. : 6.1572e+00
Ele. 30 - MAX. ABS. : 6.0809e+00
Ele. 31 - MAX. ABS. : 3.8792e+00
Ele. 32 - MAX. ABS. : 8.1535e-01
Ele. 33 - MAX. ABS. : 2.8083e+00
Ele. 34 - MAX. ABS. : 9.2895e-01
Ele. 35 - MAX. ABS. : 1.9604e+00
Ele. 36 - MAX. ABS. : 6.9718e-01
Ele. 37 - MAX. ABS. : 2.6238e+00
Ele. 38 - MAX. ABS. : 3.9969e-01
Ele. 39 - MAX. ABS. : 3.3180e+00
Ele. 40 - MAX. ABS. : 5.3670e-01
Ele. 41 - MAX. ABS. : 2.8866e+00
Ele. 42 - MAX. ABS. : 4.1654e-01
Ele. 43 - MAX. ABS. : 2.1998e+00
Ele. 44 - MAX. ABS. : 5.2500e-01
Ele. 45 - MAX. ABS. : 2.4449e+00
Ele. 46 - MAX. ABS. : 7.0791e-01
Ele. 47 - MAX. ABS. : 3.5728e+00
Ele. 48 - MAX. ABS. : 1.0567e+00
Ele. 49 - MAX. ABS. : 4.0197e+00
Ele. 50 - MAX. ABS. : 2.7011e+00
Ele. 51 - MAX. ABS. : 4.8024e+00
Ele. 52 - MAX. ABS. : 1.2198e+00
Ele. 53 - MAX. ABS. : 4.1695e+00
Ele. 54 - MAX. ABS. : 7.8073e-01
Ele. 55 - MAX. ABS. : 2.8144e+00
Ele. 56 - MAX. ABS. : 9.9628e-01
Ele. 57 - MAX. ABS. : 2.2635e+00
Ele. 58 - MAX. ABS. : 7.5079e-01
Ele. 59 - MAX. ABS. : 2.9301e+00
Ele. 60 - MAX. ABS. : 9.8772e-01
Ele. 61 - MAX. ABS. : 5.1827e+00


elements Strain in X Dir. vs Time for Bridge Structure During Seismic Analysis
1e−5
Element Strain [mm]
Time [s]
3
2
1
0
−1
−2
−3
0.0
2.5
5.0
7.5
10.0
Ele. 1 - MAX. ABS. : 1.7623e-05
Ele. 2 - MAX. ABS. : 1.7403e-05
Ele. 3 - MAX. ABS. : 2.2903e-05
Ele. 4 - MAX. ABS. : 2.2372e-05
Ele. 5 - MAX. ABS. : 1.2812e-05
Ele. 6 - MAX. ABS. : 1.3196e-05
Ele. 7 - MAX. ABS. : 1.8275e-05
Ele. 8 - MAX. ABS. : 1.8975e-05
Ele. 9 - MAX. ABS. : 1.9425e-05
Ele. 10 - MAX. ABS. : 2.0195e-05
Ele. 11 - MAX. ABS. : 1.5714e-05
Ele. 12 - MAX. ABS. : 1.4588e-05
Ele. 13 - MAX. ABS. : 1.3077e-05
Ele. 14 - MAX. ABS. : 1.3844e-05
Ele. 15 - MAX. ABS. : 7.3098e-06
Ele. 16 - MAX. ABS. : 7.4236e-06
Ele. 17 - MAX. ABS. : 2.5164e-05
Ele. 18 - MAX. ABS. : 2.7650e-05
Ele. 19 - MAX. ABS. : 2.2208e-05
Ele. 20 - MAX. ABS. : 2.2429e-05
Ele. 21 - MAX. ABS. : 2.4666e-05
Ele. 22 - MAX. ABS. : 2.4410e-05
Ele. 23 - MAX. ABS. : 1.6609e-05
Ele. 24 - MAX. ABS. : 1.6485e-05
Ele. 25 - MAX. ABS. : 2.1714e-05
Ele. 26 - MAX. ABS. : 2.1325e-05
Ele. 27 - MAX. ABS. : 2.2609e-05
Ele. 28 - MAX. ABS. : 2.3991e-05
Ele. 29 - MAX. ABS. : 3.0786e-05
Ele. 30 - MAX. ABS. : 3.0405e-05
Ele. 31 - MAX. ABS. : 1.9396e-05
Ele. 32 - MAX. ABS. : 4.0767e-06
Ele. 33 - MAX. ABS. : 1.4041e-05
Ele. 34 - MAX. ABS. : 4.6448e-06
Ele. 35 - MAX. ABS. : 9.8022e-06
Ele. 36 - MAX. ABS. : 3.4859e-06
Ele. 37 - MAX. ABS. : 1.3119e-05
Ele. 38 - MAX. ABS. : 1.9985e-06
Ele. 39 - MAX. ABS. : 1.6590e-05
Ele. 40 - MAX. ABS. : 2.6835e-06
Ele. 41 - MAX. ABS. : 1.4433e-05
Ele. 42 - MAX. ABS. : 2.0827e-06
Ele. 43 - MAX. ABS. : 1.0999e-05
Ele. 44 - MAX. ABS. : 2.6250e-06
Ele. 45 - MAX. ABS. : 1.2224e-05
Ele. 46 - MAX. ABS. : 3.5396e-06
Ele. 47 - MAX. ABS. : 1.7864e-05
Ele. 48 - MAX. ABS. : 5.2835e-06
Ele. 49 - MAX. ABS. : 2.0098e-05
Ele. 50 - MAX. ABS. : 1.3505e-05
Ele. 51 - MAX. ABS. : 2.4012e-05
Ele. 52 - MAX. ABS. : 6.0990e-06
Ele. 53 - MAX. ABS. : 2.0847e-05
Ele. 54 - MAX. ABS. : 3.9036e-06
Ele. 55 - MAX. ABS. : 1.4072e-05
Ele. 56 - MAX. ABS. : 4.9814e-06
Ele. 57 - MAX. ABS. : 1.1318e-05
Ele. 58 - MAX. ABS. : 3.7539e-06
Ele. 59 - MAX. ABS. : 1.4650e-05
Ele. 60 - MAX. ABS. : 4.9386e-06
Ele. 61 - MAX. ABS. : 2.5913e-05

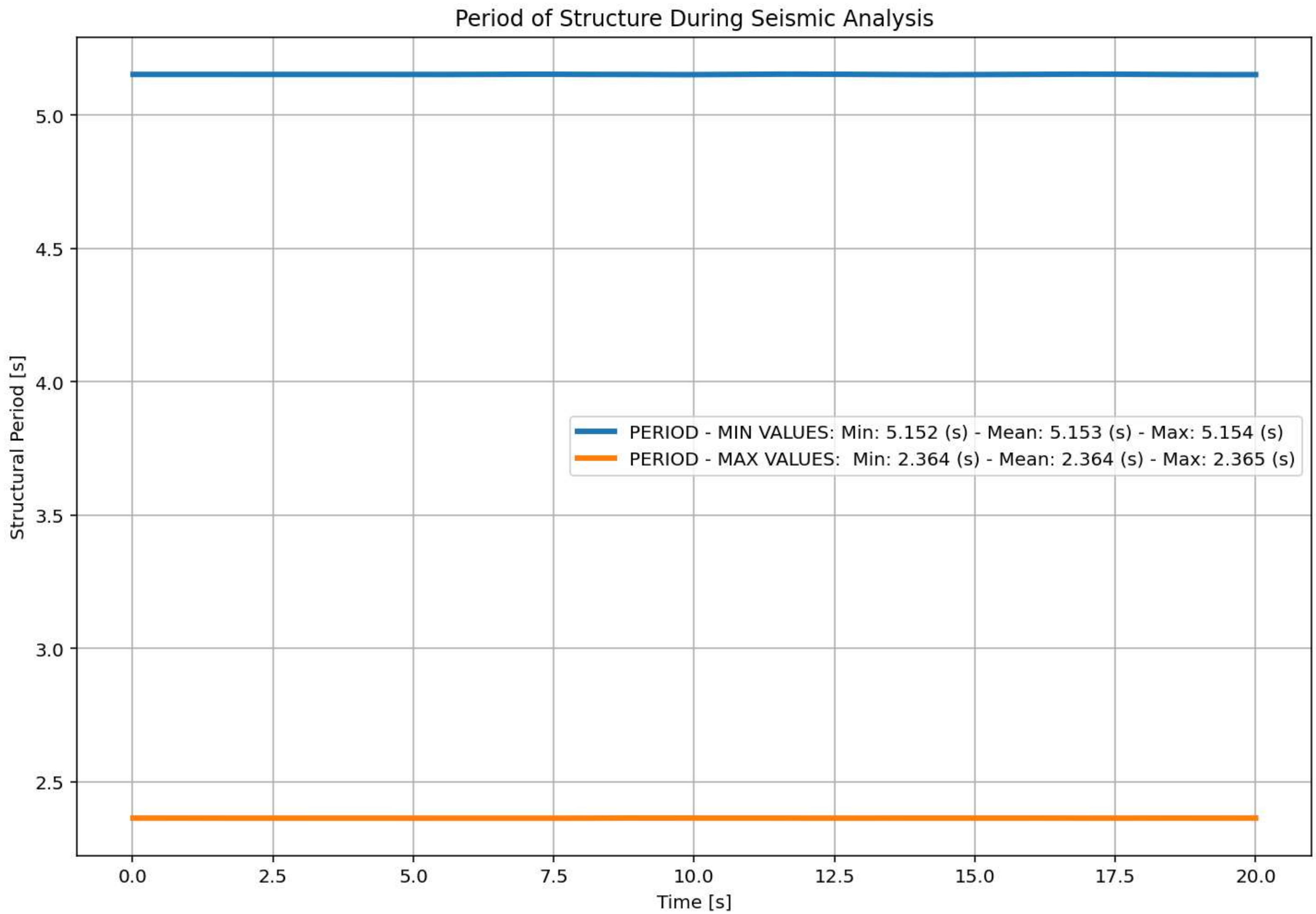


elements Damage-index [%] in X Dir. vs Time for Bridge Structure During Seismic Analysis
1e−5
Element Damage-index [%]
Time [s]
Ele. 1 - MAX. ABS. : 1.7623e-05
Ele. 2 - MAX. ABS. : 1.7403e-05
Ele. 3 - MAX. ABS. : 2.2903e-05
Ele. 4 - MAX. ABS. : 2.2372e-05
Ele. 5 - MAX. ABS. : 1.2812e-05
Ele. 6 - MAX. ABS. : 1.3196e-05
Ele. 7 - MAX. ABS. : 1.8275e-05
Ele. 8 - MAX. ABS. : 1.8975e-05
Ele. 9 - MAX. ABS. : 1.9425e-05
Ele. 10 - MAX. ABS. : 2.0195e-05
Ele. 11 - MAX. ABS. : 1.5714e-05
Ele. 12 - MAX. ABS. : 1.4588e-05
Ele. 13 - MAX. ABS. : 1.3077e-05
Ele. 14 - MAX. ABS. : 1.3844e-05
Ele. 15 - MAX. ABS. : 7.3098e-06
Ele. 16 - MAX. ABS. : 7.4236e-06
Ele. 17 - MAX. ABS. : 2.5164e-05
Ele. 18 - MAX. ABS. : 2.7650e-05
Ele. 19 - MAX. ABS. : 2.2208e-05
Ele. 20 - MAX. ABS. : 2.2429e-05
Ele. 21 - MAX. ABS. : 2.4666e-05
Ele. 22 - MAX. ABS. : 2.4410e-05
Ele. 23 - MAX. ABS. : 1.6609e-05
Ele. 24 - MAX. ABS. : 1.6485e-05
Ele. 25 - MAX. ABS. : 2.1714e-05
Ele. 26 - MAX. ABS. : 2.1325e-05
Ele. 27 - MAX. ABS. : 2.2609e-05
Ele. 28 - MAX. ABS. : 2.3991e-05
Ele. 29 - MAX. ABS. : 3.0786e-05
Ele. 30 - MAX. ABS. : 3.0405e-05
Ele. 31 - MAX. ABS. : 1.9396e-05
Ele. 32 - MAX. ABS. : 4.0767e-06
Ele. 33 - MAX. ABS. : 1.4041e-05
Ele. 34 - MAX. ABS. : 4.6448e-06
Ele. 35 - MAX. ABS. : 9.8022e-06
Ele. 36 - MAX. ABS. : 3.4859e-06
Ele. 37 - MAX. ABS. : 1.3119e-05
Ele. 38 - MAX. ABS. : 1.9985e-06
Ele. 39 - MAX. ABS. : 1.6590e-05
Ele. 40 - MAX. ABS. : 2.6835e-06
Ele. 41 - MAX. ABS. : 1.4433e-05
Ele. 42 - MAX. ABS. : 2.0827e-06
Ele. 43 - MAX. ABS. : 1.0999e-05
Ele. 44 - MAX. ABS. : 2.6250e-06
Ele. 45 - MAX. ABS. : 1.2224e-05
Ele. 46 - MAX. ABS. : 3.5396e-06
Ele. 47 - MAX. ABS. : 1.7864e-05
Ele. 48 - MAX. ABS. : 5.2835e-06
Ele. 49 - MAX. ABS. : 2.0098e-05
Ele. 50 - MAX. ABS. : 1.3505e-05
Ele. 51 - MAX. ABS. : 2.4012e-05
Ele. 52 - MAX. ABS. : 6.0990e-06
Ele. 53 - MAX. ABS. : 2.0847e-05
Ele. 54 - MAX. ABS. : 3.9036e-06
Ele. 55 - MAX. ABS. : 1.4072e-05
Ele. 56 - MAX. ABS. : 4.9814e-06
Ele. 57 - MAX. ABS. : 1.1318e-05
Ele. 58 - MAX. ABS. : 3.7539e-06
Ele. 59 - MAX. ABS. : 1.4650e-05
Ele. 60 - MAX. ABS. : 4.9386e-06
Ele. 61 - MAX. ABS. : 2.5913e-05

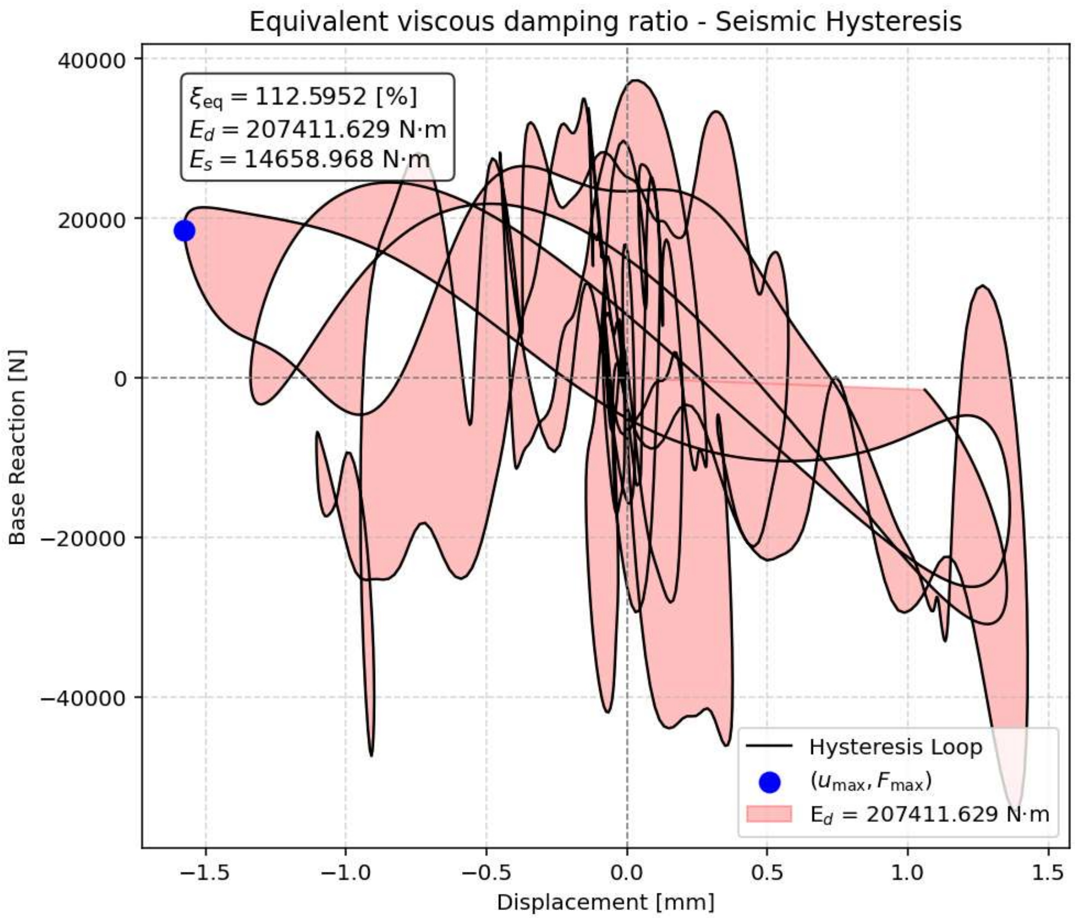

Node Displacements in X Dir. vs Time for Bridge Structure During Seismic Analysis
Displacement [mm]
Time [s]
0.4
0.2
0.0
−0.2
−0.4
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 9.2302e-02
Node 3 - MAX. ABS. : 1.8346e-01
Node 4 - MAX. ABS. : 2.4952e-01
Node 5 - MAX. ABS. : 3.6437e-01
Node 6 - MAX. ABS. : 3.7758e-01
Node 7 - MAX. ABS. : 3.8598e-01
Node 8 - MAX. ABS. : 2.9043e-01
Node 9 - MAX. ABS. : 2.0751e-01
Node 10 - MAX. ABS. : 1.0577e-01
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 8.2302e-02
Node 13 - MAX. ABS. : 1.5862e-01
Node 14 - MAX. ABS. : 1.0215e-01
Node 15 - MAX. ABS. : 7.7147e-02
Node 16 - MAX. ABS. : 3.8875e-02
Node 17 - MAX. ABS. : 0.0000e+00
Node 18 - MAX. ABS. : 5.3306e-01
Node 19 - MAX. ABS. : 4.3020e-01
Node 20 - MAX. ABS. : 3.0786e-01
Node 21 - MAX. ABS. : 1.8966e-01
Node 22 - MAX. ABS. : 8.5694e-02
Node 23 - MAX. ABS. : 1.2459e-01
Node 24 - MAX. ABS. : 2.5786e-01
Node 25 - MAX. ABS. : 3.4676e-01
Node 26 - MAX. ABS. : 4.4212e-01
Node 27 - MAX. ABS. : 4.1553e-01
Node 28 - MAX. ABS. : 3.0302e-01
Node 29 - MAX. ABS. : 1.6859e-01
Node 30 - MAX. ABS. : 8.5530e-02
Node 31 - MAX. ABS. : 1.8346e-01
Node 32 - MAX. ABS. : 2.8730e-01



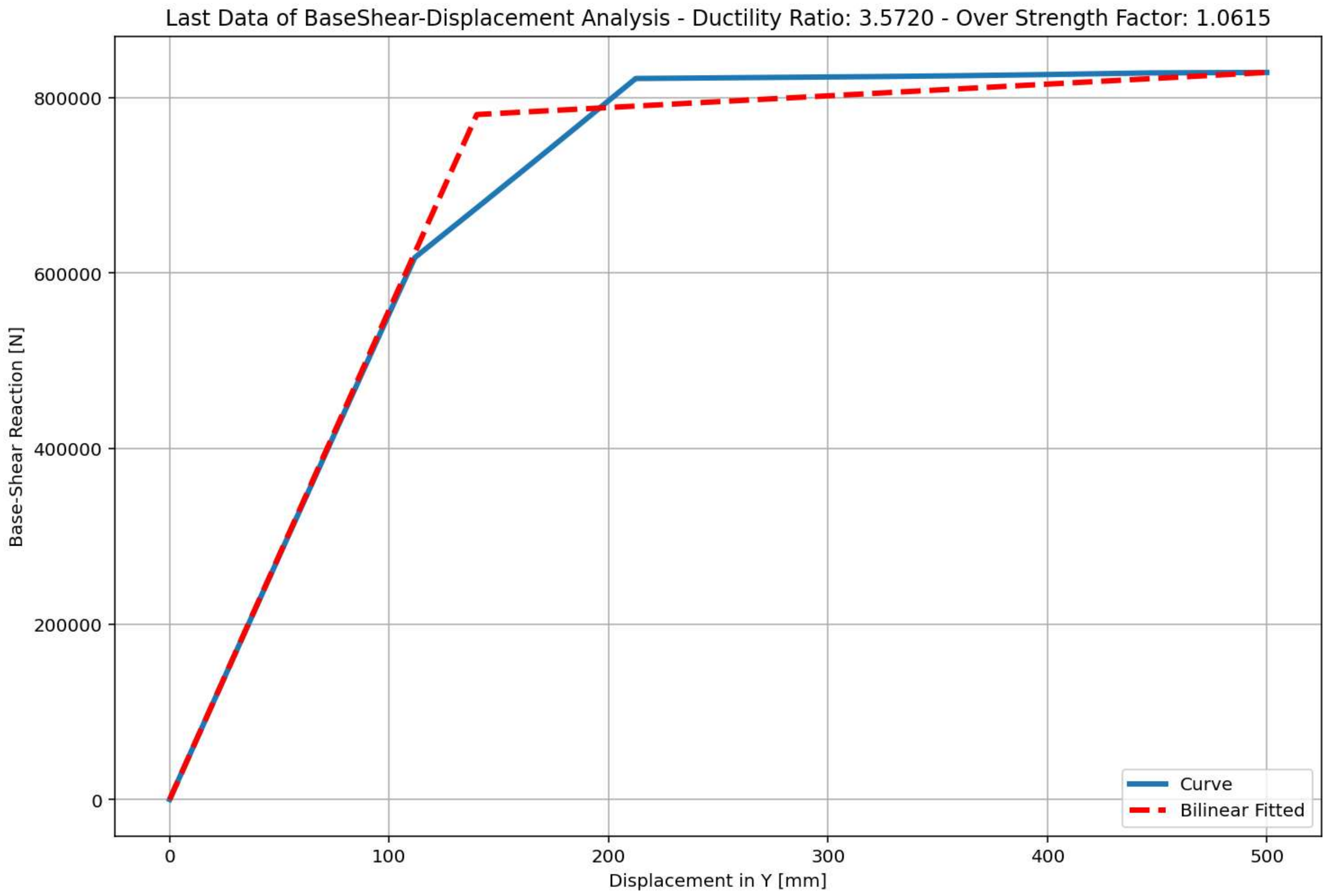
Node Displacements in Y Dir. vs Time for Bridge Structure During Seismic Analysis
Displacement [mm]
Time [s]
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 8.9765e-01
Node 3 - MAX. ABS. : 1.5733e+00
Node 4 - MAX. ABS. : 1.8408e+00
Node 5 - MAX. ABS. : 1.7571e+00
Node 6 - MAX. ABS. : 1.5759e+00
Node 7 - MAX. ABS. : 1.7764e+00
Node 8 - MAX. ABS. : 1.6554e+00
Node 9 - MAX. ABS. : 1.2463e+00
Node 10 - MAX. ABS. : 6.2980e-01
Node 11 - MAX. ABS. : 0.0000e+00
Node 12 - MAX. ABS. : 5.3531e-01
Node 13 - MAX. ABS. : 1.0121e+00
Node 14 - MAX. ABS. : 1.1585e+00
Node 15 - MAX. ABS. : 1.1684e+00
Node 16 - MAX. ABS. : 6.8468e-01
Node 17 - MAX. ABS. : 0.0000e+00
Node 18 - MAX. ABS. : 8.9265e-01
Node 19 - MAX. ABS. : 1.5814e+00
Node 20 - MAX. ABS. : 1.8324e+00
Node 21 - MAX. ABS. : 1.7631e+00
Node 22 - MAX. ABS. : 1.5707e+00
Node 23 - MAX. ABS. : 1.7861e+00
Node 24 - MAX. ABS. : 1.6441e+00
Node 25 - MAX. ABS. : 1.2561e+00
Node 26 - MAX. ABS. : 6.2355e-01
Node 27 - MAX. ABS. : 8.1032e-02
Node 28 - MAX. ABS. : 5.1620e-01
Node 29 - MAX. ABS. : 1.0232e+00
Node 30 - MAX. ABS. : 1.1535e+00
Node 31 - MAX. ABS. : 1.1791e+00
Node 32 - MAX. ABS. : 6.7293e-01



Period of Structure During Seismic Analysis
Structural Period [s]
5.0
4.5
4.0
3.5
3.0
2.5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
PERIOD - MIN VALUES: Min: 5.152 (s) - Mean: 5.153 (s) - Max: 5.154 (s)
PERIOD - MAX VALUES: Min: 2.364 (s) - Mean: 2.364 (s) - Max: 2.365 (s)

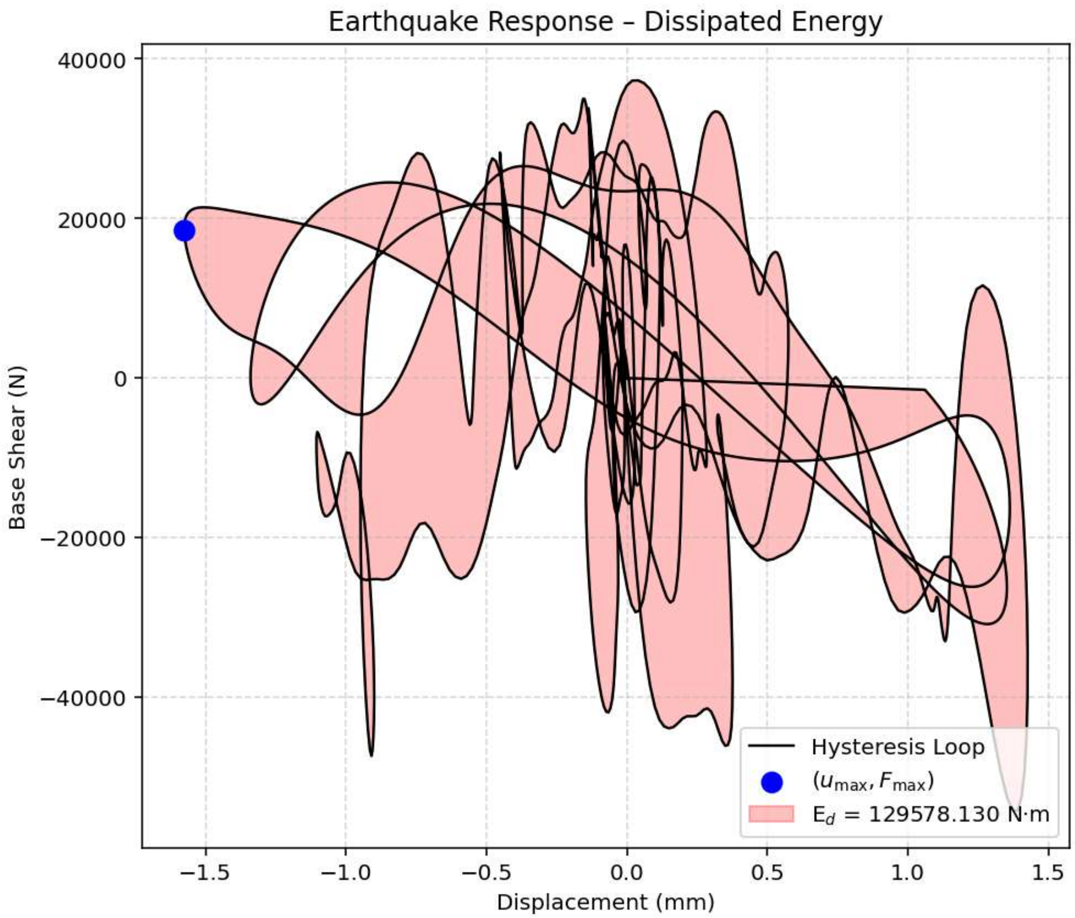


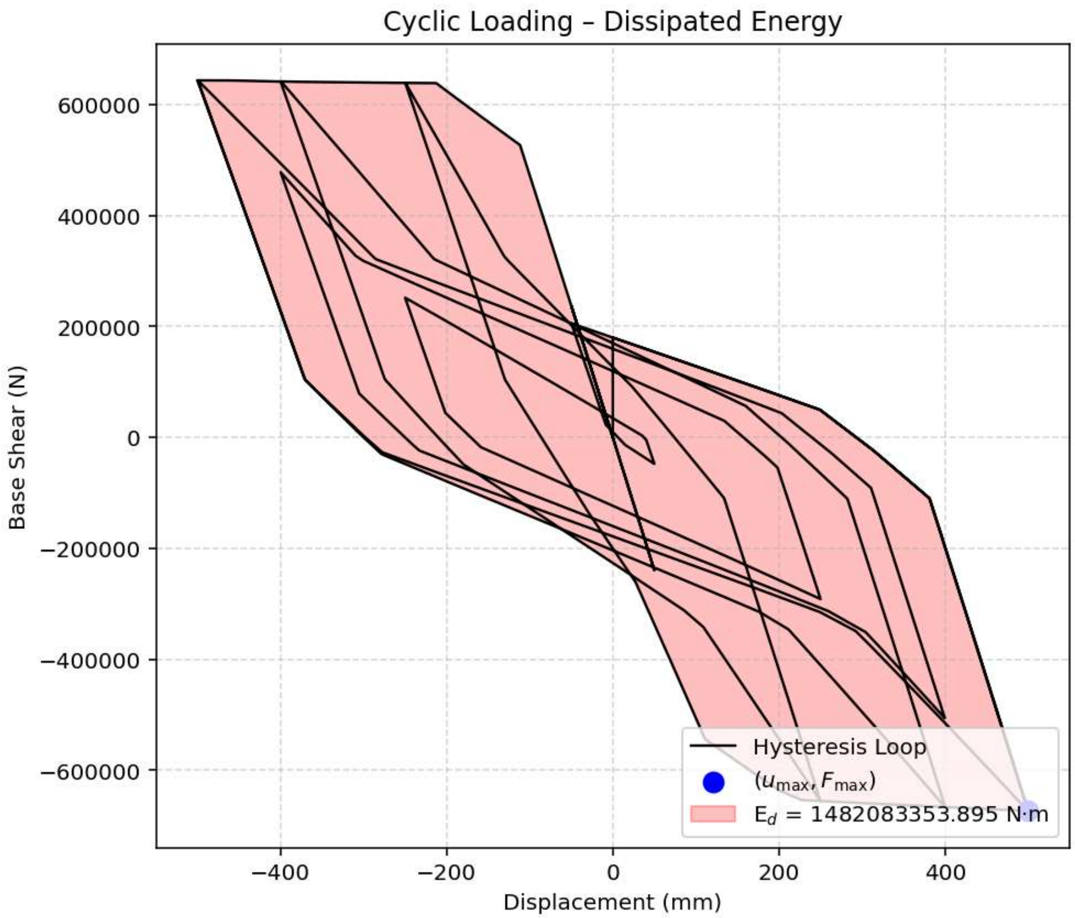
Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 112.5952$ [%]
$E_d = 207411.629$ N·m
$E_s = 14658.968$ N·m
Hysteresis Loop
(u_{max}, F_{max})
E$_d$ = 207411.629 N·m
Base Reaction [N]
Displacement [mm]
40000
20000
0
−20000
−40000
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5



Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 3.5720 - Over Strength Factor: 1.0615
Base-Shear Reaction [N]
800000
600000
400000
200000
0
0
100
200
300
400
500
Displacement in Y [mm]
Curve
Bilinear Fitted

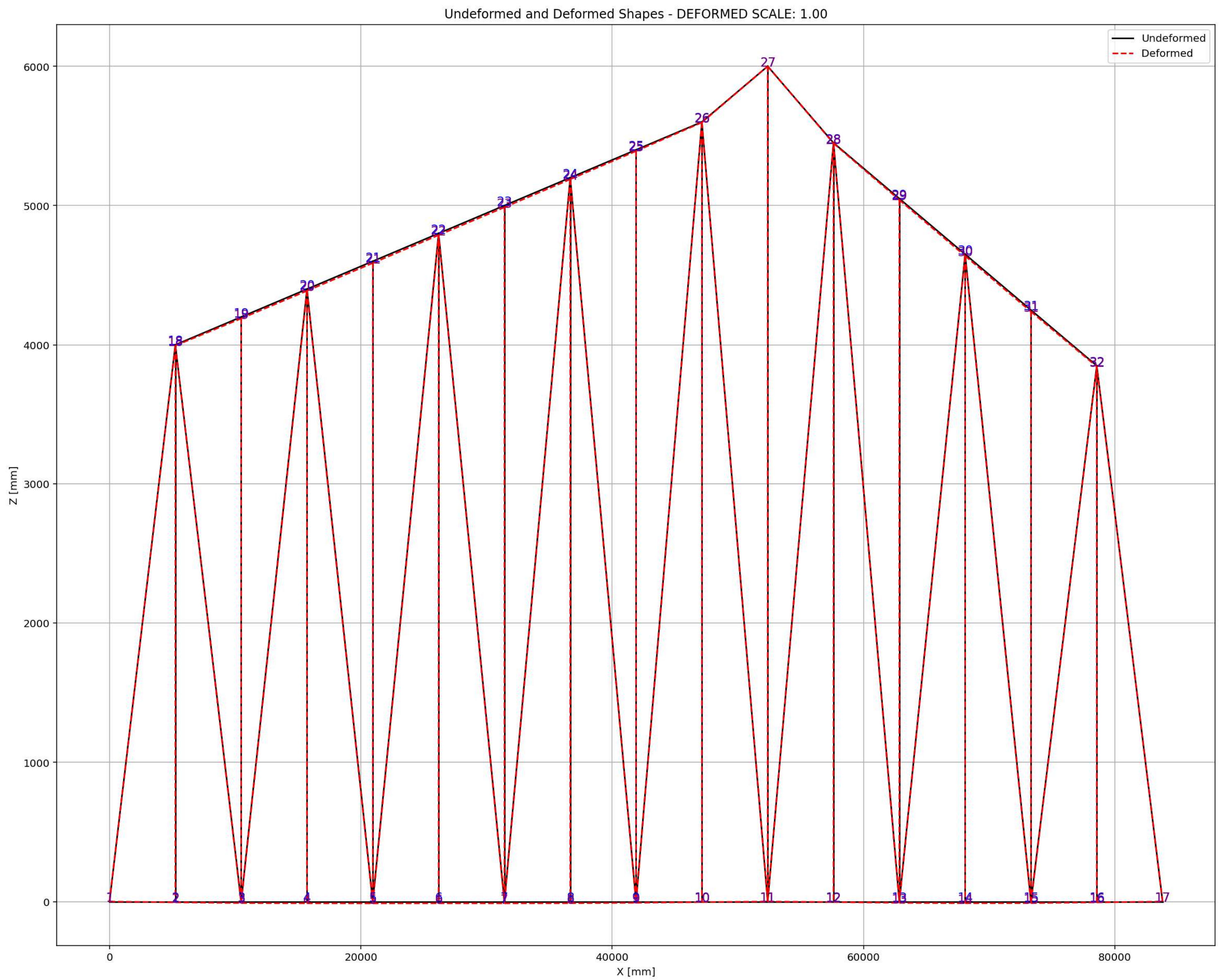
```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
----------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.39977481e+02 7.80605664e+05]
 [5.00000000e+02 8.28585660e+05]]
+==========================+
+--------------------------------------------------+
 Structure Elastic Stiffness :    5576.65
 Structure Plastic Stiffness :    1657.17
 Structure Tangent Stiffness :    133.27
 Structure Ductility Ratio :      3.57
 Structure Over Strength Factor:   1.06
+--------------------------------------------------+
Over Strength Coefficient (Ω0):      1.0615
Displacement Ductility Ratio (μ):    3.5720
Ductility Coefficient (Rμ):          3.5720
Structural Behavior Coefficient (R): 3.7916
Structural Ductility Damage Index in Y Direction: -38.4425 (%)
```


Earthquake Response – Dissipated Energy
40000
20000
0
−20000
−40000
Base Shear (N)
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Displacement (mm)
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 129578.130 N·m


Cyclic Loading – Dissipated Energy
Base Shear (N)
Displacement (mm)
600000
400000
200000
0
−200000
−400000
−600000
−400
−200
0
200
400
Hysteresis Loop
(u_{max}, F_{max})
E_d = 1482083353.895 N·m

Dissipated Energy from Earthquake= 129578.13 N·mm
Dissipated Energy from Cyclic Displacement= 1482083353.89 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.009 [%]
ZONE 1: VERY MINOR DAMAGE


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
0
20000
40000
60000
80000
1000
2000
3000
4000
5000
6000
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
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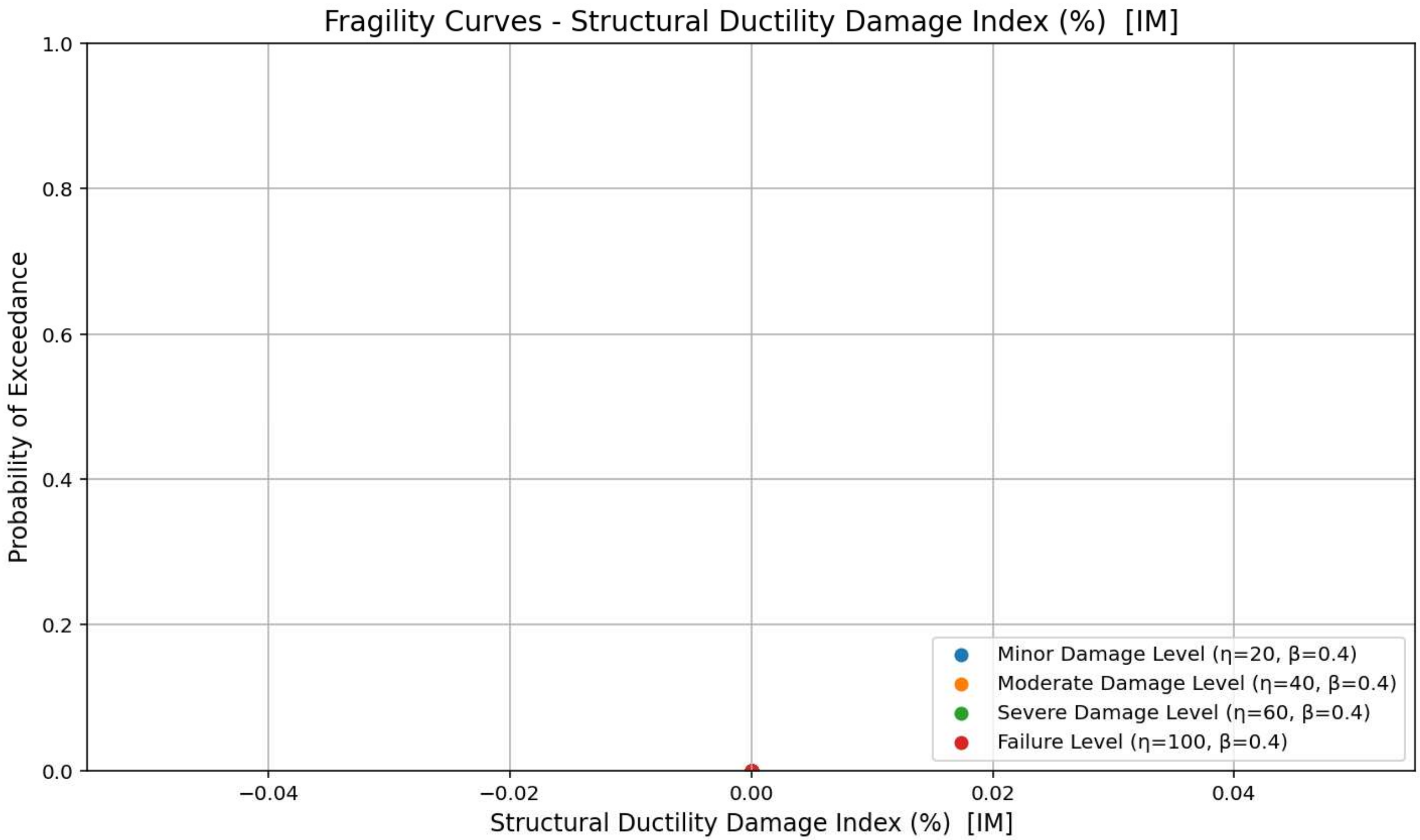
27
28
29
30
31
32



Fragility Curves - Structural Ductility Damage Index (%) [IM]
Probability of Exceedance
Structural Ductility Damage Index (%) [IM]
0.0
0.2
0.4
0.6
0.8
1.0
−0.04
−0.02
0.00
0.02
0.04
Minor Damage Level (η=20, β=0.4)
Moderate Damage Level (η=40, β=0.4)
Severe Damage Level (η=60, β=0.4)
Failure Level (η=100, β=0.4)